

Contents

1. Overview	13
1.1 Scope	13
1.2 Purpose	13
1.3 Word usage	13
1.4 Machine-readable YANG modules	14
1.5 Summary of YANG-based management framework	14
1.6 Security considerations	14
1.7 YANG module syntax validation	14
2. Normative references	16
3. Definitions	17
4. Abbreviations	18
5. Ethernet YANG modules	19
5.1 YANG module structure	19
5.2 Mapping of IEEE Std 802.3, Clause 30 managed objects	19
5.3 YANG module definition	33
5.3.1 Tree hierarchy	33
5.3.1.1 ieee802-ethernet-interface	33
5.3.1.2 ieee802-ethernet-interface-half-duplex	34
5.3.1.3 ieee802-ethernet-mac-merge	35
5.3.1.4 ieee802-ethernet-lldp	35
5.3.1.5 ieee802-ethernet-phy	38
5.3.2 YANG modules	39
5.3.2.1 Ethernet interface module	39
5.3.2.2 Ethernet interface (half-duplex) module	54
5.3.2.3 Ethernet MAC merge module	59
5.3.2.4 Ethernet LLDP module	63
5.3.2.5 Ethernet PHY module	99
5.3.2.6 Ethernet PHY type module	111
6. YANG module for Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) and Power over Data Lines (PoDL)	162
6.1 Introduction	162
6.2 YANG module structure	162
6.3 Security considerations for the Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) and Power over Data Lines module	162
6.4 Mapping of IEEE Std 802.3, Clause 30 managed objects	162
6.5 YANG module definition	165
6.5.1 Tree hierarchy	165
6.5.1.1 ieee802-ethernet-pse	165
6.5.1.2 ieee802-ethernet-pse-2	165
6.5.2 YANG module	166
6.5.2.1 Ethernet PSE module	166
6.5.2.2 Ethernet PSE-2 Module	178

7. YANG module for Ethernet passive optical network (EPON).....	190
7.1 Introduction.....	190
7.2 YANG module structure.....	190
7.3 Mapping of IEEE Std 802.3, Clause 30 managed objects.....	190
7.4 YANG module definition.....	196
7.4.1 Tree hierarchy.....	196
7.4.1.1 ieee802-ethernet-pon.....	196
7.4.2 YANG module.....	199
8. YANG module for Ethernet link OAM.....	237
8.1 Introduction.....	237
8.2 Overview.....	237
8.2.1 Remote fault indication.....	237
8.2.2 Link monitoring.....	237
8.2.3 Remote loopback.....	237
8.2.4 Ethernet OAM protocol data units.....	238
8.3 Security considerations for the Ethernet operations, administration, and maintenance (OAM) module.....	238
8.4 Mapping of IEEE 802.3 managed objects.....	239
8.5 YANG module definition.....	244
8.5.1 Tree hierarchy.....	244
8.5.1.1 ieee802-ethernet-link-oam.....	244
8.5.2 YANG module.....	247
 Annex A (informative) Bibliography	268 267

IEEE Standard for Ethernet YANG Data Model Definitions

1. Overview

This standard defines YANG modules for various Ethernet functions specified in IEEE Std 802.3.

1.1 Scope

This standard defines YANG data models for IEEE Std 802.3 Ethernet.

1.2 Purpose

The purpose of the standard is to define YANG modules for IEEE Std 802.3 and publish these modules in a machine-readable format.

1.3 Word usage

The word *shall* indicates mandatory requirements strictly to be followed in order to conform to the standard and from which no deviation is permitted (*shall* equals *is required to*).^{6,7}

The word *should* indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should* equals *is recommended that*).

The word *may* is used to indicate a course of action permissible within the limits of the standard (*may* equals *is permitted to*).

The word *can* is used for statements of possibility and capability, whether material, physical, or causal (*can* equals *is able to*).

⁶ The use of the word *must* is deprecated and cannot be used when stating mandatory requirements, *must* is used only to describe unavoidable situations.

⁷ The use of *will* is deprecated and cannot be used when stating mandatory requirements, *will* is only used in statements of fact.

1.4 Machine-readable YANG modules

The authoritative machine-readable YANG files are attached to this standard. A copy is also available for download at the following URL: <https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3> as text files with a *.yang* extension, for example, *ieee802-ethernet-interface.yang*. The use of specialized tools to view YANG modules may be useful to create tree, UML image, and HTML outputs from the YANG modules.

Like other languages, YANG (see IETF RFC 7950) has an accepted style for machine-readable files, which was followed during the development of this standard. This formatting may not be preserved when importing the machine-readable YANG modules into the PDF. In case of any discrepancies, the machine-readable files attached to this standard take precedence.

1.5 Summary of YANG-based management framework

The structure of YANG-based management framework closely resembles the structure of the Internet-Standard Management Framework, described in detail in section 7 of IETF RFC 3410.

Managed objects defined using the YANG modeling language are hosted on the managed device and accessed through NETCONF (see IETF RFC 7803) or RESTCONF (see IETF RFC 8040). This standard specifies YANG modules that are compliant to YANG 1.1 (see IETF RFC 7950).

1.6 Security considerations

The YANG modules defined in this standard are designed to be accessed via network management protocols, including NETCONF (see IETF RFC 7803) or RESTCONF (see IETF RFC 8040). The lowest NETCONF layer is the secure transport layer, and the mandatory-to-implement secure transport is Secure Shell (SSH) (see IETF RFC 6242) or TLS (see IETF RFC 8446). The lowest RESTCONF layer is HTTPS, and the mandatory-to-implement secure transport is TLS (see IETF RFC 8446).

The NETCONF access control model (see IETF RFC 8341) provides the means to restrict access for particular NETCONF or RESTCONF users to a pre-configured subset of all available NETCONF or RESTCONF protocol operations and content.

There are a number of data nodes defined in these YANG modules that are writable/creatable/deletable, that is, have the *config* property set to true, which is the default setting. These data nodes can be considered sensitive or vulnerable in some network environments. Write operations (e.g., *edit-config*) to these data nodes without proper protection can have a negative effect on network operations.

Some of the readable data nodes in these YANG modules can be considered sensitive or vulnerable in some network environments. It is thus important to control read access (e.g., via *get*, *get-config*, or *notification*) to these data nodes.

Some of the RPC operations in these YANG modules can be considered sensitive or vulnerable in some network environments. Therefore, it is important to control access to these operations.

1.7 YANG module syntax validation

All YANG modules included in this standard are YANG 1.1 (see IETF RFC 7950) compliant and pass automated checks using tools available at the time of publication.



The following open source and/or free versions of YANG validation tools may be used: Pyang (see <https://github.com/mbj4668/pyang>), ~~ConfD (see <http://www.tail-f.com/confd-basics>),~~ as well as other YANG model validation tools listed at <https://www.yangvalidator.com>.

5. Ethernet YANG modules

5.1 YANG module structure

Six modules defined in this clause are focused on the configuration and monitoring of IEEE Std 802.3 Ethernet interfaces.

The *ieee802-ethernet-interface* YANG module contains definitions of current attributes used widely in the industry in current products.

The *ieee802-ethernet-interface-half-duplex* YANG module contains definitions of half-duplex attributes.

The *ieee802-ethernet-lldp* YANG module contains definitions for configuring LLDP for IEEE Std 802.3 compliant interfaces.

The *ieee802-ethernet-mac-merge* module contains definitions for configuration of MAC Merge for IEEE Std 802.1Qcw [B2] frame preemption.

The *ieee802-ethernet-phy* YANG module contains definitions for Ethernet Phys, derived from IANA-MAU-MIB REVISION 202309070000Z, the YANG and definitions correct as per release of 2024-11-09.

The *ieee802-ethernet-phy-type* YANG module contains definitions for Ethernet Phy types, derived from IANA-MAU-MIB REVISION 202309070000Z using smidump, the YANG and definitions are correct as per 2024-11-09.

NOTE—The *ieee802-ethernet-phy-type* module may, in future, be maintained by an external organization (IANA).¹²

This standard does not have a normative requirement for data nodes of the base ietf-interfaces YANG module, but the following data nodes are supported: name, description, type, enabled, admin-status, oper-status, if-index, and phys-address.

5.2 Mapping of IEEE Std 802.3, Clause 30 managed objects

This subclause contains the mapping between YANG data nodes included in the *ieee802-ethernet-interface* (see Table 5–1), *ieee802-ethernet-interface-half-duplex* (see Table 5–4), *ieee802-ethernet-mac-merge* (see Table 5–5), and *ieee802-ethernet-lldp* (see Table 5–6) YANG modules, and the managed objects and attributes defined in IEEE Std 802.3, Clause 30.

¹² Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement this standard.

Table 5–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oAutoNegotiation	acAutoNegAdminControl	30.6.1.2.2	interfaces/interface/ethernet/	auto-negotiation/enable	R/W
	aAutoNegAutoConfig	30.6.1.1.4		negotiation-status	R
N/A	N/A			flow-control/pause/direction	R/W
oMACControl-FunctionEntity	aPAUSEMACCtrlFramesReceived	30.3.4.3		flow-control/pause/statistics/in-frames-pause	R
	aPAUSEMACCtrlFramesTransmitted	30.3.4.2		flow-control/pause/statistics/out-frames-pause	R
N/A	dot3HCOutPFCFrames				
N/A	N/A			flow-control/force-flow-control	R/W
N/A	N/A			speed	R/W
oMACEntity	aDuplexStatus	30.3.1.1.32		duplex	R/W
	aMaxFrameLength	30.3.1.1.37		max-frame-length	R
	aSlowProtocolFrameLimit	30.3.1.1.38		frame-limit-slow-protocol	R
oEXTENSION	aEXTENSIONMACCtrlStatus	30.3.8.3		mac-control-extension-control	R
N/A	N/A			capabilities/auto-negotiation	R

Table 5–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oMACEntity	aFramesReceivedOK	30.3.1.1.5	interfaces/interface/ethernet/statistics/frame	in-frames	R
	aMulticastFramesReceivedOK	30.3.1.1.21		in-multicast-frames	R
	aBroadcastFramesReceivedOK	30.3.1.1.22		in-broadcast-frames	R
	aFrameCheckSequenceErrors + aAlignmentErrors	30.4.3.1.6, 30.4.3.1.7		in-error-fcs-frames	R
oMACEntity	aFrameTooLongErrors	30.3.1.1.25		in-error-oversize-frames	R
	aFramesLostDueToIntMACRcvError	30.3.1.1.15		in-error-mac-internal-frames	R
	aFramesTransmittedOK	30.3.1.1.2		out-frames	R
	aMulticastFramesXmittedOK	30.3.1.1.18		out-multicast-frames	R
	aBroadcastFramesXmittedOK	30.3.1.1.19		out-broadcast-frames	R
	aFramesLostDueToIntMACXmitError	30.3.1.1.12		out-error-mac-internal-frames	R
oPHYEntity	aSymbolErrorDuringCarrier	30.3.2.1.5	interfaces/interface/ethernet/statistics/phy	in-error-symbol	R
	aReceiveLPITransitions	30.3.2.1.11	interfaces/interface/ethernet/statistics/phy/lpi	in-lpi-transitions	R
	aReceiveLPIMicroseconds	30.3.2.1.9		in-lpi-time	R
	aTransmitLPITransitions	30.3.2.1.10		out-lpi-transitions	R
	aTransmitLPIMicroseconds	30.3.2.1.8		out-lpi-time	R

Table 5–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oMACControlEntity	aUnsupportedOpcodesReceived	30.3.3.5	interfaces/interface/ethernet/statistics/mac-control	in-frames-mac-control-unknown	R
oEXTENSION	aEXTENSIONMACCtrlFramesReceived	30.3.8.2		in-frames-mac-control-extension	R
	aEXTENSIONMACCtrlFramesTransmitted	30.3.8.1		out-frames-mac-control-extension	R

Table 5–2—Mapping between IETF RFC 2819 [B3] managed objects and *ieee802-ethernet-interface* YANG data nodes

IETF RFC 2819 [B3] Attribute(s)	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
	Container(s)	Data node(s)	R/W
no direct object ^a	interfaces/interface/ethernet/statistics/frame	in-total-frames	R
etherStatsOctets		in-total-octets	R
etherStatsUndersizePkts + etherStatsFragments		in-error-undersize-frames	R

^a Can be calculated as: aFramesReceivedOK + aFrameCheckSequenceErrors + aAlignmentErrors + aFrameTooLongErrors + aFramesLostDueToIntMACRevError.

Table 5–3—Mapping between IETF RFC 3635 [B5] managed objects and *ieee802-ethernet-interface* YANG data nodes

ETHERLIKE MIB Attribute(s)	Corresponding <i>ieee802-ethernet-interface</i> YANG data nodes		
	Container(s)	Data node(s)	R/W
dot3HCInPFCFrames	interfaces/interface/ethernet/	flow-control/pfc {ethernet-pfc} /statistics/in-frames-pfc	R
dot3HCOutPFCFrames		flow-control/pfc {ethernet-pfc} /statistics/out-frames-pfc	R

Table 5–4—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-interface-half-duplex* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-interface-half-duplex</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oMACEntity	aRateControlAbility	30.3.1.1.33	interfaces/interface/ethernet	dynamic-rate-control	R/W
			interfaces/interface/ethernet/capability	dynamic-rate-control-supported	R
oPHYEntity	aSQETestErrors	30.3.2.1.4	interfaces/interface/ethernet/statistics/frame/csmacd{csma-cd}	in-errors-sqe-test	R
oMACEntity	aSingleCollisionFrames	30.3.1.1.3		out-frames-collision-single	R
	aMultipleCollisionFrames	30.3.1.1.4		out-frames-collision-multiple	R
	aFramesWithDeferredXmissions	30.3.1.1.9		out-frames-deferred	R
	aFramesAbortedDueToXSColls	30.3.1.1.11		out-frames-collisions-excessive	R
	aLateCollisions	30.3.1.1.10		out-collisions-late	R
	aCarrierSenseErrors	30.3.1.1.13		out-errors-carrier-sense	R
	aCollisionFrames	30.3.1.1.30		collision-histogram/collision-count	R
collision-histogram/collision-count-frames				R	

Table 5–5—Mapping between IEEE Std 802.3, 30.14 managed objects and *ieee802-ethernet-mac-merge* YANG data nodes

IEEE Std 802.3, 30.14		Reference	Corresponding <i>ieee802-ethernet-mac-merge</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oMacMergeEntity	aMACMergeSupport	30.14.1.1	Interfaces/interface/ethernet/mac-merge/admin-status	merge-support	R
	aMACMergeStatusVerify	30.14.1.2		verify-status	R
	aMACMergeStatusTx	30.14.1.5		status-tx	R
	aMACMergeEnableTx	30.14.1.3	Interfaces/interface/ethernet/mac-merge/admin-control	merge-enable-tx	R/W
	aMACMergeVerifyDisableTx	30.14.1.4		verify-disable-tx	R/W
	aMACMergeVerifyTime	30.14.1.6		verify-time	R/W
	aMACMergeAddFragSize	30.14.1.7		frag-size	R/W
	aMACMergeFrameAssErrorCount	30.14.1.8	Interfaces/interface/ethernet/mac-merge/statistics	assembly-error-count	R
	aMACMergeFrameSmdErrorCount	30.14.1.9		smd-error-count	R
	aMACMergeFrameAssOkCount	30.14.1.10		assembly-ok-count	R
	aMACMergeFragCountRx	30.14.1.11		fragment-count-rx	R
	aMACMergeFragCountTx	30.14.1.12		fragment-count-tx	R
	aMACMergeHoldCount	30.14.1.13		hold-count	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-lldp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oLldpXdot3Config	aLldpXdot3PortConfigTLVsTxEnable	30.12.1.1.1	lldp/port	tlvs-port-config-enable	R/W
oLldpXdot3LocSystemsGroup	aLldpXdot3LocPortAutoNegSupported	30.12.2.1.1		auto-negotiation-supported	R
	aLldpXdot3LocPortAutoNegEnabled	30.12.2.1.2		auto-negotiation-enabled	R
	aLldpXdot3LocPortAutoNegAdvertisedCap	30.12.2.1.3		auto-negotiation-cap	R
	aLldpXdot3LocPortOperMauType	30.12.2.1.4		operational-mau-type	R
	aLldpXdot3LocPowerPortClass	30.12.2.1.5		power-port-class	R
	aLldpXdot3LocPowerMDISupported	30.12.2.1.6		mdi-power-supported	R
	aLldpXdot3LocPowerMDIEnabled	30.12.2.1.7		mdi-power-enabled	R
	aLldpXdot3LocPowerPairControllable	30.12.2.1.8		power-pair-controlable	R
	aLldpXdot3LocPowerPairs	30.12.2.1.9		power-pairs	R
	aLldpXdot3LocPowerClass	30.12.2.1.10		local-power-class	R
	aLldpXdot3LocLinkAggStatus	30.12.2.1.11		link-aggregation-status	R
	aLldpXdot3LocLinkAggPortId	30.12.2.1.12		aggregation-port-id	R
	aLldpXdot3LocMaxFrameSize	30.12.2.1.13		local-max-frame-size	R
	aLldpXdot3LocPowerType	30.12.2.1.14		power-type	R
	aLldpXdot3LocPowerSource	30.12.2.1.15		power-source	R
	aLldpXdot3LocPowerPriority	30.12.2.1.16		local-power-priority	R/W
	aLldpXdot3LocPDRequestedPowerValue	30.12.2.1.17		pd-requested-power-value	R
	aLldpXdot3LocPDRequestedPowerValueA	30.12.2.1.18		pd-requested-power-value-a	R
	aLldpXdot3LocPDRequestedPowerValueB	30.12.2.1.19		pd-requested-power-value-b	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-lldp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3LocPSEAllocatedPowerValue	30.12.2.1.20		pse-allocated-power-value	R
	aLldpXdot3LocPSEAllocatedPowerValueA	30.12.2.1.21		pse-allocated-power-value-a	R
	aLldpXdot3LocPSEAllocatedPowerValueB	30.12.2.1.22		pse-allocated-power-value-b	R
	aLldpXdot3LocPSEPoweringStatus	30.12.2.1.23		pse-powering-status	R
	aLldpXdot3LocPDPoweredStatus	30.12.2.1.24		pd-powered-status	R
	aLldpXdot3LocPowerPairsExt	30.12.2.1.25		power-pairs-ext	R
	aLldpXdot3LocPowerClassExtA	30.12.2.1.26		power-class-ext-A	R
	aLldpXdot3LocPowerClassExtB	30.12.2.1.27		power-class-ext-B	R
	aLldpXdot3LocPowerClassExt	30.12.2.1.28		power-class-ext	R
	aLldpXdot3LocPowerTypeExt	30.12.2.1.29		power-type-ext	R
	aLldpXdot3LocPDLoad	30.12.2.1.30		pd-load	R
	aLldpXdot3LocPD4PID	30.12.2.1.31		pd-4pid	R
	aLldpXdot3LocPSEMaxAvailPower	30.12.2.1.32		pse-max-avail-power	R
	aLldpXdot3LocPSEAutoclassSupport	30.12.2.1.33		pse-autoclass-support	R
	aLldpXdot3LocAutoclassCompleted	30.12.2.1.34		autoclass-completed	R
	aLldpXdot3LocAutoclassRequest	30.12.2.1.35		autoclass-request	R
	aLldpXdot3LocPowerDownRequest	30.12.2.1.36		power-down-request	R
	aLldpXdot3LocPowerDownTime	30.12.2.1.37		power-down-time	R
	aLldpXdot3LocMeasVoltageSupport	30.12.2.1.38		meas-voltage-support	R
	aLldpXdot3LocMeasCurrentSupport	30.12.2.1.39		meas-current-support	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-lldp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3LocMeasPowerSupport	30.12.2.1.40		meas-power-support	R
	aLldpXdot3LocMeasEnergySupport	30.12.2.1.41		meas-energy-support	R
	aLldpXdot3LocMeasurementSource	30.12.2.1.42		measurement-source	R
	aLldpXdot3LocMeasVoltageRequest	30.12.2.1.43		meas-voltage-request	R
	aLldpXdot3LocMeasCurrentRequest	30.12.2.1.44		meas-current-request	R
	aLldpXdot3LocMeasPowerRequest	30.12.2.1.45		meas-power-request	R
	aLldpXdot3LocMeasEnergyRequest	30.12.2.1.46		meas-energy-request	R
	aLldpXdot3LocMeasVoltageValid	30.12.2.1.47		meas-voltage-valid	R
	aLldpXdot3LocMeasCurrentValid	30.12.2.1.48		meas-current-valid	R
	aLldpXdot3LocMeasPowerValid	30.12.2.1.49		meas-power-valid	R
	aLldpXdot3LocMeasEnergyValid	30.12.2.1.50		meas-energy-valid	R
	aLldpXdot3LocMeasVoltageUncertainty	30.12.2.1.51		meas-voltage-uncertainty	R
	aLldpXdot3LocMeasCurrentUncertainty	30.12.2.1.52		meas-current-uncertainty	R
	aLldpXdot3LocMeasPowerUncertainty	30.12.2.1.53		meas-power-uncertainty	R
	aLldpXdot3LocMeasEnergyUncertainty	30.12.2.1.54		meas-energy-uncertainty	R
	aLldpXdot3LocVoltageMeasurement	30.12.2.1.55		voltage-measurement	R
	aLldpXdot3LocCurrentMeasurement	30.12.2.1.56		current-measurement	R
	aLldpXdot3LocPowerMeasurement	30.12.2.1.57		power-measurement	R
	aLldpXdot3LocEnergyMeasurement	30.12.2.1.58		energy-measurement	R
	aLldpXdot3LocPSEPowerPriceIndex	30.12.2.1.59		pse-power-price-index	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3LocResponseTime	30.12.2.1.60		local-response	R
	aLldpXdot3LocReady	30.12.2.1.61		local-system-ready	R
	aLldpXdot3LocTxTwSys	30.12.2.1.62		tx-system-value	R
	aLldpXdot3LocTxTwSysEcho	30.12.2.1.63		tx-system-value-echo	R
	aLldpXdot3LocRxTwSys	30.12.2.1.64		rx-system-value	R
	aLldpXdot3LocRxTwSysEcho	30.12.2.1.65		rx-system-value-echo	R
	aLldpXdot3LocFbTwSys	30.12.2.1.66		fallback-system-value	R
	aLldpXdot3TxDllReady	30.12.2.1.67		tx-dll-ready	R
	aLldpXdot3RxDllReady	30.12.2.1.68		rx-dll-ready	R
	aLldpXdot3LocDllEnabled	30.12.2.1.69		dll-ready	R
	aLldpXdot3LocTxFw	30.12.2.1.70		tx-system-fw	R
	aLldpXdot3LocTxFwEcho	30.12.2.1.71		tx-system-fw-echo	R
	aLldpXdot3LocRxFw	30.12.2.1.72		rx-system-fw	R
	aLldpXdot3LocRxFwEcho	30.12.2.1.73		rx-system-fw-echo	R
	aLldpXdot3LocPreemptSupported	30.12.2.1.74		preemption-supported	R
	aLldpXdot3LocPreemptEnabled	30.12.2.1.75		preemption-enabled	R
	aLldpXdot3LocPreemptActive	30.12.2.1.76		preemption-active	R
	aLldpXdot3LocAddFragSize	30.12.2.1.77		additional-fragment-size	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llDP</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oLldpXdot3RemSystemsGroup	aLldpXdot3RemPortAutoNegSupported	30.12.3.1.1	lldp/port/remote-systems-data	auto-negotiation-supported	R
	aLldpXdot3RemPortAutoNegEnabled	30.12.3.1.2		auto-negotiation-enabled	R
	aLldpXdot3RemPortAutoNegAdvertisedCap	30.12.3.1.3		auto-negotiation-cap	R
	aLldpXdot3RemPortOperMauType	30.12.3.1.4		operational-mau-type	R
	aLldpXdot3RemPowerPortClass	30.12.3.1.5		power-port-class	R
	aLldpXdot3RemPowerMDISupported	30.12.3.1.6		mdi-power-supported	R
	aLldpXdot3RemPowerMDIEnabled	30.12.3.1.7		mdi-power-enabled	R
	aLldpXdot3RemPowerPairControllable	30.12.3.1.8		power-pair-controlable	R
	aLldpXdot3RemPowerPairs	30.12.3.1.9		power-pairs	R
	aLldpXdot3RemPowerClass	30.12.3.1.10		power-class	R
	aLldpXdot3RemLinkAggStatus	30.12.3.1.11		link-aggregation-status	R
	aLldpXdot3RemLinkAggPortId	30.12.3.1.12		aggregation-port-id	R
	aLldpXdot3RemMaxFrameSize	30.12.3.1.13		local-max-frame-size	R
	aLldpXdot3RemPowerType	30.12.3.1.14		power-type	R
	aLldpXdot3RemPowerSource	30.12.3.1.15		power-source	R
	aLldpXdot3RemPowerPriority	30.12.3.1.16		power-priority	RW
	aLldpXdot3RemPDRequestedPowerValue	30.12.3.1.17		pd-requested-power-value	R
	aLldpXdot3RemPDRequestedPowerValueA	30.12.3.1.18		pd-requested-power-value-a	R
	aLldpXdot3RemPDRequestedPowerValueB	30.12.3.1.19		pd-requested-power-value-b	R
	aLldpXdot3RemPSEAllocatedPowerValue	30.12.3.1.20		pse-allocated-power-value	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llDP</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3RemPSEAllocatedPowerValueA	30.12.3.1.21		pse-allocated-power-value-a	R
	aLldpXdot3RemPSEAllocatedPowerValueB	30.12.3.1.22		pse-allocated-power-value-b	R
	aLldpXdot3RemPSEPoweringStatus	30.12.3.1.23		pse-powering-status	R
	aLldpXdot3RemPDPoweredStatus	30.12.3.1.24		pd-powered-status	R
	aLldpXdot3RemPowerPairsExt	30.12.3.1.25		power-pairs-ext	R
	aLldpXdot3RemPowerClassExtA	30.12.3.1.26		power-class-ext-A	R
	aLldpXdot3RemPowerClassExtB	30.12.3.1.27		power-class-ext-B	R
	aLldpXdot3RemPowerClassExt	30.12.3.1.28		power-class-ext	R
	aLldpXdot3RemPowerTypeExt	30.12.3.1.29		power-type-ext	R
	aLldpXdot3RemPDLoad	30.12.3.1.30		pd-load	R
	aLldpXdot3RemPD4PID	30.12.3.1.31		pd-4pid	R
	aLldpXdot3RemPSEMaxAvailPower	30.12.3.1.32		pse-max-avail-power	R
	aLldpXdot3RemPSEAutoclassSupport	30.12.3.1.33		pse-autoclass-support	R
	aLldpXdot3RemAutoclassCompleted	30.12.3.1.34		autoclass-completed	R
	aLldpXdot3RemAutoclassRequest	30.12.3.1.35		autoclass-request	R
	aLldpXdot3RemPowerDownRequest	30.12.3.1.36		power-down-request	R
	aLldpXdot3RemPowerDownTime	30.12.3.1.37		power-down-time	R
	aLldpXdot3RemMeasVoltageSupport	30.12.3.1.38		meas-voltage-support	R
	aLldpXdot3RemMeasCurrentSupport	30.12.3.1.39		meas-current-support	R
	aLldpXdot3RemMeasPowerSupport	30.12.3.1.40		meas-power-support	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-llp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3RemMeasEnergySupport	30.12.3.1.41		meas-energy-support	R
	aLldpXdot3RemMeasurementSource	30.12.3.1.42		measurement-source	R
	aLldpXdot3RemMeasVoltageRequest	30.12.3.1.43		meas-voltage-request	R
	aLldpXdot3RemMeasCurrentRequest	30.12.3.1.44		meas-current-request	R
	aLldpXdot3RemMeasPowerRequest	30.12.3.1.45		meas-power-request	R
	aLldpXdot3RemMeasEnergyRequest	30.12.3.1.46		meas-energy-request	R
	aLldpXdot3RemMeasVoltageValid	30.12.3.1.47		meas-voltage-valid	R
	aLldpXdot3RemMeasCurrentValid	30.12.3.1.48		meas-current-valid	R
	aLldpXdot3RemMeasPowerValid	30.12.3.1.49		meas-power-valid	R
	aLldpXdot3RemMeasEnergyValid	30.12.3.1.50		meas-energy-valid	R
	aLldpXdot3RemMeasVoltageUncertainty	30.12.3.1.51		meas-voltage-uncertainty	R
	aLldpXdot3RemMeasCurrentUncertainty	30.12.3.1.52		meas-current-uncertainty	R
	aLldpXdot3RemMeasPowerUncertainty	30.12.3.1.53		meas-power-uncertainty	R
	aLldpXdot3RemMeasEnergyUncertainty	30.12.3.1.54		meas-energy-uncertainty	R
	aLldpXdot3RemVoltageMeasurement	30.12.3.1.55		voltage-measurement	R
	aLldpXdot3RemCurrentMeasurement	30.12.3.1.56		current-measurement	R
	aLldpXdot3RemPowerMeasurement	30.12.3.1.57		power-measurement	R
	aLldpXdot3RemEnergyMeasurement	30.12.3.1.58		energy-measurement	R
	aLldpXdot3RemPSEPowerPriceIndex	30.12.3.1.59		pse-power-price-index	R
	aLldpXdot3RemTxTwSys	30.12.3.1.60		tx-system-value	R

Table 5–6—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-lldp* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-lldp</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aLldpXdot3RemTxTwSysEcho	30.12.3.1.61		tx-system-value-echo	R
	aLldpXdot3RemRxTwSys	30.12.3.1.62		rx-system-value	R
	aLldpXdot3RemRxTwSysEcho	30.12.3.1.63		rx-system-value-echo	R
	aLldpXdot3RemFbTwSys	30.12.3.1.64		fallback-system-value	R
	aLldpXdot3RemTxFw	30.12.3.1.65 ^a		tx-system-fw	R
	aLldpXdot3RemTxFwEcho	30.12.3.1.66 ^a		tx-system-fw-echo	R
	aLldpXdot3RemRxFw	30.12.3.1.67 ^a		rx-system-fw	R
	aLldpXdot3RemRxFwEcho	30.12.3.1.68 ^a		rx-system-fw-echo	R
	aLldpXdot3RemPreemptSupported	30.12.3.1.69 ^a		preemption-supported	R
	aLldpXdot3RemPreemptEnabled	30.12.3.1.70 ^a		preemption-enabled	R
	aLldpXdot3RemPreemptActive	30.12.3.1.71 ^a		preemption-active	R
	aLldpXdot3RemAddFragSize	30.12.3.1.72 ^a		additional-fragment-size	R

^a Subclauses 30.12.3.1.65 through 30.12.3.1.72 in IEEE Std 802.3-2022 can be found as the second instance of 30.12.3.1.24 through 30.12.3.1.31.

5.3 YANG module definition¹³

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

5.3.1 Tree hierarchy

5.3.1.1 ieee802-ethernet-interface

module: ieee802-ethernet-interface

```
augment /if:interfaces/if:interface:
  +--rw ethernet
    +--rw auto-negotiation!
      | +--rw enable?          boolean
      | +--ro negotiation-status? enumeration
    +--ro phy-type?            identityref
    +--ro pmd-type?            identityref
    +--rw duplex?              duplex-type
    o--rw speed?               eth-if-speed-type
    x--rw flow-control
      | +--rw pause {ethernet-pause}?
      | | +--rw direction?      pause-fc-direction-type
      | | +--ro statistics
      | |   +--ro in-frames-pause? yang:counter64
      | |   +--ro out-frames-pause? yang:counter64
      | +--rw pfc {ethernet-pfc}?
      | | +--rw enable?         boolean
      | | x--ro statistics
      | |   +--ro in-frames-pfc?  yang:counter64
      | |   +--ro out-frames-pfc? yang:counter64
      | +--rw force-flow-control? boolean
    +--ro max-frame-length?     uint16
    +--ro mac-control-extension-control? boolean
    +--ro frame-limit-slow-protocol? uint64
    +--ro capabilities
      | +--ro auto-negotiation?  boolean
    +--rw ethernet-pause {ethernet-pause}?
      | +--rw control-and-status
      | | +--rw pause-admin-control? pause-fc-direction-type
      | | +--ro pause-oper-status?   pause-fc-direction-type
      | | +--rw link-delay-allowance? uint32
      | | +--ro pfc-enable-status?   boolean
      | +--ro statistics
      |   +--ro in-frames-pause?     yang:counter64
      |   +--ro out-frames-pause?    yang:counter64
    +--ro statistics
      +--ro frame
        | +--ro in-total-frames?      yang:counter64
        | +--ro in-total-octets?      yang:counter64
        | +--ro in-frames?            yang:counter64
        | +--ro in-multicast-frames?  yang:counter64
        | +--ro in-broadcast-frames?  yang:counter64
        | +--ro in-error-fcs-frames?  yang:counter64
        | x--ro in-error-undersize-frames? yang:counter64
        | +--ro in-error-oversize-frames? yang:counter64
```

¹³ Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```

|   +--ro in-error-mac-internal-frames?   yang:counter64
|   +--ro out-frames?                     yang:counter64
|   +--ro out-multicast-frames?           yang:counter64
|   +--ro out-broadcast-frames?           yang:counter64
|   +--ro out-error-mac-internal-frames?  yang:counter64
+--ro phy
|   +--ro in-error-symbol?                 yang:counter64
|   +--ro lpi
|       +--ro in-lpi-transitions?         yang:counter64
|       +--ro in-lpi-time?                decimal64
|       +--ro out-lpi-transitions?        yang:counter64
|       +--ro out-lpi-time?               decimal64
+--ro mac-control
    +--ro in-frames-mac-control-unknown?   yang:counter64
    +--ro in-frames-mac-control-extension?  yang:counter64
    +--ro out-frames-mac-control-extension? yang:counter64

```

5.3.1.2 ieee802-ethernet-interface-half-duplex

module: ieee802-ethernet-interface-half-duplex

```

augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
  o--rw dynamic-rate-control?  dynamic-rate-control-type {dynamic-rate-control}?
augment /if:interfaces/if:interface/ieee802-eth-if:ethernet/ieee802-eth-if:capabilities:
  o--ro dynamic-rate-control-supported?  boolean {dynamic-rate-control}?
augment /if:interfaces/if:interface/ieee802-eth-if:ethernet/ieee802-eth-if:statistics/ieee802-eth-if:frame:
  +--ro csma-cd {csma-cd}?
    +--ro in-errors-sqe-test?                yang:counter64
    +--ro out-frames-collision-single?        yang:counter64
    +--ro out-frames-collision-multiple?      yang:counter64
    +--ro out-frames-deferred?                yang:counter64
    +--ro out-frames-collisions-excessive?    yang:counter64
    +--ro out-collisions-late?                yang:counter64
    +--ro out-errors-carrier-sense?           yang:counter64
    +--ro collision-histogram* [collision-count]
      +--ro collision-count                   yang:counter64
      +--ro collision-count-frames?           yang:counter64

```

5.3.1.3 ieee802-ethernet-mac-merge

module: ieee802-ethernet-mac-merge

```
augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
  +--rw mac-merge {mac-merge}?
    +--rw admin-control
      | +--rw merge-enable-tx?      enumeration
      | +--rw verify-disable-tx?    enumeration
      | +--rw verify-time?          uint16
      | +--rw frag-size?            uint8
    +--ro admin-status
      | +--ro merge-support?        enumeration
      | +--ro verify-status?        enumeration
      | +--ro status-tx?            enumeration
    +--ro statistics
      +--ro assembly-error-count?   yang:counter64
      +--ro smd-error-count?         yang:counter64
      +--ro assembly-ok-count?       yang:counter64
      +--ro fragment-count-rx?       yang:counter64
      +--ro fragment-count-tx?       yang:counter64
      +--ro hold-count?              yang:counter64
```

5.3.1.4 ieee802-ethernet-lldp

module: ieee802-ethernet-lldp

```
augment /lldp:lldp/lldp:port:
  +--ro auto-negotiation-supported?  boolean
  +--ro auto-negotiation-enabled?    boolean
  x--ro auto-negotiation-cap?         binary
  x--ro operational-mau-type?         int32
  +--rw auto-negotiation-cap-bits?    ieee802-phy:auto-neg-ability-bits
  +--rw operational-pmd-type?         identityref
  +--rw power-port-class?             port-class-type
  +--rw mdi-power-supported?          boolean
  +--rw mdi-power-enabled?            boolean
  +--rw power-pair-controlable?       boolean
  +--rw power-pairs?                  pse-pinout-type
  +--rw local-power-class?            pse-power-class-type
  o--ro link-aggregation-status?      bits
  o--ro aggregation-port-id?          int32
  o--ro local-max-frame-size?         int32
  +--ro power-type?                   bits
  +--rw power-support?                bits
  +--ro power-source?                 power-source-type
  +--ro pd-requested-power-value?     int32
  +--ro pd-requested-power-value-a?   int32
  +--ro pd-requested-power-value-b?   int32
  +--ro pse-allocated-power-value?    int32
  +--ro pse-allocated-power-value-a?  int32
  +--ro pse-allocated-power-value-b?  int32
  +--ro pse-powering-status?          powering-status-type
  +--ro pd-powered-status?            powered-status-type
  +--ro power-pairs-ext?              power-pairs-type
  +--ro power-class-ext-A?            power-class-ext-AB-type
  +--ro power-class-ext-B?            power-class-ext-AB-type
```

```

+--ro power-class-ext?           power-class-ext-type
+--ro power-type-ext?           power-type
+--ro pd-load?                   boolean
+--ro pd-4pid?                   boolean
+--ro pse-max-avail-power?       int32
+--ro pse-autoclass-support?     boolean
+--ro autoclass-completed?      boolean
+--ro autoclass-request?        boolean
+--rw power-down-request?       int32
+--rw power-down-time?          int32
+--rw meas-voltage-support?      boolean
+--rw meas-current-support?      boolean
+--rw meas-power-support?        boolean
+--rw meas-energy-support?        boolean
o--rw measurement-source?        bits
+--ro meas-voltage-request?      boolean
+--ro meas-current-request?      boolean
+--ro meas-power-request?        boolean
+--ro meas-energy-request?        boolean
+--ro meas-voltage-valid?        boolean
+--ro meas-current-valid?        boolean
+--ro meas-power-valid?          boolean
+--ro meas-energy-valid?          boolean
+--ro meas-voltage-uncertainty?  int32
+--ro meas-current-uncertainty?  int32
+--ro meas-power-uncertainty?    int32
+--ro meas-energy-uncertainty?    int32
+--ro voltage-measurement?       int32
+--ro current-measurement?       int32
+--ro power-measurement?         int32
+--ro energy-measurement?        int32
+--ro pse-power-price-index?     int32
+--ro local-response?            int32
+--ro local-system-ready?        boolean
+--ro tx-system-value?           int32
+--ro tx-system-value-echo?      int32
+--ro rx-system-value?           int32
+--ro rx-system-value-echo?      int32
+--ro fallback-system-value?     int32
+--ro tx-dll-ready?              boolean
+--ro rx-dll-ready?              boolean
+--ro dll-enabled?               boolean
+--ro tx-system-fw?              boolean
+--ro tx-system-fw-echo?         boolean
+--ro rx-system-fw?              boolean
+--ro rx-system-fw-echo?         boolean
+--ro preemption-supported?      boolean
+--ro preemption-enabled?        boolean
+--ro preemption-active?         boolean
+--ro additional-fragment-size?  int32
augment /lldp:lldp/lldp:port:
+--rw tlvs-port-config-enable?   bits
+--rw local-power-priority?       power-priority-type
+--rw local-measurement-source?   measurement-source-type
augment /lldp:lldp/lldp:port/lldp:remote-systems-data:
+--ro auto-negotiation-supported? boolean
+--ro auto-negotiation-enabled?   boolean
o--ro auto-negotiation-cap?        binary
o--ro operational-mau-type?        int32

```

+--ro auto-negotiation-cap-bits?	ieee802-phy:auto-neg-ability-bits
+--ro operational-pmd-type?	identityref
+--ro power-port-class?	port-class-type
+--ro mdi-power-supported?	boolean
+--ro mdi-power-enabled?	boolean
+--ro power-pair-controlable?	boolean
+--ro power-pairs?	pse-pinout-type
+--ro power-class?	pse-power-class-type
o--ro link-aggregation-status?	bits
o--ro aggregation-port-id?	int32
o--ro local-max-frame-size?	int32
o--ro power-type?	bits
+--ro power-source?	power-source-type
+--ro power-priority?	power-priority-type
+--ro pd-requested-power-value?	int32
+--ro pd-requested-power-value-a?	int32
+--ro pd-requested-power-value-b?	int32
+--ro pse-allocated-power-value?	int32
+--ro pse-allocated-power-value-a?	int32
+--ro pse-allocated-power-value-b?	int32
+--ro pse-powering-status?	powering-status-type
+--ro pd-powered-status?	powered-status-type
+--ro power-pairs-ext?	power-pairs-type
+--ro power-class-ext-A?	power-class-ext-AB-type
+--ro power-class-ext-B?	power-class-ext-AB-type
+--ro power-class-ext?	power-class-ext-type
+--ro power-type-ext?	power-type
+--ro pd-load?	boolean
+--ro pd-4pid?	boolean
+--ro pse-max-avail-power?	int32
+--ro pse-autoclass-support?	boolean
+--ro autoclass-completed?	boolean
+--ro autoclass-request?	boolean
+--ro power-down-request?	int32
+--ro power-down-time?	int32
+--ro meas-voltage-support?	boolean
+--ro meas-current-support?	boolean
+--ro meas-power-support?	boolean
+--ro meas-energy-support?	boolean
o--ro measurement-source?	bits
+--ro rem-measurement-source?	measurement-source-type
+--ro meas-voltage-request?	boolean
+--ro meas-current-request?	boolean
+--ro meas-power-request?	boolean
+--ro meas-energy-request?	boolean
+--ro meas-voltage-valid?	boolean
+--ro meas-current-valid?	boolean
+--ro meas-power-valid?	boolean
+--ro meas-energy-valid?	boolean
+--ro meas-voltage-uncertainty?	int32
+--ro meas-current-uncertainty?	int32
+--ro meas-power-uncertainty?	int32
+--ro meas-energy-uncertainty?	int32
+--ro voltage-measurement?	int32
+--ro current-measurement?	int32
+--ro power-measurement?	int32
+--ro energy-measurement?	int32
+--ro pse-power-price-index?	int32
+--ro tx-system-value?	int32


```

+--ro tx-system-value-echo?          int32
+--ro rx-system-value?               int32
+--ro rx-system-value-echo?          int32
+--ro fallback-system-value?         int32
+--ro tx-system-fw?                  boolean
+--ro tx-system-fw-echo?             boolean
+--ro rx-system-fw?                  boolean
+--ro rx-system-fw-echo?             boolean
+--ro preemption-supported?          boolean
+--ro preemption-enabled?            boolean
+--ro preemption-active?             boolean
+--ro additional-fragment-size?      int32

```

5.3.1.5 ieee802-ethernet-phy

module: ieee802-ethernet-phy

```

augment /if:interfaces/if:interface:
  +--rw phy-table
  | +--rw control
  | | +--rw phy-reset?          boolean
  | | +--rw phy-admin-control?  enumeration
  | | +--rw fec-control! {fec-support}?
  | | | +--rw fec-mode-control?  enumeration
  | | | +--rw rs-fec-bypass-control?  enabled-state
  | +--ro status
  | | +--ro phy-type?           identityref
  | | +--ro pmd-type?           identityref
  | | +--ro pmd-type-list?      ieee802-phy:pmd-type-list
  | | +--ro phy-status?         enumeration
  | | +--ro phy-media-available? ieee802-phy:media-available-type
  | | +--ro jabber-state?       enumeration
  | | +--ro jabbering-state-enters? yang:counter64
  | | +--ro fec-status! {fec-support}?
  | | | +--ro fec-ability?       bits
  | | | +--ro fec-mode-status?   enumeration
  | | | +--ro rs-fec-bypass-enabled-status?  enabled-state
  | | +--ro baset-snr! {baset-snr-support}?
  | | | +--ro snr-op-margin-chnl-a?  snr-value
  | | | +--ro snr-op-margin-chnl-b?  snr-value
  | | | +--ro snr-op-margin-chnl-c?  snr-value
  | | | +--ro snr-op-margin-chnl-d?  snr-value
  | | +--ro time-sync! {time-sync-support}?
  | | | +--ro time-sync-ability?    bits
  | | | +--ro time-sync-delay-tx-max? uint32
  | | | +--ro time-sync-delay-tx-min? uint32
  | | | +--ro time-sync-delay-rx-max? int32
  | | | +--ro time-sync-delay-rx-min? int32
  | +--ro statistics
  | | +--ro phy-media-losts?      yang:counter64
  | | +--ro jabbering-state-enters? yang:counter64
  | | +--ro false-carriers?       yang:counter64
  | | +--ro pcs-coding-violations? yang:counter64
  | | +--ro fast-retrain-count-local? counter-1000-second
  | | +--ro fast-retrain-count-peer? counter-1000-second
  | +--ro pcs-lane-statistics {pcs-lane-statistics-support}?
  | | +--ro pcs-lane* [pcs-lane-index]

```

```

|         +--ro pcs-lane-index          uint32
|         +--ro fec-corrected-blocks?   counter-pcs-lane
|         +--ro fec-uncorrected-blocks? counter-pcs-lane
+--rw auto-neg-table {auto-negotiation-support}?
  +--rw control
  |   +--rw auto-neg-control?   boolean
  |   +--rw auto-neg-restart?   boolean
  +--ro status
  |   +--ro auto-neg-enabled?     boolean
  |   +--ro auto-neg-remote-signaling? boolean
  |   +--ro auto-neg-status?     enumeration
  |   +--ro auto-neg-capability-bits? ieee802-phy:auto-neg-ability-bits
  |   +--ro auto-neg-cap-advertised-bits? ieee802-phy:auto-neg-ability-bits
  |   +--ro auto-neg-cap-received-bits? ieee802-phy:auto-neg-ability-bits

```

5.3.2 YANG modules

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 5.2 through 5.3 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the Ethernet YANG module can be found at the following URL:
<https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3>.¹⁴

5.3.2.1 Ethernet interface module

```

module ieee802-ethernet-interface {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-interface";
  prefix ieee802-eth-if;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
      iana-if-type@2023-01-26.yang";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }
}

```

¹⁴ Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
organization
  "IEEE Std 802.3 Ethernet Working Group
  Web URL: http://www.ieee802.org/3/";
contact
  "Web URL: http://www.ieee802.org/3/";
description
  "This module contains YANG definitions for configuring IEEE Std
  802.3 Ethernet Interfaces.
  In this YANG module, 'Ethernet interface' can be interpreted
  as referring to 'IEEE Std 802.3 compliant Ethernet
  interfaces'.";

revision 2025-09-10 {
  description
    "Updates under IEEE Std 802.3.2-2025";
  reference
    "IEEE Std 802.3-2022 and IEEE Std 802.3.1-2024, unless dated
    explicitly";
}

feature ethernet-pfc {
  description
    "This device supports Ethernet priority flow control.";
}

feature ethernet-pause {
  description
    "This device supports Ethernet PAUSE.";
}

typedef eth-if-speed-type {
  type decimal64 {
    fraction-digits 3;
  }
  units "Gb/s";
  status obsolete;
  description
    "Used to represent the configured, negotiated, or actual
    speed of an Ethernet interface in Gigabits per second
    (Gb/s), accurate to 3 decimal places (i.e., accurate to
    1 Mb/s).";
}

typedef duplex-type {
  type enumeration {
    enum full {
      description
        "Full duplex.";
    }
    enum half {
      description
        "Half duplex.";
    }
    enum unknown {
      description
        "Link is currently disconnected or initializing.";
    }
  }
  default "full";
}
```

```
description
  "Used to represent the configured, negotiated, or actual
  duplex mode of an Ethernet interface.";
reference
  "IEEE Std 802.3, 30.3.1.1.32, aDuplexStatus";
}

typedef pause-fc-direction-type {
  type enumeration {
    enum disabled {
      description
        "Flow control disabled in both ingress and egress
        directions.";
    }
    enum ingress-only {
      description
        "PAUSE frame-based flow control is enabled in the ingress
        direction only.";
    }
    enum egress-only {
      description
        "PAUSE frame-based flow control is enabled in the egress
        direction only.";
    }
    enum bi-directional {
      description
        "PAUSE frame-based flow control is enabled in both
        ingress and egress directions.";
    }
    enum undefined {
      description
        "Link is currently disconnected or initializing.";
    }
  }
  description
    "Used to represent the configured, negotiated, or actual
    PAUSE frame-based flow control setting.";
  reference
    "IEEE Std 802.3.1, dot3PauseAdminMode and dot3PauseOperMode";
}

augment "/if:interfaces/if:interface" {
  when "derived-from-or-self(if:type, 'ianaift:ethernetCsmacd')" {
    description
      "Applies to all Ethernet interfaces.";
  }
  description
    "Augment interface model with Ethernet interface
    specific configuration nodes.";
  container ethernet {
    description
      "Contains all Ethernet interface related configuration.";
    container auto-negotiation {
      presence "The presence of this container indicates that
        auto-negotiation is supported on this Ethernet
        interface.";
      description
        "Contains auto-negotiation transmission parameters.
```

This container contains a data node that allows the advertised duplex value in the negotiation to be restricted.

If not specified then the default behavior for the duplex data node is to negotiate all available values for the particular type of Ethernet PHY associated with the interface.";

```
reference
  "IEEE Std 802.3, Clause 28 and Annexes 28A-D";
leaf enable {
  type boolean;
  default "true";
  description
    "Controls whether auto-negotiation is enabled or
    disabled.
    For interface types that support auto-negotiation then
    it defaults to being enabled.

    For interface types that do not support
    auto-negotiation, the related configuration data is
    ignored.";
}
leaf negotiation-status {
  when "../enable = 'true'";
  type enumeration {
    enum in-progress {
      description
        "The auto-negotiation protocol is running and
        negotiation is currently in-progress.";
    }
    enum complete {
      description
        "The auto-negotiation protocol has completed
        successfully.";
    }
    enum failed {
      description
        "The auto-negotiation protocol has failed.";
    }
    enum unknown {
      description
        "The auto-negotiation status is not currently known,
        this could be because it is still negotiating or the
        protocol cannot run
        (e.g., if no medium is present).";
    }
    enum no-negotiation {
      description
        "No auto-negotiation is executed.
        The auto-negotiation function is either not
        supported on this interface or has not been
        enabled.";
    }
  }
}
config false;
description
  "The status of the auto-negotiation protocol.";
reference
```

```
        "IEEE 802.3, 30.6.1.1.4, aAutoNegAutoConfig";
    }
}
leaf phy-type {
    type identityref {
        base ieee802-phy:phy-type;
    }
    config false;
    description
        "A value that uniquely identifies the IEEE 802.3 PHY type
        of the interface.";
    reference
        "IEEE Std 802.3, 30.3.2.1.2 aPhyType";
}
leaf pmd-type {
    type identityref {
        base ieee802-phy:pmd-type;
    }
    config false;
    description
        "A value that uniquely identifies the IEEE 802.3 PMD type
        of the interface.";
    reference
        "IEEE Std 802.3, 30.5.1.1.2 aMAUType";
}
leaf duplex {
    type duplex-type;
    description
        "Operational duplex mode of the Ethernet interface.";
    reference
        "IEEE Std 802.3, 30.3.1.1.32 aDuplexStatus";
}
leaf speed {
    type eth-if-speed-type;
    units "Gb/s";
    status obsolete;
    description
        "Operational speed (data rate) of the Ethernet interface.
        The default value is implementation dependent.
        Superseded by speed in /if:interfaces/if:interface";
}
/* deprecated flow-control container */
container flow-control {
    status deprecated;
    description
        "Holds the different types of Ethernet PAUSE frame-based
        flow control that can be enabled.
        Superseded by new container - pause.";
    container pause {
        if-feature "ethernet-pause";
        description
            "IEEE Std 802.3 PAUSE frame-based flow control.";
        reference
            "IEEE Std 802.3, Annex 31B";
        leaf direction {
            type pause-fc-direction-type;
            description
                "Indicates which direction PAUSE frame-based flow
                control is enabled in, or whether it is disabled.
```

```

    The default flow-control settings are vendor
    specific. If auto-negotiation is enabled, then PAUSE
    based flow control is negotiated by default.
    The default value is implementation dependent.";
}
container statistics {
  config false;
  description
    "Contains the number of PAUSE frames received or
    transmitted.

    Discontinuities in the values of counters in
    this container can occur at re-initialization of
    the management system, and at other times as
    indicated by the value of the 'discontinuity-time'
    leaf defined in the ietf-interfaces YANG module
    (IETF RFC 8343).";
  leaf in-frames-pause {
    type yang:counter64;
    units "frames";
    description
      "A count of PAUSE MAC Control frames transmitted on
      this Ethernet interface.";
    reference
      "IEEE Std 802.3, 30.3.4.3
      aPAUSEMACCtrlFramesReceived";
  }
  leaf out-frames-pause {
    type yang:counter64;
    units "frames";
    description
      "A count of PAUSE MAC Control frames transmitted on
      this Ethernet interface.";
    reference
      "IEEE Std 802.3, 30.3.4.2
      aPAUSEMACCtrlFramesTransmitted";
  }
}
}
container pfc {
  if-feature "ethernet-pfc";
  description
    "IEEE Std 802.3 Priority-based flow control.";
  reference
    "IEEE Std 802.3, Annex 31D";
  leaf enable {
    type boolean;
    description
      "True indicates that IEEE Std 802.3 priority-based
      flow control is enabled, false indicates that
      IEEE Std 802.3 priority-based flow control is
      disabled. For interfaces that have auto-negotiation,
      the priority-based flow control is enabled by
      default.";
  }
}
container statistics {
  config false;
  status deprecated;
  description
```

"This container collects all statistics for Ethernet interfaces.

Discontinuities in the values of counters in this container can occur at re-initialization of the management system, and at other times as indicated by the value of the 'discontinuity-time' leaf defined in the ietf-interfaces YANG module (IETF RFC 8343).";

```
leaf in-frames-pfc {
  type yang:counter64;
  units "frames";
  description
    "Deprecated in-frames-pfc as not defined in base
     standard. A count of PFC MAC Control frames
     received on this Ethernet interface.";
  reference
    "IEEE Std 802.3.1, dot3HCInPFCFrames";
}
```

```
leaf out-frames-pfc {
  type yang:counter64;
  units "frames";
  description
    "Deprecated out-frames-pfc as not defined in base
     standard. A count of PFC MAC Control frames
     transmitted on this interface.";
  reference
    "IEEE Std 802.3.1, dot3HCInPFCFrames";
}
```

```
}
}
leaf force-flow-control {
  type boolean;
  default "false";
  description
    "Explicitly forces the local PAUSE frame-based flow
     control settings regardless of what has been
     negotiated.
```

Since the auto-negotiation of flow control settings does not allow all sane combinations to be negotiated (e.g., consider a device that is only capable of sending PAUSE frames connected to a peer device that is only capable of receiving and acting on PAUSE frames) and failing to agree on the flow control settings does not cause the auto-negotiation to fail completely, then it is sometimes useful to be able to explicitly enable particular PAUSE frame-based flow control settings on the local device regardless of what is being advertised or negotiated.";

```
reference
  "IEEE Std 802.3, Table 28B-3";
}
```

```
}
leaf max-frame-length {
  type uint16;
  units "octets";
  config false;
  description
```



```
        "This indicates the MAC frame length (including FCS bytes)
        at which frames are dropped for being too long.";
    reference
        "IEEE Std 802.3, 30.3.1.1.37 aMaxFrameLength";
}
leaf mac-control-extension-control {
    type boolean;
    config false;
    description
        "A value that identifies the current EXTENSION
        MAC Control function, as specified in
        IEEE Std 802.3, Annex 31C.";
    reference
        "IEEE Std 802.3, 30.3.8.3 aEXTENSIONMACCtrlStatus
        IEEE Std 802.3.1, dot3ExtensionMacCtrlStatus ";
}
leaf frame-limit-slow-protocol {
    type uint64;
    units "f/s";
    default "10";
    config false;
    description
        "The maximum number of Slow Protocol frames of a given
        subtype that can be transmitted in a one second
        interval.
        The default value is 10.";
    reference
        "IEEE Std 802.3, 30.3.1.1.38 aSlowProtocolFrameLimit";
}
container capabilities {
    config false;
    description
        "Contains all Ethernet interface specific capabilities.";
    leaf auto-negotiation {
        type boolean;
        description
            "Indicates whether auto-negotiation may be configured on
            this interface.";
    }
}
container ethernet-pause {
    if-feature "ethernet-pause";
    description
        "IEEE Std 802.3 PAUSE flow control.

        Discontinuities in the values of counters in
        this container can occur at re-initialization of
        the management system, and at other times as
        indicated by the value of the 'discontinuity-time'
        leaf defined in the ietf-interfaces YANG module
        (IETF RFC 8343).";
    reference
        "IEEE Std 802.3, 30.3.4 PAUSE entity managed object class
        and Clause 31, Annex 31B)";
    container control-and-status {
        description
            "PAUSE function control and status objects.";
        leaf pause-admin-control {
            type pause-fc-direction-type;
```

```
default "disabled";
description
  "What PAUSE functionality will run on this interface
  when Auto-Negotiation is not implemented
  or disabled.";
reference
  "IEEE Std 802.3.1, ????dot3PauseAdminMode";
}
leaf pause-oper-status {
  type pause-fc-direction-type;
  config false;
  description
    "What PAUSE functionality is running on this
    interface, as a result of Auto-Negotiation or
    setting pause-admin-control.";
  reference
    "IEEE Std 802.3.1, ????dot3PauseOperMode";
}
leaf link-delay-allowance {
  type uint32;
  description
    "The value in bit-times of the allowance made by the
    MAC Control PAUSE entity for round-trip propagation
    delay.
    The default value is implementation dependent.";
}
leaf pfc-enable-status {
  type boolean;
  config false;
  description
    "Is IEEE 802.1 Priority-based Flow Control (PFC)
    enabled on the interface. If PFC is enabled, then
    802.3 PAUSE flow control is disabled.";
  reference
    "IEEE Std 802.3, 30.3.3.6 aPFCEnableStatus";
}
}
container statistics {
  config false;
  description
    "PAUSE frame counters.";
  leaf in-frames-pause {
    type yang:counter64;
    units "frames";
    description
      "A count of MAC Control PAUSE frames transmitted on
      this Ethernet interface.";
    reference
      "IEEE Std 802.3, 30.3.4.3
      aPAUSEMACCtrlFramesReceived";
  }
  leaf out-frames-pause {
    type yang:counter64;
    units "frames";
    description
      "A count of MAC Control PAUSE frames transmitted on
      this Ethernet interface.";
    reference
      "IEEE Std 802.3, 30.3.4.2
```

```
        aPAUSEMACCtrlFramesTransmitted";
    }
}
}
container statistics {
    config false;
    description
        "Contains statistics specific to Ethernet interfaces.

        Discontinuities in the values of counters in the
        container can occur at re-initialization of the
        management system, and at other times as indicated by
        the value of the 'discontinuity-time' leaf defined in
        the ietf-interfaces YANG module (IETF RFC 8343).";
    container frame {
        description
            "Contains frame statistics specific to Ethernet
            interfaces.

            All octet frame lengths include the 4-byte FCS.

            Error counters are only reported once. The count
            represented by an instance of this object is
            incremented when the frameCheckError status is
            returned by the MAC service to the MAC Client.
            Received frames for which multiple error conditions
            pertain are, according to the conventions of
            IEEE Std 802.3 Layer Management, counted exclusively
            according to the error status presented to the MAC
            Client.

            A frame that is counted by an instance of this object
            is also counted by the corresponding instance of
            'in-errors' leaf defined in the ietf-interfaces YANG
            module (IETF RFC 8343).";
        leaf in-total-frames {
            type yang:counter64;
            units "frames";
            description
                "The total number of frames (including bad frames)
                received on the Ethernet interface.

                This counter is calculated by summing the following
                IEEE Std 802.3, Clause 30 counters:
                aFramesReceivedOK +
                aFrameCheckSequenceErrors +
                aAlignmentErrors +
                aFrameTooLongErrors +
                aFramesLostDueToIntMACRcvError";
            reference
                "IEEE Std 802.3, Clause 30 counters, as specified
                in the description above.";
        }
        leaf in-total-octets {
            type yang:counter64;
            units "octets";
            description
                "The total number of octets of data (including those
                in bad frames) received on the Ethernet interface.
```

```
        Includes the 4-octet FCS.";
    reference
        "IETF RFC 2819, etherStatsOctets";
}
leaf in-frames {
    type yang:counter64;
    units "frames";
    description
        "A count of frames (including unicast, multicast, and
        broadcast) that have been successfully received on
        the Ethernet interface.

        This count does not include frames received with
        frame-too-long, FCS, length, or alignment errors, or
        frames lost due to internal MAC sublayer error.";
    reference
        "IEEE Std 802.3, 30.3.1.1.5 aFramesReceivedOK";
}
leaf in-multicast-frames {
    type yang:counter64;
    units "frames";
    description
        "A count of multicast frames that have been
        successfully received on the Ethernet interface.

        This counter represents a subset of the frames
        counted by in-frames.

        This count does not include frames received with
        frame-too-long, FCS, length, or alignment errors, or
        frames lost due to internal MAC sublayer error.";
    reference
        "IEEE Std 802.3, 30.3.1.1.21
        aMulticastFramesReceivedOK";
}
leaf in-broadcast-frames {
    type yang:counter64;
    units "frames";
    description
        "A count of broadcast frames that have been
        successfully received on the Ethernet interface.

        This counter represents a subset of the frames
        counted by in-frames.

        This count does not include frames received with
        frame-too-long, FCS, length, or alignment errors, or
        frames lost due to internal MAC sublayer error.";
    reference
        "IEEE Std 802.3, 30.3.1.1.22
        aBroadcastFramesReceivedOK";
}
leaf in-error-fcs-frames {
    type yang:counter64;
    units "frames";
    description
        "A count of receive frames that are of valid length,
        but do not pass the FCS check, regardless of whether
```

or not the frames are an integral number of octets in length.

This counter is calculated by summing the following counters:

aFrameCheckSequenceErrors +
aAlignmentErrors";

reference

"IEEE Std 802.3, 30.3.1.1.6 aFrameCheckSequenceErrors;
IEEE Std 802.3, 30.3.1.1.7 aAlignmentErrors";

}

leaf in-error-undersize-frames {

type yang:counter64;

units "frames";

status deprecated;

description

"Deprecated in-error-undersize-frames as not defined in base standard. A count of frames received on a particular Ethernet interface that are less than 64 bytes in length, and are discarded.

This counter is incremented regardless of whether the frame passes the FCS check.";

reference

"IETF RFC 2819, etherStatsUndersizePkts and
etherStatsFragments";

}

leaf in-error-oversize-frames {

type yang:counter64;

units "frames";

description

"A count of frames received on a particular Ethernet interface that exceed the maximum permitted frame size, that is specified in max-frame-length, and are discarded.

This counter is incremented regardless of whether the frame passes the FCS check.";

reference

"IEEE Std 802.3, 30.3.1.1.25
aFrameTooLongErrors";

}

leaf in-error-mac-internal-frames {

type yang:counter64;

units "frames";

description

"A count of frames for which reception on a particular Ethernet interface fails due to an internal MAC sublayer receive error.

A frame is only counted by an instance of this object if it is not counted by the corresponding instance of either the in-error-fcs-frames, in-error-undersize-frames, or in-error-oversize-frames. The precise meaning of the count represented by an instance of this object is implementation-specific.

In particular, an instance of this object may

```
        represent a count of receive errors on a particular
        Ethernet interface that are not otherwise counted.";
reference
    "IEEE Std 802.3, 30.3.1.1.15
    aFramesLostDueToIntMACRcvError";
}
leaf out-frames {
    type yang:counter64;
    units "frames";
    description
        "A count of frames (including unicast, multicast, and
        broadcast) that have been successfully transmitted
        on the Ethernet interface.";
    reference
        "IEEE Std 802.3, 30.3.1.1.2 aFramesTransmittedOK";
}
leaf out-multicast-frames {
    type yang:counter64;
    units "frames";
    description
        "A count of multicast frames that have been
        successfully transmitted on the Ethernet interface.

        This counter represents a subset of the frames
        counted by out-frames.";
    reference
        "IEEE Std 802.3, 30.3.1.1.18
        aMulticastFramesXmittedOK";
}
leaf out-broadcast-frames {
    type yang:counter64;
    units "frames";
    description
        "A count of broadcast frames that have been
        successfully transmitted on the Ethernet interface.

        This counter represents a subset of the frames
        counted by out-frames.";
    reference
        "IEEE Std 802.3, 30.3.1.1.19
        aBroadcastFramesXmittedOK";
}
leaf out-error-mac-internal-frames {
    type yang:counter64;
    units "frames";
    description
        "A count of frames for which transmission on a
        particular Ethernet interface fails due to an
        internal MAC sublayer transmit error.

        The precise meaning of the count represented by an
        instance of this object is implementation-specific.
        In particular, an instance of this object may
        represent a count of transmission errors on a
        particular Ethernet interface that are not otherwise
        counted.";
    reference
        "IEEE Std 802.3, 30.3.1.1.12
        aFramesLostDueToIntMACXmitError";
}
```

```
}
}
container phy {
  description
    "Ethernet statistics related to the PHY layer.";
  leaf in-error-symbol {
    type yang:counter64;
    units "errors";
    description
      "A count of the number of symbol errors that have
      occurred.

      For the precise definition of when the symbol error
      counter is incremented, please see the 'description'
      text associated with aSymbolErrorDuringCarrier,
      specified in IEEE Std 802.3, 30.3.2.1.5.";
    reference
      "IEEE Std 802.3, 30.3.2.1.5
      aSymbolErrorDuringCarrier";
  }
  container lpi {
    description
      "Physical Ethernet statistics for the energy
      efficiency related low power idle indications.";
    leaf in-lpi-transitions {
      type yang:counter64;
      units "transitions";
      description
        "The number of times the link partner transitioned to
        Low Power Idle.
        This counter has a maximum per second increment rate
        of:
          50 thousand for 100 Mb/s;
          90 thousand for 1000 Mb/s
          230 thousand for 10 Gb/s.";
      reference
        "IEEE Std 802.3, 30.3.2.1.11
        aReceiveLPITransitions";
    }
    leaf in-lpi-time {
      type decimal64 {
        fraction-digits 6;
      }
      units "seconds";
      description
        "The amount of time the link partner has
        been in Low Power Idle.";
      reference
        "IEEE Std 802.3, 30.3.2.1.9
        aReceiveLPIMicroseconds";
    }
    leaf out-lpi-transitions {
      type yang:counter64;
      units "transitions";
      description
        "The number of times this port transitioned to
        Low Power Idle.
        This counter has a maximum per second increment rate
        of:
```

```

        50 thousand for 100 Mb/s;
        90 thousand for 1000 Mb/s
        230 thousand for 10 Gb/s.";
    reference
        "IEEE Std 802.3, 30.3.2.1.10
        aTransmitLPITransitions";
}
leaf out-lpi-time {
    type decimal64 {
        fraction-digits 6;
    }
    units "seconds";
    description
        "The amount of time in this port has
        been in Low Power Idle.";
    reference
        "IEEE Std 802.3, 30.3.2.1.8
        aTransmitLPIMicroseconds";
}
}
container mac-control {
    description
        "A group of statistics specific to MAC Control
        operation of selected Ethernet interfaces.";
    reference
        "IEEE Std 802.3.1, dot3ExtensionTable";
    leaf in-frames-mac-control-unknown {
        type yang:counter64;
        units "frames";
        description
            "A count of MAC Control frames with an unsupported
            opcode received on this Ethernet interface.

            Frames counted against this counter are also counted
            against in-discards defined in the ietf-interfaces
            YANG module (IETF RFC 8343).";
        reference
            "IEEE Std 802.3, 30.3.3.5
            aUnsupportedOpcodesReceived";
    }
    leaf in-frames-mac-control-extension {
        type yang:counter64;
        units "frames";
        description
            "The count of Extension MAC Control frames received
            on this Ethernet interface.";
        reference
            "IEEE Std 802.3, 30.3.8.2
            aEXTENSIONMACCtrlFramesReceived";
    }
    leaf out-frames-mac-control-extension {
        type yang:counter64;
        units "frames";
        description
            "The count of Extension MAC Control frames
            transmitted on this Ethernet interface.";
        reference
            "IEEE Std 802.3, 30.3.8.1

```



```
        aEXTENSIONMACCtrlFramesTransmitted";
    }
}
}
}
}
```

5.3.2.2 Ethernet interface (half-duplex) module

```
module ieee802-ethernet-interface-half-duplex {
  yang-version 1.1;
  namespace
    "urn:ieee:std:802.3:yang:ieee802-ethernet-interface-half-duplex";
  prefix ieee802-eth-half-duplex;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "http://www.iana.org/assignments/yang-parameters/
      iana-if-type@2023-01-26.yang";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group
    Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for half duplex
    Ethernet interfaces, e.g., 10BASE-T1S.";

  revision 2025-09-10 {
    description
      "Updates under IEEE Std 802.3.2-2025";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  feature dynamic-rate-control {
    status obsolete;
    description
      "This feature indicates that the device supports Ethernet
      interfaces lowering the average data rate of the MAC
```

```
    sublayer, with frame granularity, by using Rate Control to
    dynamically increase the interpacket gap for some types of
    Ethernet interface.
    Only valid for Ethernet interfaces operating at speeds
    (data rates) above 1000 Mb/s.";
  reference
    "IEEE Std 802.3, 30.3.1.1.33 aRateControlAbility";
}

feature csma-cd {
  description
    "This feature indicates that the device supports Ethernet
    interfaces running at half-duplex using CSMA/CD.";
}

typedef dynamic-rate-control-type {
  type enumeration {
    enum disabled {
      description
        "Dynamic rate control is disabled";
    }
    enum sonet-oc192 {
      value 2;
      description
        "Dynamic rate control is enabled for a 10 Gb/s Ethernet
        interface to SONET/SDH OC192/STM64.";
    }
  }
  default "disabled";
  status obsolete;
  description
    "Allowed values for dynamic-rate-control.";
  reference
    "IEEE Std 802.3, 4.4.2 ipgStretchRatio and 30.3.1.1.34
    aRateControlStatus";
}

augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
  when "derived-from-or-self(..if:type, 'ianaift:ethernetCsmacd')
    and ieee802-eth-if:duplex = 'half'" {
    description
      "Applies to half-duplex Ethernet interfaces.";
  }
  status obsolete;
  description
    "Augment with Ethernet interface configuration parameters
    for half-duplex operation.";
  leaf dynamic-rate-control {
    if-feature "dynamic-rate-control";
    type dynamic-rate-control-type;
    status obsolete;
    description
      "Enables dynamic rate control and specifies what speed
      (data rate) the dynamic rate control is operating at.
      The value of this attribute is constrained by the MAC
      data rate and hardware support.
      The default value is implementation dependent.";
    reference
      "IEEE Std 802.3, 30.3.1.1.34 aRateControlStatus";
  }
}
```

```
}  
}  
  
augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet/"  
  + "ieee802-eth-if:capabilities" {  
    when "derived-from-or-self(..../if:type,  
      'ianaift:ethernetCsmacd')  
      and ../ieee802-eth-if:duplex = 'half'" {  
      description  
        "Applies to half-duplex Ethernet interfaces";  
    }  
    status obsolete;  
    description  
      "Augment with configuration capabilities for half-duplex  
      Ethernet interface.";  
    leaf dynamic-rate-control-supported {  
      if-feature "dynamic-rate-control";  
      type boolean;  
      default "false";  
      status obsolete;  
      description  
        "Indicates whether the Ethernet interface supports lowering  
        the average data rate of the MAC sublayer, with frame  
        granularity, by using Rate Control to dynamically increase  
        the interpacket gap.  
        Only valid for Ethernet interfaces operating at speeds  
        (data rates) above 1000 Mb/s.";  
      reference  
        "IEEE Std 802.3, 30.3.1.1.33 aRateControlAbility";  
    }  
  }  
}  
  
augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet/"  
  + "ieee802-eth-if:statistics/ieee802-eth-if:frame" {  
    when "derived-from-or-self(..../if:type,  
      'ianaift:ethernetCsmacd') and  
      ../ieee802-eth-if:duplex = 'half'" {  
      description  
        "Applies to half-duplex Ethernet interfaces.";  
    }  
    description  
      "Augment with statistics for half-duplex Ethernet interface.";  
    container csma-cd {  
      if-feature "csma-cd";  
      description  
        "Holds counters that are specific to CDMA/CD half-duplex  
        operation of Ethernet interfaces.  
        This counter does not increment on Ethernet interfaces  
        operating at speeds (data rates) greater than 10 Mb/s, or  
        on Ethernet interfaces operating in full-duplex mode.  
        Discontinuities in the value of this counter can occur at  
        re-initialization of the management system, and at other  
        times as indicated by the value of the  
        'discontinuity-time' leaf defined in the ietf-interfaces  
        YANG module (IETF RFC 8343).";  
      leaf in-errors-sqe-test {  
        type yang:counter64;  
        units "errors";  
        description
```

```
"A count of times that the SQE TEST ERROR is received on
a particular interface. The SQE TEST ERROR is set in
accordance with the rules for verification of the SQE
detection mechanism in the PLS Carrier Sense function as
described in IEEE Std 802.3, 7.2.4.6.";
reference
  "IEEE Std 802.3, 7.2.4.6, and 30.3.2.1.4 aSQETestErrors";
}
leaf out-frames-collision-single {
  type yang:counter64;
  units "frames";
  description
    "A count of frames that are involved in a single
    collision, and are subsequently transmitted
    successfully. A frame that is counted by an instance of
    this object is also counted by the corresponding
    instance of either 'out-unicast-frames',
    'out-broadcast-frames', or 'out-multicast-frames',
    and is not counted by the corresponding instance of the
    'out-frames-collision-multiple'.";
  reference
    "IEEE Std 802.3, 30.3.1.1.3 aSingleCollisionFrames";
}
leaf out-frames-collision-multiple {
  type yang:counter64;
  units "frames";
  description
    "A count of frames that are involved in multiple
    collisions, and are subsequently transmitted
    successfully. A frame that is counted by an instance of
    this object is also counted by the corresponding
    instance of either 'out-unicast-frames',
    'out-broadcast-frames', or 'out-multicast-frames', and
    is not counted by the corresponding instance of the
    'out-frames-collision-single'.";
  reference
    "IEEE Std 802.3, 30.3.1.1.4 aMultipleCollisionFrames";
}
leaf out-frames-deferred {
  type yang:counter64;
  units "frames";
  description
    "A count of frames for which the first transmission
    attempt on a particular Ethernet interface is delayed
    because the medium is busy.
    A deferred frame that is not subject to any number of
    collisions is not counted by an instance of
    'out-frames-collision-single' or
    'out-frames-collision-multiple' objects.";
  reference
    "IEEE Std 802.3, 30.3.1.1.9
    aFramesWithDeferredXmissions";
}
leaf out-frames-collisions-excessive {
  type yang:counter64;
  units "frames";
  description
    "A count of frames for which transmission on a particular
    Ethernet interface fails due to excessive collisions.";
```

```
reference
  "IEEE Std 802.3, 30.3.1.1.11 aFramesAbortedDueToXSColls";
}
leaf out-collisions-late {
  type yang:counter64;
  units "collisions";
  description
    "The number of times that a collision is detected on a
    particular Ethernet interface later than one slotTime
    into the transmission of a packet.
    A (late) collision included in a count represented by an
    instance of this object is also considered as a
    (generic) collision for purposes of other
    collision-related statistics.";
  reference
    "IEEE Std 802.3, 30.3.1.1.10 aLateCollisions";
}
leaf out-errors-carrier-sense {
  type yang:counter64;
  units "errors";
  description
    "The number of times that the carrier sense condition was
    lost or never asserted when attempting to transmit a
    frame on a particular Ethernet interface.
    The count represented by an instance of this object is
    incremented at most once per transmission attempt, even
    if the carrier sense condition fluctuates during a
    transmission attempt.";
  reference
    "IEEE Std 802.3, 30.3.1.1.13 aCarrierSenseErrors";
}
list collision-histogram {
  key "collision-count";
  description
    "A collection of collision histograms for a particular
    interface.";
  reference
    "IEEE Std 802.3, 30.3.1.1.30 aCollisionFrames";
  leaf collision-count {
    type yang:counter64;
    units "collisions";
    description
      "The number of per-frame media collisions for which a
      particular collision histogram cell represents the
      frequency on a particular interface.";
  }
  leaf collision-count-frames {
    type yang:counter64;
    units "frames";
    description
      "A count of individual MAC frames for which the
      transmission (successful or otherwise) on a particular
      interface occurs after the frame has experienced
      exactly the number of collisions in the associated
      dot3CollCount object.
      For example, a frame which is transmitted on an
      interface after experiencing exactly 4 collisions
      would be indicated by incrementing only the
      collision-count-frames object associated with the
```

```

        collision-count value of 4. No other instance of
        collision-count-frames would be incremented in this
        example.";
    }
}
}
}
}
}

```

5.3.2.3 Ethernet MAC merge module

```

module ieee802-ethernet-mac-merge {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-mac-merge";
  prefix mac-merge;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
    reference
      "IEEE Std 802.3.2-2019";
  }

  organization
    "IEEE Std 802.3 Ethernet Working Group
    Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "The Yang model for managing devices that support the MAC Merge
    (aka preemption) sublayer as defined in Clause 99. Unless
    otherwise indicated, the references in this model module are to
    IEEE Std 802.3-2022.";

  revision 2025-09-10 {
    description
      "Updates under IEEE Std 802.3.2-2025";
    reference
      "IEEE Std 802.3-2022, unless dated explicitly";
  }

  feature mac-merge {
    description
      "MAC Merge is supported.";
    reference
      "IEEE Std 802.3-2022";
  }
}

```



```
augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
  if-feature "mac-merge";
  description
    "Interfaces supporting MAC Merge.";
  container mac-merge {
    description
      "Container for MAC Merge objects.";
    container admin-control {
      description
        "Container for MAC Merge control objects.";
      leaf merge-enable-tx {
        type enumeration {
          enum Disabled {
            description
              "Transmit preemption is disabled";
          }
          enum Enabled {
            description
              "Transmit preemption is enabled";
          }
        }
        default "Disabled";
        description
          "Is MAC Merge enabled when transmitting?
          It may be modified using an edit operation.";
        reference
          "IEEE Std 802.3, 30.14.1.3";
      }
      leaf verify-disable-tx {
        type enumeration {
          enum Disabled {
            description
              "Verify is disabled";
          }
          enum Enabled {
            description
              "Verify is enabled";
          }
        }
        default "Disabled";
        description
          "Does MAC Merge check to make sure the link partner
          supports MAC Merge?
          The check is performed by sending a verify request and
          getting a verify response.
          It may be modified using an edit operation.";
        reference
          "IEEE Std 802.3, 30.14.1.4";
      }
    }
    leaf verify-time {
      type uint16 {
        range "1..128";
      }
      units "milliseconds";
      default "10";
      description
        "The retry timer for verification attempts.
        The valid range is 1 to 128 inclusive.
        The default value is 10.";
```

```
        reference
            "IEEE Std 802.3, 30.14.1.6";
    }
    leaf frag-size {
        type uint8 {
            range "0..3";
        }
        default "0";
        description
            "This controls minimum frame size for packet fragments.
            It's specified in units of 64 bytes above the minimum
            of 64 bytes. The equation is:
            minimum-size = 64 + (frag-size*64)-4";
        reference
            "IEEE Std 802.3, 30.14.1.7";
    }
}
container admin-status {
    config false;
    description
        "Container for MAC Merge status objects.";
    leaf merge-support {
        type enumeration {
            enum Supported {
                description
                    "MAC Merge is supported on the device";
            }
            enum NotSupported {
                description
                    "MAC Merge is not supported on the device";
            }
        }
        description
            "Is MAC Merge is supported?";
        reference
            "IEEE Std 802.3, 30.14.1.1";
    }
    leaf verify-status {
        type enumeration {
            enum unknown {
                description
                    "unknown";
            }
            enum initial {
                description
                    "initial";
            }
            enum verifying {
                description
                    "verifying";
            }
            enum succeeded {
                description
                    "succeeded";
            }
            enum failed {
                description
                    "failed";
            }
        }
    }
}
```



```
        enum disabled {
            description
                "disabled";
        }
    }
    description
        "The status of MAC Merge Verification.";
    reference
        "IEEE Std 802.3, 30.14.1.2";
}
leaf status-tx {
    type enumeration {
        enum unknown {
            description
                "unknown";
        }
        enum inactive {
            description
                "inactive";
        }
        enum active {
            description
                "active";
        }
    }
    description
        "The status of MAC Merge in the transmit direction.";
    reference
        "IEEE Std 802.3, 30.14.1.5";
}
}
container statistics {
    config false;
    description
        "Container for MAC Merge counters.
        Discontinuities in the values of counters in
        this container can occur at re-initialization of
        the management system, and at other times as
        indicated by the value of the 'discontinuity-time'
        leaf defined in the ietf-interfaces YANG module
        (IETF RFC 8343).";
    leaf assembly-error-count {
        type yang:counter64;
        description
            "A count of MAC frames with reassembly errors.";
        reference
            "IEEE Std 802.3, 30.14.1.8";
    }
    leaf smd-error-count {
        type yang:counter64;
        description
            "A count of received MAC frames / MAC frame fragments
            rejected with an unexpected preemption header.";
        reference
            "IEEE Std 802.3, 30.14.1.9";
    }
    leaf assembly-ok-count {
        type yang:counter64;
        description
```

```
        "A count of preemptable MAC frames that were successfully
        reassembled and delivered to the MAC.";
    reference
        "IEEE Std 802.3, 30.14.1.10";
}
leaf fragment-count-rx {
    type yang:counter64;
    description
        "The number of preemptable frame fragments recieved.";
    reference
        "IEEE Std 802.3, 30.14.1.11";
}
leaf fragment-count-tx {
    type yang:counter64;
    description
        "The number of preemptable frame fragments transmitted.";
    reference
        "IEEE Std 802.3, 30.14.1.12";
}
leaf hold-count {
    type yang:counter64;
    description
        "The number of MAC Merge hold requests (i.e., to stop
        transmissions of preemptable traffic) received from
        the MAC";
    reference
        "IEEE Std 802.3, 30.14.1.13";
}
}
}
}
}
```

5.3.2.4 Ethernet LLDP module

```
module ieee802-ethernet-lldp {
    yang-version 1.1;
    namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-lldp";
    prefix ieee802-eth-lldp;

    import ieee802-dot1ab-lldp {
        prefix lldp;
        reference
            "IEEE Std 802.1ABcu-2021";
    }
    import ieee802-ethernet-phy-type {
        prefix ieee802-phy;
        reference
            "IEEE Std 802.3-2022";
    }

    organization
        "IEEE Std 802.3 Ethernet Working Group
        Web URL: http://www.ieee802.org/3/";
    contact
        "Web URL: http://www.ieee802.org/3/";
    description
```

```
"This module contains YANG definitions for configuring LLDP for
802.3 Ethernet interfaces.
In this YANG module, 'Ethernet interface' can be interpreted
as referring to 'IEEE Std 802.3 compliant Ethernet
interfaces'.";

revision 2025-09-10 {
  description
    "Updates under IEEE Std 802.3.2-2025";
  reference
    "IEEE Std 802.3-2022, unless dated explicitly";
}

typedef port-class-type {
  type enumeration {
    enum p-class-pse {
      value 0;
      description
        "Power Sourcing Equipment";
    }
    enum p-class-pd {
      value 1;
      description
        "Powered Device";
    }
  }
  description
    "Enumeration for the power port class";
  reference
    "IEEE Std 802.3, 30.12.2.1.5";
}

typedef pse-pinout-type {
  type enumeration {
    enum signal {
      value 0;
      description
        "PSE Pinout Alternative A";
    }
    enum spare {
      value 1;
      description
        "PSE Pinout Alternative B";
    }
  }
  description
    "Enumeration for the pinout alternatives used for PD
    detection and power ";
  reference
    "IEEE Std 802.3, 30.12.2.1.9";
}

typedef pse-power-class-type {
  type enumeration {
    enum class0 {
      value 0;
      description
        "Class 0 PD";
    }
  }
}
```

```
enum class1 {
    value 1;
    description
        "Class 1 PD";
}
enum class2 {
    value 2;
    description
        "Class 2 PD";
}
enum class3 {
    value 3;
    description
        "Class 3 PD";
}
enum class4 {
    value 4;
    description
        "Class 4 PD";
}
}
description
    "Enumeration for the PD class";
reference
    "IEEE Std 802.3, 30.12.2.1.10";
}

typedef power-class-ext-AB-type {
    type enumeration {
        enum singlesig {
            value 0;
            description
                "Single-signature PD or 2-pair only PSE";
        }
        enum class1 {
            value 1;
            description
                "Class 1";
        }
        enum class2 {
            value 2;
            description
                "Class 2";
        }
        enum class3 {
            value 3;
            description
                "Class 3";
        }
        enum class4 {
            value 4;
            description
                "Class 4";
        }
        enum class5 {
            value 5;
            description
                "Class 5";
        }
    }
}
```

```
}
description
  "Enumeration for the assigned power class ";
reference
  "IEEE Std 802.3, 30.12.3.1.26";
}

typedef power-class-ext-type {
  type enumeration {
    enum dualsig {
      value 0;
      description
        "Dual-signature PD";
    }
    enum class1 {
      value 1;
      description
        "Class 1";
    }
    enum class2 {
      value 2;
      description
        "Class 2";
    }
    enum class3 {
      value 3;
      description
        "Class 3";
    }
    enum class4 {
      value 4;
      description
        "Class 4";
    }
    enum class5 {
      value 5;
      description
        "Class 5";
    }
    enum class6 {
      value 6;
      description
        "Class 6";
    }
    enum class7 {
      value 7;
      description
        "Class 7";
    }
    enum class8 {
      value 8;
      description
        "Class 8";
    }
  }
  description
    "Enumeration for the assigned power class ";
  reference
    "IEEE Std 802.3, 30.12.3.1.28";
}
```

```
}  
  
typedef power-type {  
  type enumeration {  
    enum type4dualsigPD {  
      value 0;  
      description  
        "Type 4 dual-signature PD";  
    }  
    enum type4singlesigPD {  
      value 1;  
      description  
        "Type 4 single-signature PD";  
    }  
    enum type3dualsigPD {  
      value 2;  
      description  
        "Type 3 dual-signature PD";  
    }  
    enum type3singlesigPD {  
      value 3;  
      description  
        "Type 3 single-signature PD";  
    }  
    enum type4PSE {  
      value 4;  
      description  
        "Type 4 PSE";  
    }  
    enum type3PSE {  
      value 5;  
      description  
        "Type 3 PSE";  
    }  
  }  
  description  
    "Enumeration for the PD class";  
  reference  
    "IEEE Std 802.3, 30.12.2.1.29";  
}
```

```
typedef power-priority-type {  
  type enumeration {  
    enum low {  
      value 0;  
      description  
        "low priority PD";  
    }  
    enum high {  
      value 1;  
      description  
        "high priority PD";  
    }  
    enum critical {  
      value 2;  
      description  
        "critical priority PD";  
    }  
    enum unknown {
```

```
        value 3;
        description
            "priority unknown";
    }
}
description
    "Enumeration for possible priorities of a PD system";
reference
    "IEEE Std 802.3, 30.12.2.1.16";
}

typedef power-source-type {
    type enumeration {
        enum pse-primary {
            value 0;
            description
                "PSE powered by a primary power source";
        }
        enum pse-backup {
            value 1;
            description
                "PSE powered by a backup power source";
        }
        enum pse-unknown {
            value 2;
            description
                "PSE powered by an unknown power source";
        }
        enum pd-pse-and-local {
            value 3;
            description
                "PD powered by a PSE and locally";
        }
        enum pd-local-only {
            value 4;
            description
                "PD powered only locally";
        }
        enum pd-pse-only {
            value 5;
            description
                "PD powered by PD only";
        }
        enum pd-unknown {
            value 6;
            description
                "PD powered by an unknown source";
        }
    }
}
description
    "Enumeration for the power sources of the
    remote system. When the remote system is a PSE, it
    indicates whether it is being powered by a primary
    power source; a backup power source; or unknown.
    When the remote system is a PD, it indicates whether
    it is being powered by a PSE and locally;
    locally only; by a PSE only; or unknown.";
reference
    "IEEE Std 802.3, 30.12.2.1.15";
```

```
}

typedef powering-status-type {
  type enumeration {
    enum 4PdualsigPD {
      value 0;
      description
        "4-pair powering a dual-signature PD";
    }
    enum 4PsinglePD {
      value 1;
      description
        "4-pair powering a single-signature PD";
    }
    enum 2P {
      value 2;
      description
        "2-pair powering";
    }
  }
  description
    "Enumeration for the power status of the PSE";
  reference
    "IEEE Std 802.3, 30.12.2.1.23";
}
```

```
typedef powered-status-type {
  type enumeration {
    enum 4PdualsigPD {
      value 0;
      description
        "4-pair powered dual-signature PD";
    }
    enum 2PdualsigPD {
      value 1;
      description
        "2-pair powered dual-signature PD";
    }
    enum singlePD {
      value 2;
      description
        "powered single-signature PD";
    }
  }
  description
    "Enumeration for the power status of the PSE";
  reference
    "IEEE Std 802.3, 30.12.2.1.24";
}
```

```
typedef power-pairs-type {
  type enumeration {
    enum altA {
      value 0;
      description
        "Alternative A";
    }
    enum altB {
      value 1;
    }
  }
}
```



```
        description
            "Alternative B";
    }
    enum both {
        value 2;
        description
            "both";
    }
}
description
    "Enumeration for the PSE Pinout Alternative";
reference
    "IEEE Std 802.3, 30.12.2.1.25";
}

typedef measurement-source-type {
    type enumeration {
        enum noRequest {
            description
                "No Request";
        }
        enum altA {
            description
                "Pairset Alternative A / Mode A";
        }
        enum altB {
            description
                "Pairset Alternative B / Mode B";
        }
        enum total {
            description
                "Port total";
        }
    }
}
description
    "PoE measurement source";
reference
    "IEEE Std 802.3, 30.12.2.1.42 and 30.12.3.1.42";
}

augment "/lldp:lldp/lldp:port" {
    description
        "Augments port with 802.3 port read-only objects";
    leaf auto-negotiation-supported {
        type boolean;
        config false;
        description
            "True if the port supports Auto-negotiation";
        reference
            "IEEE Std 802.3, 30.12.2.1.1";
    }
    leaf auto-negotiation-enabled {
        type boolean;
        config false;
        description
            "True if Auto-negotiation is enabled";
        reference
            "IEEE Std 802.3, 30.12.2.1.2";
    }
}
```

```

/* Deprecated MAU objects */
leaf auto-negotiation-cap {
  type binary {
    length "2";
  }
  config false;
  status deprecated;
  description
    "A read-only 2-octet value that contains the value (bitmap)
    of the ifMauAutoNegCapAdvertisedBits object
    (defined in IETF RFC 4836) which is associated with the
    given port on the local system.
    Superseded by auto-negotiation-cap-bits.";
  reference
    "IEEE Std 802.3, 30.12.2.1.3";
}
leaf operational-mau-type {
  type int32;
  config false;
  status deprecated;
  description
    "32-bit integer value that indicates the operational MAU
    type of the given port.
    Superseded by operational-pmd-type.";
  reference
    "IEEE Std 802.3, 30.12.2.1.4";
}
leaf auto-negotiation-cap-bits {
  type ieee802-phy:auto-neg-ability-bits;
  description
    "The capabilities advertised by the local Auto-Negotiation
    entity.";
  reference
    "IEEE Std 802.3, 30.6.1.1.6,
    aAutoNegadvertisedTechnologyAbility.";
}
leaf operational-pmd-type {
  type identityref {
    base ieee802-phy:pmd-type;
  }
  description
    "Operational PMD type of the given port";
  reference
    "IEEE Std 802.3, 30.5.1.1.2 aMAUType";
}
/* Clause 30 & Clause 145 PoE related LLDP definitions */
leaf power-port-class {
  type port-class-type;
  description
    "A read-only value that identifies the port Class of the
    given port";
  reference
    "IEEE Std 802.3, 30.12.2.1.5";
}
leaf mdi-power-supported {
  type boolean;
  description
    "True if MDI power is supported";
  reference

```

```
        "IEEE Std 802.3, 30.12.2.1.6";
    }
    leaf mdi-power-enabled {
        type boolean;
        description
            "True if MDI power is enabled";
        reference
            "IEEE Std 802.3, 30.12.2.1.7";
    }
    leaf power-pair-controlable {
        type boolean;
        description
            "True if the pair selection can be controlled";
        reference
            "IEEE Std 802.3, 30.12.2.1.8";
    }
    leaf power-pairs {
        type pse-pinout-type;
        description
            "Indicates which pinout alternative is used for PD detection
            and power";
        reference
            "IEEE Std 802.3, 30.12.2.1.9";
    }
    leaf local-power-class {
        type pse-power-class-type;
        description
            "PD Power Class";
        reference
            "IEEE Std 802.3, 30.12.2.1.10";
    }
    /* Obsolete Link Aggregation objects
       Link aggregation has not been part of IEEE Std 802.3
       since IEEE Std 802.1AX-2008 */
    leaf link-aggregation-status {
        type bits {
            bit aggregation-capability {
                position 0;
                description
                    "IEEE Std 802.3, 79.3.3.1";
            }
            bit aggregation-status {
                position 1;
                description
                    "IEEE Std 802.3, 79.3.3.1";
            }
        }
    }
    config false;
    status obsolete;
    description
        "The bitmap value which contains the link aggregation
        capabilities and the current aggregation
        status of the link";
    reference
        "IEEE Std 802.3, 30.12.2.1.11";
}
leaf aggregation-port-id {
    type int32;
    config false;
```

```
status obsolete;
description
  "The unique identifier allocated to this Aggregation Port
  by the local system.";
reference
  "IEEE Std 802.3, 30.12.2.1.12";
}
leaf local-max-frame-size {
  type int32;
  config false;
  status obsolete;
  description
    "An integer value indicating the maximum supported frame
    size in octets on the given port of the local system.";
  reference
    "IEEE Std 802.3, 30.12.2.1.13";
}
leaf power-type {
  type bits {
    bit type1-or-greater {
      position 0;
      description
        "0-type1, 1-greater than type1";
    }
    bit pse-or-pd {
      position 1;
      description
        "0-pse, 1-pd";
    }
  }
  config false;
  description
    "A read-only attribute that returns a bit string indicating
    whether the local system is a PSE or a PD and whether it
    is Type 1 or greater than Type 1. The first bit indicates
    Type 1 or greater than Type 1. The second bit indicates
    PSE or PD. A PSE sets this bit to indicate a PSE. A PD sets
    this bit to indicate a PD. See also aLldpXdot3LocPowerTypeExt.
    Superseded by power-support";
  reference
    "IEEE Std 802.3, 30.12.2.1.14";
}
leaf power-support {
  type bits {
    bit pse-pd {
      description
        "0-PD, 1-PSE";
    }
    bit pse-power-support {
      description
        "0-not supported, 1-supported";
    }
    bit pse-power-state {
      description
        "0-disabled 1-enabled";
    }
    bit pse-pairs-selection {
      description
        "0-pair selection not controllable
```

```
        1-pair selection    controllable";
    }
}
description
    "A read-only attribute that returns a bit string indicating
    the local system PoE support.";
reference
    "IEEE Std 802.3, 79.3.2.1 MDI power support";
}
leaf power-source {
    type power-source-type;
    config false;
    description
        "Indicates the power sources of the local system. A PSE
        indicates whether it is being powered by a primary power
        source; a backup power source; or unknown. A PD indicates
        whether it is being powered by a PSE and locally; by a PSE
        only; or unknown.";
    reference
        "IEEE Std 802.3, 30.12.2.1.15";
}
leaf pd-requested-power-value {
    type int32;
    config false;
    description
        "PD requested power value. For a PD, it is the power value
        that the PD has currently requested from the remote
        system. For a PSE, it is the power value that the PSE
        mirrors back to the remote system";
    reference
        "IEEE Std 802.3, 30.12.2.1.17";
}
leaf pd-requested-power-value-a {
    type int32;
    config false;
    description
        "A read-only attribute that returns the PD requested power
        value for the Mode A pairset in units of 0.1 W. For a PD,
        it's the power value that the PD has currently requested
        from the remote system for the Mode A pairset. For a PSE,
        it is the power value for the Alternative A pairset that
        the PSE echoes back to the remote system";
    reference
        "IEEE Std 802.3, 30.12.2.1.18";
}
leaf pd-requested-power-value-b {
    type int32;
    config false;
    description
        "A read-only attribute that returns the PD requested power
        value for the Mode B pairset in units of 0.1 W.
        For a PD, it is the power value that the PD has currently
        requested from the remote system for the Mode B pairset.
        For a PSE, it is the power value for the Alternative B
        pairset that the PSE echoes back to the remote system";
    reference
        "IEEE Std 802.3, 30.12.2.1.19";
}
leaf pse-allocated-power-value {
```

```
type int32;
config false;
description
  "PSE allocated power value. For a PSE, it is the power
   value that the PSE has currently allocated to the remote
   system. For a PD, it is the power value that the PD mirrors
   back to the remote system";
reference
  "IEEE Std 802.3, 30.12.2.1.20";
}
leaf pse-allocated-power-value-a {
  type int32;
  config false;
  description
    " PSE allocated power value for the Alternative A pairset
     in units of 0.1 W. For a PSE, it is the power value for the
     Alternative A pairset that the PSE has currently allocated
     to the remote system. For a PD, it is the power value for
     the Mode A pairset that the PD echoes back to the remote
     system.";
  reference
    "IEEE Std 802.3, 30.12.2.1.21";
}
leaf pse-allocated-power-value-b {
  type int32;
  config false;
  description
    "PSE allocated power value for the Alternative B pairset
     in units of 0.1 W. For a PSE, it is the power value for the
     Alternative B pairset that the PSE has currently
     allocated to the remote system. For a PD, it is the power
     value for the Mode B pairset that the PD echoes back to the
     remote system.";
  reference
    "IEEE Std 802.3, 30.12.2.1.22";
}
leaf pse-powering-status {
  type powering-status-type;
  config false;
  description
    "A read only value that indicates the powering status of
     the PSE. For a PD, the contents of this
     attribute are undefined.";
  reference
    "IEEE Std 802.3, 30.12.2.1.23";
}
leaf pd-powered-status {
  type powered-status-type;
  config false;
  description
    "A read only value that indicates the powering status of
     the PD. For a PSE, the contents of this attribute are
     undefined";
  reference
    "IEEE Std 802.3, 30.12.2.1.24";
}
leaf power-pairs-ext {
  type power-pairs-type;
  config false;
```

```
description
  "A read-only value that identifies the supported PSE
  Pinout Alternative specified in 145.2.4. For a PSE, this
  attribute contains the value of the aPSEPowerPairs
  attribute (see 30.9.1.1.4). For a PD, the contents of this
  attribute are undefined";
reference
  "IEEE Std 802.3, 30.12.2.1.25";
}
leaf power-class-ext-A {
  type power-class-ext-AB-type;
  config false;
  description
    "For a dual-signature PD, a read-only value that indicates
    the requested Class for Mode A during Physical Layer
    Classification (see 145.3.6). For a single-signature PD, a
    read-only value set to 'singlesig'. For a PSE connected to
    a dual-signature PD, a read-only value that indicates the
    currently assigned Class for Mode A (see 145.2.8). For a
    PSE connected to a single-signature PD or a PSE that
    operates only in 2-pair mode, a read-only value set to
    'singlesig'.";
  reference
    "IEEE Std 802.3, 30.12.2.1.26";
}
leaf power-class-ext-B {
  type power-class-ext-AB-type;
  config false;
  description
    "For a dual-signature PD, a read-only value that indicates
    the requested Class for Mode B during Physical Layer
    Classification (see 145.3.6). For a single-signature PD,
    a read-only value set to 'singlesig'. For a PSE connected to
    a dual-signature PD, a read-only value that indicates the
    currently assigned Class for Mode B (see 145.2.8). For a
    PSE connected to a single-signature PD or a PSE that
    operates only in 2-pair mode, a read-only value set to
    'singlesig'.";
  reference
    "IEEE Std 802.3, 30.12.2.1.27";
}
leaf power-class-ext {
  type power-class-ext-type;
  config false;
  description
    "For a single-signature PD, a read-only value that
    indicates the requested Class during Physical
    Layer Classification (see 145.3.6). For a dual-signature
    PD, a read-only value set to 'dualsig'.
    For a PSE connected to a single-signature PD or a PSE that
    operates only in 2-pair mode, a read-only value that
    indicates the currently assigned Class (see 145.2.8).
    For a PSE connected to a dual-signature PD,
    a read-only value set to 'dualsig'.";
  reference
    "IEEE Std 802.3, 30.12.2.1.28";
}
leaf power-type-ext {
  type power-type;
```

```
    config false;
    description
        "A read-only attribute that returns a value to indicate if
        the local system is a Type 3 or Type 4 PSE or PD and, in
        the case of a Type 3 or Type 4 PD, if it is a
        single-signature PD or a dual-signature
        PD";
    reference
        "IEEE Std 802.3, 30.12.2.1.29";
}
leaf pd-load {
    type boolean;
    config false;
    description
        "For a dual-signature PD, a read-only attribute that
        returns whether the load of a dual-signature PD is
        electrically isolated, as defined in 79.3.2.10.2. For a
        single-signature PD or a PSE, the value of this attribute
        is FALSE";
    reference
        "IEEE Std 802.3, 30.12.2.1.30";
}
leaf pd-4pid {
    type boolean;
    config false;
    description
        "A read-only Boolean attribute indicating whether the local
        PD system supports powering of both PD Modes.";
    reference
        "IEEE Std 802.3, 30.12.2.1.31";
}
leaf pse-max-avail-power {
    type int32;
    config false;
    description
        "A read-only attribute that returns the local PSE maximum
        available power value in units of 0.1 W";
    reference
        "IEEE Std 802.3, 30.12.2.1.32";
}
leaf pse-autoclass-support {
    type boolean;
    config false;
    description
        "Indicates whether the local PSE system supports Autoclass.";
    reference
        "IEEE Std 802.3, 30.12.2.1.33";
}
leaf autoclass-completed {
    type boolean;
    config false;
    description
        "Indicates whether the local PSE system has completed the
        Autoclass measurement.";
    reference
        "IEEE Std 802.3, 30.12.2.1.34";
}
leaf autoclass-request {
    type boolean;
```



```
    config false;
    description
      "A read-only Boolean attribute indicating whether the local
      PD system is requesting an Autoclass measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.35";
  }
  leaf power-down-request {
    type int32;
    description
      "A read-write attribute that indicates the local PD system
      is requesting a power down when the value is 0x1D.";
    reference
      "IEEE Std 802.3, 30.12.2.1.36";
  }
  leaf power-down-time {
    type int32;
    description
      "A read-write attribute that indicates the number of
      seconds the PD requests to stay powered off. A value of
      zero indicates an indefinite amount of time.";
    reference
      "IEEE Std 802.3, 30.12.2.1.37";
  }
  /* PoE measurement objects */
  leaf meas-voltage-support {
    type boolean;
    description
      "A read-only attribute that indicates the local device is
      capable of providing a voltage measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.38";
  }
  leaf meas-current-support {
    type boolean;
    description
      "A read-only attribute that indicates the local device is
      capable of providing a current measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.39";
  }
  leaf meas-power-support {
    type boolean;
    description
      "A read-only attribute that indicates the local device is
      capable of providing a power measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.40";
  }
  leaf meas-energy-support {
    type boolean;
    description
      "A read-only attribute that indicates the local device is
      capable of providing a energy measurement.";
    reference
      "IEEE Std 802.3, 30.12.2.1.41";
  }
  leaf measurement-source {
    type bits {
```

```
    bit bit1 {
      position 0;
      description
        "???";MSB of Measurement Source. Used with the LSB of
        Measurement Source to determine where the measurement
        is to be taken. See IEEE Std 802.3, Table 79-21.";
    }
    bit bit2 {
      position 1;
      description
        "LSB of Measurement Source. Used with the MSB of
        ""; Measurement Source to determine where the measurement
        is to be taken. See IEEE Std 802.3, Table 79-21.";
    }
  }
  status obsolete;
  description
    "A read-write attribute value that indicates to local
    device on which Alternative or Mode the measurement
    is to be taken";
  reference
    "IEEE Std 802.3, 30.12.2.1.42";
}
leaf meas-voltage-request {
  type boolean;
  config false;
  description
    "A read-only attribute that indicates the local device is
    requesting a voltage measurement from the remote device.";
  reference
    "IEEE Std 802.3, 30.12.2.1.43";
}
leaf meas-current-request {
  type boolean;
  config false;
  description
    "A read-only attribute that indicates the local device is
    requesting a current measurement from the remote device.";
  reference
    "IEEE Std 802.3, 30.12.2.1.44";
}
leaf meas-power-request {
  type boolean;
  config false;
  description
    "A read-only attribute that indicates the local device is
    requesting a power measurement from the remote device.";
  reference
    "IEEE Std 802.3, 30.12.2.1.45";
}
leaf meas-energy-request {
  type boolean;
  config false;
  description
    "A read-only attribute that indicates the local device is
    requesting an energy measurement from the remote device.";
  reference
    "IEEE Std 802.3, 30.12.2.1.46";
}
```

```
leaf meas-voltage-valid {
  type boolean;
  config false;
  description
    "A read-only attribute that indicates the local device's
    voltage measurement is valid.";
  reference
    "IEEE Std 802.3, 30.12.2.1.47";
}
leaf meas-current-valid {
  type boolean;
  config false;
  description
    "A read-only attribute that indicates the local device's
    current measurement is valid.";
  reference
    "IEEE Std 802.3, 30.12.2.1.48";
}
leaf meas-power-valid {
  type boolean;
  config false;
  description
    "A read-only attribute that indicates the local device's
    power measurement is valid.";
  reference
    "IEEE Std 802.3, 30.12.2.1.49";
}
leaf meas-energy-valid {
  type boolean;
  config false;
  description
    "A read-only attribute that indicates the local device's
    energy measurement is valid.";
  reference
    "IEEE Std 802.3, 30.12.2.1.50";
}
leaf meas-voltage-uncertainty {
  type int32;
  config false;
  description
    "A read-only attribute that indicates the expanded
    uncertainty (coverage factor k = 2) for the device's
    voltage measurement.";
  reference
    "IEEE Std 802.3, 30.12.2.1.51";
}
leaf meas-current-uncertainty {
  type int32;
  config false;
  description
    "A read-only attribute that indicates the expanded
    uncertainty (coverage factor k = 2) for the device's
    current measurement.";
  reference
    "IEEE Std 802.3, 30.12.2.1.52";
}
leaf meas-power-uncertainty {
  type int32;
  config false;
```

```
description
  "A read-only attribute that indicates the expanded
  uncertainty (coverage factor k = 2) for the device's
  power measurement.";
reference
  "IEEE Std 802.3, 30.12.2.1.53";
}
leaf meas-energy-uncertainty {
  type int32;
  config false;
  description
    "A read-only attribute that indicates the expanded
    uncertainty (coverage factor k = 2) for the device's
    energy measurement.";
  reference
    "IEEE Std 802.3, 30.12.2.1.54";
}
leaf voltage-measurement {
  type int32;
  config false;
  description
    "A read-only attribute that returns the measured device
    voltage.";
  reference
    "IEEE Std 802.3, 30.12.2.1.55";
}
leaf current-measurement {
  type int32;
  config false;
  description
    "A read-only attribute that returns the measured device
    current.";
  reference
    "IEEE Std 802.3, 30.12.2.1.56";
}
leaf power-measurement {
  type int32;
  config false;
  description
    "A read-only attribute that returns the measured device
    power.";
  reference
    "IEEE Std 802.3, 30.12.2.1.57";
}
leaf energy-measurement {
  type int32;
  config false;
  description
    "A read-only attribute that returns the measured device
    energy.";
  reference
    "IEEE Std 802.3, 30.12.2.1.58";
}
leaf pse-power-price-index {
  type int32;
  config false;
  description
    "A read-only attribute that returns an index of the price
    of power being sourced by the PSE. For a PD, this value
```

```
        is undefined";
    reference
        "IEEE Std 802.3, 30.12.2.1.59";
}
leaf local-response {
    type int32;
    config false;
    description
        "The maximum time required to update
        pse-allocated-power-value";
    reference
        "IEEE Std 802.3, 30.12.2.1.60";
}
leaf local-system-ready {
    type boolean;
    config false;
    description
        "Initialization status of the Data Link Layer
        classification engine on the local system";
    reference
        "IEEE Std 802.3, 30.12.2.1.61";
}
leaf tx-system-value {
    type int32;
    config false;
    description
        "Returns the value of Tw_sys_tx that the local system can
        support in the transmit direction.";
    reference
        "IEEE Std 802.3, 30.12.2.1.62";
}
leaf tx-system-value-echo {
    type int32;
    config false;
    description
        "Returns the value of Tw_sys_tx that the remote system is
        advertising that it can support in the transmit direction
        and is echoed by the local system under the control of the
        EEE DLL receiver state diagram.";
    reference
        "IEEE Std 802.3, 30.12.2.1.63";
}
leaf rx-system-value {
    type int32;
    config false;
    description
        "Returns the value of Tw_sys_tx that the local system is
        requesting in the receive direction.";
    reference
        "IEEE Std 802.3, 30.12.2.1.64";
}
leaf rx-system-value-echo {
    type int32;
    config false;
    description
        "Returns the value of Tw_sys_tx that the remote system is
        advertising that it is requesting in the receive direction
        and is echoed by the local system under the control of the
        EEE DLL transmitter state diagram.";
```

```
        reference
            "IEEE Std 802.3, 30.12.2.1.65";
    }
    leaf fallback-system-value {
        type int32;
        config false;
        description
            "Returns the value of the fallback Tw_sys_tx that the local
            system is advertising to the remote system.";
        reference
            "IEEE Std 802.3, 30.12.2.1.66";
    }
    /* LDP EEE objects */
    leaf tx-dll-ready {
        type boolean;
        config false;
        description
            "Returns the initialization status of the EEE transmit Data
            Link Layer management function on the local system.";
        reference
            "IEEE Std 802.3, 30.12.2.1.67";
    }
    leaf rx-dll-ready {
        type boolean;
        config false;
        description
            "Returns the initialization status of the EEE receive Data
            Link Layer management function on the local system.";
        reference
            "IEEE Std 802.3, 30.12.2.1.68";
    }
    leaf dll-enabled {
        type boolean;
        config false;
        description
            "Returns the status of the EEE capability negotiation on
            the local system.";
        reference
            "IEEE Std 802.3, 30.12.2.1.69";
    }
    leaf tx-system-fw {
        type boolean;
        config false;
        description
            "Returns the value of LPI_FW that the local system can
            support in the transmit direction.";
        reference
            "IEEE Std 802.3, 30.12.2.1.70";
    }
    leaf tx-system-fw-echo {
        type boolean;
        config false;
        description
            "Returns the value of LPI_FW that the remote system is
            advertising that it can support in the transmit direction
            and is echoed by the local system under the control of the
            EEE DLL receiver state diagram.";
        reference
            "IEEE Std 802.3, 30.12.2.1.71";
    }
```

```
}
leaf rx-system-fw {
  type boolean;
  config false;
  description
    "Returns the value of LPI_FW that the local system is
    requesting in the receive direction.";
  reference
    "IEEE Std 802.3, 30.12.2.1.72";
}
leaf rx-system-fw-echo {
  type boolean;
  config false;
  description
    "Returns the value of LPI_FW that the remote system is
    advertising that it is requesting in the receive direction
    and is echoed by the local system under the control of the
    IEEE DLL transmitter state diagram.";
  reference
    "IEEE Std 802.3, 30.12.2.1.73";
}
/* LLDP preemption objects */
leaf preemption-supported {
  type boolean;
  config false;
  description
    "Indicates whether the given port (associated with the local
    system) supports the preemption capability.";
  reference
    "IEEE Std 802.3, 30.12.2.1.74";
}
leaf preemption-enabled {
  type boolean;
  config false;
  description
    "Indicates whether the preemption capability is enabled on
    the given port associated with the local system.";
  reference
    "IEEE Std 802.3, 30.12.2.1.75";
}
leaf preemption-active {
  type boolean;
  config false;
  description
    "Indicates whether the preemption capability is active on
    the given port associated with the local system.";
  reference
    "IEEE Std 802.3, 30.12.2.1.76";
}
leaf additional-fragment-size {
  type int32;
  config false;
  description
    "Indicates the minimum size of non-final fragments supported
    by the receiver on the given port associated with the local
    system. This value is expressed in units of 64 octets of
    additional fragment length.";
  reference
    "IEEE Std 802.3, 30.12.2.1.77";
}
```

```
    }  
  }  
  
  augment "/lldp:lldp/lldp:port" {  
    description  
      "Augments port with IEEE 802.3 port read-write objects";  
    leaf tlvs-port-config-enable {  
      type bits {  
        bit mac-phy-config-status {  
          position 0;  
          description  
            "IEEE Std 802.3, 30.12.1.1.1";  
        }  
        bit power-via-mdi {  
          position 1;  
          description  
            "IEEE Std 802.3, 30.12.1.1.1";  
        }  
        bit unused {  
          position 2;  
          description  
            "IEEE Std 802.3, 30.12.1.1.1";  
        }  
        bit max-frame-size {  
          position 3;  
          description  
            "IEEE Std 802.3, 30.12.1.1.1";  
        }  
        bit eee-tlv {  
          position 4;  
          description  
            "IEEE Std 802.3, 30.12.1.1.1";  
        }  
        bit eee-fast-wake-tlv {  
          position 5;  
          description  
            "IEEE Std 802.3, 30.12.1.1.1";  
        }  
        bit additional-ethernet-capabilities-tlv {  
          position 6;  
          description  
            "IEEE Std 802.3, 30.12.1.1.1";  
        }  
      }  
    }  
    description  
      "Bitmap that corresponds to an IEEE 802.3 subtype  
       associated with a specific IEEE 802.3 port config TLV";  
    reference  
      "IEEE Std 802.3, 30.12.1.1.1";  
  }  
  
  leaf local-power-priority {  
    type power-priority-type;  
    description  
      "Priority of a PD system. For a PSE, this is the priority  
       that the PSE assigns to the PD. For a PD, this is the  
       priority that the PD requests from the PSE.";  
    reference  
      "IEEE Std 802.3, 30.12.2.1.16";  
  }  
}
```



```
leaf local-measurement-source {
  type measurement-source-type;
  description
    "A read-write attribute value that indicates to local
    device on which Alternative or Mode the measurement
    is to be taken.";
  reference
    "IEEE Std 802.3, 30.12.2.1.42";
}
}

/* NOTE - lldp:remote-systems-data container is "config false;" */

augment "/lldp:lldp/lldp:port/lldp:remote-systems-data" {
  description
    "Augments port with IEEE 802.3 port config tlvs";
  leaf auto-negotiation-supported {
    type boolean;
    description
      "True if the port supports Auto-negotiation";
    reference
      "IEEE Std 802.3, 30.12.3.1.1";
  }
  leaf auto-negotiation-enabled {
    type boolean;
    description
      "True if Auto-negotiation is enabled";
    reference
      "IEEE Std 802.3, 30.12.3.1.2";
  }
}
/* Obsolete */
leaf auto-negotiation-cap {
  type binary {
    length "2";
  }
  status obsolete;
  description
    "A read-only 2-octet value that contains the value (bitmap)
    of the ifMauAutoNegCapAdvertisedBits object (defined in
    IETF RFC 4836) which is associated with the given port on
    the local system.
    Superseded by auto-negotiation-cap-bits.";
  reference
    "IEEE Std 802.3, 30.12.3.1.3";
}
leaf operational-mau-type {
  type int32;
  status obsolete;
  description
    "32-bit integer value that indicates the operational MAU
    type of the given port.
    Superseded by operational-pmd-type.";
  reference
    "IEEE Std 802.3, 30.12.3.1.4";
}
leaf auto-negotiation-cap-bits {
  type ieee802-phy:auto-neg-ability-bits;
  description
    "The capabilities advertised by the remote Auto-Negotiation
```

```
        entity.";
    reference
        "IEEE Std 802.3, 30.12.3.1.3,
        aAutoNegadvertisedTechnologyAbility.";
}
leaf operational-pmd-type {
    type identityref {
        base ieee802-phy:pmd-type;
    }
    description
        "The operational PMD type of the given port";
    reference
        "IEEE Std 802.3, 30.12.3.1.4";
}
leaf power-port-class {
    type port-class-type;
    description
        "A read-only value that identifies the port Class of the
        given port";
    reference
        "IEEE Std 802.3, 30.12.3.1.5";
}
leaf mdi-power-supported {
    type boolean;
    description
        "True if MDI power is supported";
    reference
        "IEEE Std 802.3, 30.12.3.1.6";
}
leaf mdi-power-enabled {
    type boolean;
    description
        "True if MDI power is enabled";
    reference
        "IEEE Std 802.3, 30.12.3.1.7";
}
leaf power-pair-controlable {
    type boolean;
    description
        "True if pair selection can be controlled";
    reference
        "IEEE Std 802.3, 30.12.3.1.8";
}
leaf power-pairs {
    type pse-pinout-type;
    description
        "Indicates which pinout alternative is used for PD
        detection and power";
    reference
        "IEEE Std 802.3, 30.12.3.1.9";
}
leaf power-class {
    type pse-power-class-type;
    description
        "PD Power Class";
    reference
        "IEEE Std 802.3, 30.12.3.1.10";
}
/* Obsolete Link Aggregation objects
```

```
Link aggregation has not been part of IEEE Std 802.3
since IEEE Std 802.1AX-2008 */
leaf link-aggregation-status {
  type bits {
    bit aggregation-capability {
      position 0;
      description
        "IEEE Std 802.3, 79.3.3.1";
    }
    bit aggregation-status {
      position 1;
      description
        "IEEE Std 802.3, 79.3.3.1";
    }
    bit bit2-reserved {
      position 2;
      description
        "IEEE Std 802.3, 79.3.3.1";
    }
    bit bit3-reserved {
      position 3;
      description
        "IEEE Std 802.3, 79.3.3.1";
    }
    bit bit4-reserved {
      position 4;
      description
        "IEEE Std 802.3, 79.3.3.1";
    }
    bit bit5-reserved {
      position 5;
      description
        "IEEE Std 802.3, 79.3.3.1";
    }
    bit bit6-reserved {
      position 6;
      description
        "IEEE Std 802.3, 79.3.3.1";
    }
    bit bit7-reserved {
      position 7;
      description
        "IEEE Std 802.3, 79.3.3.1";
    }
  }
  status obsolete;
  description
    "The bitmap value which contains the link aggregation
    capabilities and the current aggregation status
    of the link";
  reference
    "IEEE Std 802.3, 30.12.3.1.11";
}
leaf aggregation-port-id {
  type int32;
  status obsolete;
  description
    "The unique identifier allocated to this Aggregation Port
    by the local system.";
```

```
reference
  "IEEE Std 802.3, 30.12.3.1.12";
}
leaf local-max-frame-size {
  type int32;
  status obsolete;
  description
    "An integer value indicating the maximum supported frame
    size in octets on the given port of the local system.";
  reference
    "IEEE Std 802.3, 30.12.3.1.13";
}
leaf power-type {
  type bits {
    bit type1-or-greater {
      position 0;
      description
        "0-type1, 1-greater than type1";
    }
    bit pse-or-pd {
      position 1;
      description
        "0-pse, 1-pd";
    }
  }
  status obsolete;
  description
    "A read-only attribute that returns a bit string indicating
    whether the local system is a PSE or a PD and whether it
    is Type 1 or greater than Type 1. The first bit indicates
    Type 1 or greater than Type 1. The second bit indicates
    PSE or PD. A PSE sets this bit to indicate a PSE. A PD
    sets this bit to indicate a PD. See also
    aLldpXdot3LocPowerTypeExt.";
  reference
    "IEEE Std 802.3, 30.12.3.1.14";
}
leaf power-source {
  type power-source-type;
  description
    "Indicates the power sources of the remote system. A PSE
    indicates whether it is being powered by a primary power
    source; a backup power source; or unknown. A PD indicates
    whether it is being powered by a PSE and locally; by a PSE
    only; or unknown.";
  reference
    "IEEE Std 802.3, 30.12.3.1.15";
}
leaf power-priority {
  type power-priority-type;
  description
    "The priority of the PD system received from the remote
    system";
  reference
    "IEEE Std 802.3, 30.12.3.1.16";
}
leaf pd-requested-power-value {
  type int32;
  description
```

```
"PD requested power value that was used by the remote
system to compute the power value that is has currently
allocated to the PD.";
reference
  "IEEE Std 802.3, 30.12.3.1.17";
}
leaf pd-requested-power-value-a {
  type int32;
  description
    "A read-only attribute that returns the PD requested power
    value for the Mode A pairset that was used by the remote
    system to compute the power value that it has currently
    allocated to the PD. For a PSE, it is the PD requested
    power value for the Alternative A pairset received from
    the remote system. For a PD, it is the PD requested power
    value for the Alternative A pairset that the PSE echoes
    back to the remote system. The definition and encoding of
    PD requested power value for the Mode A pairset is the same
    as described in aLldpXdot3LocPDRequestedPowerValueA";
  reference
    "IEEE Std 802.3, 30.12.3.1.18";
}
leaf pd-requested-power-value-b {
  type int32;
  description
    "A read-only attribute that returns the PD requested power
    value for the Mode B pairset that was used by the remote
    system to compute the power value that it has currently
    allocated to the PD. For a PSE, it is the PD requested
    power value for the Alternative B pairset received from
    the remote system. For a PD, it is the PD requested power
    value for the Alternative B pairset that the PSE echoes
    back to the remote system. The definition and encoding of
    PD requested power value for the Mode B pairset is the
    same as described in aLldpXdot3LocPDRequestedPowerValueB";
  reference
    "IEEE Std 802.3, 30.12.3.1.19";
}
leaf pse-allocated-power-value {
  type int32;
  description
    "PSE allocated power value. For a PSE, it is the power value
    that the PSE has currently allocated to the remote system.
    For a PD, it is the power value that the PD mirrors back to
    the remote system";
  reference
    "IEEE Std 802.3, 30.12.3.1.20";
}
leaf pse-allocated-power-value-a {
  type int32;
  description
    "A read-only attribute that returns the PSE allocated power
    value for the Alternative A pairset received from the
    remote system. For a PSE, it is the PSE allocated power
    value for the Alternative A pairset that was echoed back
    by the remote PD. For a PD, it is the PSE allocated power
    value for the Mode A pairset received from the remote
    system. The definition and encoding of PSE allocated power
    value for the Alternative A pairset is the same as
```

```
        described in aLldpXdot3LocPSEAllocatedPowerValueA";
    reference
        "IEEE Std 802.3, 30.12.3.1.21";
}
leaf pse-allocated-power-value-b {
    type int32;
    description
        "A read-only attribute that returns the PSE allocated power
        value for the Alternative B pairset received from the
        value for the Alternative B pairset that was echoed back
        by the remote PD. For a PD, it is the PSE allocated power
        value for the Mode B pairset received from the remote
        system. The definition and encoding of PSE allocated power
        value for the Alternative B pairset is the same as
        described in aLldpXdot3LocPSEAllocatedPowerValueB";
    reference
        "IEEE Std 802.3, 30.12.3.1.22";
}
leaf pse-powering-status {
    type powering-status-type;
    description
        "A read only value that indicates the powering status of
        the remote PSE. For a PD, the contents of this attribute
        are undefined.";
    reference
        "IEEE Std 802.3, 30.12.3.1.23";
}
leaf pd-powered-status {
    type powered-status-type;
    description
        "A read only value that indicates the powering status of
        the PD. For a PSE, the contents of this attribute are
        undefined";
    reference
        "IEEE Std 802.3, 30.12.3.1.24";
}
leaf power-pairs-ext {
    type power-pairs-type;
    description
        "A read-only value that identifies the supported PSE
        Pinout Alternative specified in 145.2.4. For a PD, this
        attribute contains the value of the aPSEPowerPairs
        attribute (see 30.9.1.1.4). For a PSE, the contents of
        this attribute are undefined";
    reference
        "IEEE Std 802.3, 30.12.3.1.25";
}
leaf power-class-ext-A {
    type power-class-ext-AB-type;
    description
        "For a dual-signature PD, a read-only value that indicates
        the currently assigned Class for Mode A by the remote
        4-pair PSE. For a single-signature PD or a dual-signature
        PD connected to a 2-pair only PSE, a read-only value set
        to 'singlesig' by the remote PSE. For a PSE connected to a
        dual-signature PD, a read-only value that indicates the
        requested Class for Mode A during Physical Layer
        classification (see 145.2.8) by the remote PD. For a PSE
        connected to a single-signature PD, a read-only value set
```

```
        to 'singlesig' by the remote PD";
    reference
        "IEEE Std 802.3, 30.12.3.1.26";
}
leaf power-class-ext-B {
    type power-class-ext-AB-type;
    description
        "For a dual-signature PD, a read-only value that indicates
        the currently assigned Class for Mode B by the remote
        4-pair PSE. For a single-signature PD or a dual-signature
        PD connected to a 2-pair only PSE, a read-only value set
        to 'singlesig' by the remote PSE. For a PSE connected to a
        dual-signature PD, a read-only value that indicates the
        requested Class for Mode B during Physical Layer
        classification (see 145.2.8) by the remote PD. For a PSE
        connected to a single-signature PD, a read-only value set
        to 'singlesig' by the remote PD";
    reference
        "IEEE Std 802.3, 30.12.3.1.27";
}
leaf power-class-ext {
    type power-class-ext-type;
    description
        "For a single-signature PD or a dual-signature PD connected
        to a 2-pair only PSE, a read-only value that indicates the
        currently assigned Class by the remote PSE. For a
        dual-signature PD connected to a 4-pair capable PSE, a
        read-only value set to 'dualsig' by the remote PSE. For a
        PSE connected to a single-signature PD, a read-only value
        that indicates the requested Class during Physical Layer
        classification (see 145.2.8) by the remote PD. For a PSE
        connected to a dual-signature PD, a read-only value set to
        'dualsig' by the remote PD.";
    reference
        "IEEE Std 802.3, 30.12.3.1.28";
}
leaf power-type-ext {
    type power-type;
    description
        "A read-only attribute that returns a value to indicate if
        the remote system is a Type 3 or Type 4 PSE or PD and, in
        the case of a Type 3 or Type 4 PD, if it is a
        single-signature PD or dual-signature PD.";
    reference
        "IEEE Std 802.3, 30.12.3.1.29";
}
leaf pd-load {
    type boolean;
    description
        "For a PSE, a read-only attribute that returns whether the
        load of the remote dual-signature PD is electrically
        isolated, as defined in 79.3.2.10.2. For a PD, this
        attribute is set to FALSE.";
    reference
        "IEEE Std 802.3, 30.12.3.1.30";
}
leaf pd-4pid {
    type boolean;
    description
```

```
        "A read-only Boolean attribute indicating whether the
        remote PD system supports powering of both PD Modes.";
    reference
        "IEEE Std 802.3, 30.12.3.1.31";
}
leaf pse-max-avail-power {
    type int32;
    description
        "A read-only attribute that returns the remote PSE maximum
        available power value in units of 0.1 W";
    reference
        "IEEE Std 802.3, 30.12.3.1.32";
}
leaf pse-autoclass-support {
    type boolean;
    description
        "Indicates whether the remote PSE system supports
        Autoclass.";
    reference
        "IEEE Std 802.3, 30.12.3.1.33";
}
leaf autoclass-completed {
    type boolean;
    description
        "Indicates whether the remote PSE system has completed the
        Autoclass measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.34";
}
leaf autoclass-request {
    type boolean;
    description
        "A read-only Boolean attribute indicating whether the
        remote PD system is requesting an Autoclass measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.35";
}
leaf power-down-request {
    type int32;
    description
        "A read-write attribute that indicates the remote PD system
        is requesting a power down when the value is 0x1D.";
    reference
        "IEEE Std 802.3, 30.12.3.1.36";
}
leaf power-down-time {
    type int32;
    description
        "A read-only attribute that indicates the number of seconds
        the remote PD requests to stay powered off. A value of zero
        indicates an indefinite amount of time";
    reference
        "IEEE Std 802.3, 30.12.3.1.37";
}
/* PoE measurement objects */
leaf meas-voltage-support {
    type boolean;
    description
        "A read-only attribute that indicates the remote device is
```



```
        capable of providing a voltage measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.38";
}
leaf meas-current-support {
    type boolean;
    description
        "A read-only attribute that indicates the remote device is
        capable of providing a current measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.39";
}
leaf meas-power-support {
    type boolean;
    description
        "A read-only attribute that indicates the remote device is
        capable of providing a power measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.40";
}
leaf meas-energy-support {
    type boolean;
    description
        "A read-only attribute that indicates the remote device is
        capable of providing an energy measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.41 ";
}
leaf measurement-source {
    type bits {
        bit bit1 {
            position 0;
            description
                "-";
        }
        bit bit2 {
            position 1;
            description
                "-";
        }
    }
    status obsolete;
    description
        "A read-write attribute value that indicates on which
        Alternative or Mode the measurement was taken by the remote
        device.
        Superseded by rem-measurement-source.";
    reference
        "IEEE Std 802.3, 30.12.3.1.42";
}
leaf rem-measurement-source {
    type measurement-source-type;
    description
        "A read-only attribute that indicates which Alternative or
        Mode the measurement is to be taken";
    reference
        "IEEE Std 802.3, 30.12.3.1.42";
}
leaf meas-voltage-request {
```

```
type boolean;
description
  "A read-only attribute that indicates the remote device is
  requesting a voltage measurement from the local device.";
reference
  "IEEE Std 802.3, 30.12.3.1.43";
}
leaf meas-current-request {
  type boolean;
  description
    "A read-only attribute that indicates the remote device is
    requesting a current measurement from the local device.";
  reference
    "IEEE Std 802.3, 30.12.3.1.44";
}
leaf meas-power-request {
  type boolean;
  description
    "A read-only attribute that indicates the remote device is
    requesting a power measurement from the local device.";
  reference
    "IEEE Std 802.3, 30.12.3.1.45";
}
leaf meas-energy-request {
  type boolean;
  description
    "A read-only attribute that indicates the remote device is
    requesting an energy measurement from the local device.";
  reference
    "IEEE Std 802.3, 30.12.3.1.46";
}
leaf meas-voltage-valid {
  type boolean;
  description
    "A read-only attribute that indicates the remote device's
    voltage measurement is valid.";
  reference
    "IEEE Std 802.3, 30.12.3.1.47";
}
leaf meas-current-valid {
  type boolean;
  description
    "A read-only attribute that indicates the remote device's
    current measurement is valid.";
  reference
    "IEEE Std 802.3, 30.12.3.1.48";
}
leaf meas-power-valid {
  type boolean;
  description
    "A read-only attribute that indicates the remote device's
    power measurement is valid.";
  reference
    "IEEE Std 802.3, 30.12.3.1.49";
}
leaf meas-energy-valid {
  type boolean;
  description
    "A read-only attribute that indicates the remote device's
```

```
        energy measurement is valid.";
    reference
        "IEEE Std 802.3, 30.12.3.1.50";
}
leaf meas-voltage-uncertainty {
    type int32;
    description
        "A read-only attribute that indicates the expanded
        uncertainty (coverage factor k = 2) for the remote device's
        voltage measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.51";
}
leaf meas-current-uncertainty {
    type int32;
    description
        "A read-only attribute that indicates the expanded
        uncertainty (coverage factor k = 2) for the remote device's
        current measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.52";
}
leaf meas-power-uncertainty {
    type int32;
    description
        "A read-only attribute that indicates the expanded
        uncertainty (coverage factor k = 2) for the remote device's
        power measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.53";
}
leaf meas-energy-uncertainty {
    type int32;
    description
        "A read-only attribute that indicates the expanded
        uncertainty (coverage factor k = 2) for the remote device's
        energy measurement.";
    reference
        "IEEE Std 802.3, 30.12.3.1.54";
}
leaf voltage-measurement {
    type int32;
    description
        "A read-only attribute that returns the measured remote
        device voltage.";
    reference
        "IEEE Std 802.3, 30.12.3.1.55";
}
leaf current-measurement {
    type int32;
    description
        "A read-only attribute that returns the measured remote
        device current.";
    reference
        "IEEE Std 802.3, 30.12.3.1.56";
}
leaf power-measurement {
    type int32;
    description
```

```
        "A read-only attribute that returns the measured remote
        device power.";
    reference
        "IEEE Std 802.3, 30.12.3.1.57";
}
leaf energy-measurement {
    type int32;
    description
        "A read-only attribute that returns the measured remote
        device energy.";
    reference
        "IEEE Std 802.3, 30.12.3.1.58";
}
leaf pse-power-price-index {
    type int32;
    description
        "A read-only attribute that returns an index of the price
        of power being sourced by the remote PSE. For a PSE, this
        value is undefined.";
    reference
        "IEEE Std 802.3, 30.12.3.1.59";
}
leaf tx-system-value {
    type int32;
    description
        "Returns the value of Tw_sys_tx that the remote system can
        support in the transmit direction.";
    reference
        "IEEE Std 802.3, 30.12.3.1.60";
}
leaf tx-system-value-echo {
    type int32;
    description
        "Returns the value of Tw_sys_tx that the local system is
        advertising that it can support in the transmit direction
        and is echoed by the local system under the control of the
        IEEE DLL receiver state diagram.";
    reference
        "IEEE Std 802.3, 30.12.3.1.61";
}
leaf rx-system-value {
    type int32;
    description
        "Returns the value of Tw_sys_tx that the remote system is
        requesting in the receive direction.";
    reference
        "IEEE Std 802.3, 30.12.3.1.62";
}
leaf rx-system-value-echo {
    type int32;
    description
        "Returns the value of Tw_sys_tx that the local system is
        advertising that it is requesting in the receive direction
        and is echoed by the local system under the control of the
        IEEE DLL transmitter state diagram.";
    reference
        "IEEE Std 802.3, 30.12.3.1.63 ";
}
leaf fallback-system-value {
```

```
type int32;
description
  "Returns the value of the fallback Tw_sys_tx that the
  remote system is advertising.";
reference
  "IEEE Std 802.3, 30.12.3.1.64";
}
/* LDP EEE objects */
leaf tx-system-fw {
  type boolean;
  description
    "Returns the value of LPI_FW that the remote system can
    support in the transmit direction.";
  reference
    "IEEE Std 802.3, 30.12.3.1.65 (second instance of
    30.12.3.1.24 in IEEE Std 802.3-2022)";
}
leaf tx-system-fw-echo {
  type boolean;
  description
    "Returns the value of LPI_FW that the local system is
    advertising that it can support in the transmit direction
    and is echoed by the remote system under the control of the
    EEE DLL receiver state diagram.";
  reference
    "IEEE Std 802.3, 30.12.3.1.66 (second instance of
    30.12.3.1.25 in IEEE Std 802.3-2022)";
}
leaf rx-system-fw {
  type boolean;
  description
    "Returns the value of LPI_FW that the remote system is
    requesting in the receive direction.";
  reference
    "IEEE Std 802.3, 30.12.3.1.67 (second instance of
    30.12.3.1.26 in IEEE Std 802.3-2022)";
}
leaf rx-system-fw-echo {
  type boolean;
  description
    "Returns the value of LPI_FW that the local system is
    advertising that it is requesting in the receive direction
    and is echoed by the remote system under the control of the
    EEE DLL transmitter state diagram.";
  reference
    "IEEE Std 802.3, 30.12.3.1.68 (second instance of
    30.12.3.1.27 in IEEE Std 802.3-2022)";
}
/* LLDP preemption objects */
leaf preemption-supported {
  type boolean;
  description
    "Indicates whether the given port (associated with the
    remote system) supports the preemption capability.";
  reference
    "IEEE Std 802.3, 30.12.3.1.69 (second instance of
    30.12.3.1.28 in IEEE Std 802.3-2022)";
}
leaf preemption-enabled {
```

```
    type boolean;
    description
      "Indicates whether the preemption capability is enabled on
       the given port associated with the remote system.";
    reference
      "IEEE Std 802.3, 30.12.3.1.70 (second instance of
       30.12.3.1.29 in IEEE Std 802.3-2022)";
  }
  leaf preemption-active {
    type boolean;
    description
      "Indicates whether the preemption capability is active on
       the given port associated with the remote system.";
    reference
      "IEEE Std 802.3, 30.12.3.1.71 (second instance of
       30.12.3.1.30 in IEEE Std 802.3-2022)";
  }
  leaf additional-fragment-size {
    type int32;
    description
      "Indicates the minimum size of non-final fragments supported
       by the receiver on the given port associated with the
       remote system. This value is expressed in units of 64
       octets of additional fragment length.";
    reference
      "IEEE Std 802.3, 30.12.3.1.72 (second instance of
       30.12.3.1.31 in IEEE Std 802.3-2022)";
  }
}
}
```

5.3.2.5 Ethernet PHY module

```
module ieee802-ethernet-phy {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-phy";
  prefix ieee802-eth-phy;

  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991 YANG Types";
  }
  import iana-if-type {
    prefix ianaift;
    reference
      "IANA Interface Types";
  }
  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ieee802-ethernet-phy-type {
    prefix ieee802-phy;
    reference
      "IEEE Std 802.3-2022";
  }
}
```

```
}

organization
  "IEEE Std 802.3 Ethernet Working Group";
contact
  "Web URL: http://www.ieee802.org/3/";
description
  "This module contains YANG definitions for 802.3 PHYs and PMDs.";

revision 2025-09-10 {
  description
    "Initial revision derived from IEEE8023-MAU-MIB
    using smidump.";
  reference
    "IEEE Std 802.3-2022, unless dated explicitly";
}

feature fec-support {
  description
    "This object indicates support for one or more FEC modes.";
}

feature baset-snr-support {
  description
    "This object indicates support for SNR measurement for
    BASE-T interfaces.";
}

feature time-sync-support {
  description
    "This object indicates support for time synchronisation.";
}

feature pcs-lane-statistics-support {
  description
    "This object indicates support for PCS Lane statistics.";
}

feature auto-negotiation-support {
  description
    "This object indicates support for Auto-Negotiation.";
}

typedef snr-value {
  type int32 {
    range "-127..127";
  }
  units "0.1dB";
  description
    "SNR units of 0.1 dB to an accuracy of 0.5 dB within
    the range of -12.7 dB to 12.7 dB.";
}

typedef counter-1000-second {
  type yang:counter64;
  description
    "This counter increments no more than 1000 times per second.";
}
```

```
typedef counter-5000-second {
    type yang:counter64;
    description
        "This counter increments no more than 5000 times per second.";
}

typedef counter-pcs-lane {
    type yang:counter64;
    description
        "This counter has a maximum per second increment rate of:
        1.2 million for 1000 Mb/s
        5 million for 10 Gb/s and 40 Gb/s
        2.5 million for 100 Gb/s";
}

typedef enabled-state {
    type enumeration {
        enum unknown {
            description
                "Initializing.";
        }
        enum disabled {
            description
                "Disabled";
        }
        enum enabled {
            description
                "Enabled";
        }
    }
    description
        "Enabled, disabled, or unknown.";
}

augment "/if:interfaces/if:interface" {
    when "derived-from-or-self(if:type, 'ianaift:ethernetCsmacd') " {
        description
            "Applies to all Ethernet interfaces.";
    }
    description
        "Augment interface model with Ethernet interface
        information";
    container phy-table {
        description
            "Table of descriptive and status information
            about the Ethernet interface physical layer.";
        container control {
            description
                "Control information for the Ethernet interface
                physical layer.";
            leaf phy-reset {
                type boolean;
                description
                    "Setting this to 'True' resets the interface in the
                    same manner as would a power-off, power-on cycle of
                    at least 0.5 s duration.";
                reference
                    "IEEE Std 802.3, 30.5.1.2.1 acResetMAU";
            }
        }
    }
}
```



```
leaf phy-admin-control {
  type enumeration {
    enum operational {
      description
        "Operational";
    }
    enum standby {
      description
        "Standby";
    }
    enum shutdown {
      description
        "Shutdown";
    }
  }
  description
    "The configured state of the interface. This object
    MAY be implemented as a read-only object by devices
    that do not implement software control of the interface
    state. Some devices may not support setting the value
    of this object to some of the enumerated values.

    This can be set to 'standby', 'operational', and
    'shutdown' state, except that a 'standby' action
    to a multidrop interface will cause the interface
    to enter the 'shutdown' management state.";
  reference
    "IEEE Std 802.3, 30.5.1.2.2 acMAUAdminControl";
}
container fec-control {
  if-feature "fec-support";
  presence "The port supports one or more FEC modes.";
  description
    "FEC Control.";
  leaf fec-mode-control {
    type enumeration {
      enum disabled {
        description
          "FEC disabled";
      }
      enum enabled {
        description
          "FEC enabled";
      }
      enum BASE-R-enabled {
        description
          "BASE-R FEC enabled";
      }
      enum RS-FEC-enabled {
        description
          "RS-FEC enabled";
      }
      enum RS-FEC-Int-enabled {
        description
          "RS-FEC-Int enabled";
      }
    }
  }
  description
    "The FEC mode across the MDI of the port.
```

```
        BASE-R-enabled, RS-FEC-enabled, and
        RS-FEC-Int-enabled are only used for PHYs which
        support more than one type of FEC operation.
        When Clause 73 Auto-Negotiation is enabled, this
        leaf becomes read-only.";
    reference
        "IEEE Std 802.3, 30.5.1.1.16";
}
leaf rs-fec-bypass-control {
    type enabled-state;
    description
        "Is RS-FEC bypass enabled?";
    reference
        "IEEE Std 802.3, 30.5.1.1.31";
}
}
}
container status {
    config false;
    description
        "Status for the port.";
    leaf phy-type {
        type identityref {
            base ieee802-phy:phy-type;
        }
        description
            "This object identifies the PHY type.";
        reference
            "IEEE Std 802.3, 30.3.2.1.2, aPHYType.";
    }
    leaf pmd-type {
        type identityref {
            base ieee802-phy:pmd-type;
        }
        description
            "This object identifies the PMD type.";
        reference
            "IEEE Std 802.3, 30.5.1.1.2, aMAUType.";
    }
    leaf pmd-type-list {
        type ieee802-phy:pmd-type-list;
        description
            "PMD types this interface supports.";
    }
    leaf phy-status {
        type enumeration {
            enum unknown {
                description
                    "Initializing, true state
                     not yet known";
            }
            enum operational {
                description
                    "Powered and connected";
            }
            enum standby {
                description
                    "Inactive but on";
            }
        }
    }
}
```

```
        enum shutdown {
            description
                "Inactive and powered down";
        }
    }
    description
        "The current state of the interface.";
    reference
        "IEEE Std 802.3, 30.5.1.1.7, aMAUAdminState";
}
leaf phy-media-available {
    type ieee802-phy:media-available-type;
    description
        "This object reports if the media is available.";
    reference
        "IEEE Std 802.3, 30.5.1.1.4, aMediaAvailable.";
}
/* Jabber */
leaf jabber-state {
    type enumeration {
        enum other {
            description
                "undefined";
        }
        enum unknown {
            description
                "initializing, true state
                not yet known";
        }
        enum normal {
            description
                "state is true or normal";
        }
        enum fault {
            description
                "state is false, fault, or
                abnormal";
        }
    }
}
description
    "The JABBER state of the multidrop interface.";
reference
    "IEEE Std 802.3, 30.5.1.1.6, aJabber.jabberFlag.";
}
leaf jabbering-state-enters {
    type yang:counter64;
    description
        "The number of times this interface detected JABBER.";
    reference
        "IEEE Std 802.3,
        30.5.1.1.6, aJabber.jabberCounter.";
}
/* FEC */
container fec-status {
    if-feature "fec-support";
    presence "The port supports one or more FEC modes.";
    description
        "FEC Status.";
    leaf fec-ability {
```

```
type bits {
  bit other {
    description
      "Other";
  }
  bit base-r {
    description
      "BASE-R FEC supported";
  }
  bit rs-fec {
    description
      "RS-FEC supported";
  }
  bit rs-fec-bypass {
    description
      "RS-FEC bypass supported";
  }
  bit rs-fec-bypass-indication {
    description
      "RS-FEC error indication bypass supported";
  }
  bit pcs-fec-bypass {
    description
      "PCS FEC error bypass supported";
  }
  bit pcs-fec-bypass-indication {
    description
      "PCS FEC error indication bypass supported";
  }
}
description
  "The FEC modes this port supports.";
reference
  "IEEE Std 802.3, 30.5.1.1.15, 30.5.1.1.16,
  30.5.1.1.29, 30.5.1.1.30, 30.5.1.1.33,
  30.5.1.1.34";
}
leaf fec-mode-status {
  type enumeration {
    enum unknown {
      description
        "Initializing, true state not yet known.";
    }
    enum disabled {
      description
        "FEC disabled";
    }
    enum enabled {
      description
        "FEC enabled";
    }
    enum BASE-R-enabled {
      description
        "BASE-R FEC enabled";
    }
    enum RS-FEC-enabled {
      description
        "RS-FEC enabled";
    }
  }
}
```

```
        enum RS-FEC-Int-enabled {
            description
                "RS-FEC-Int enabled";
        }
    }
    description
        "The FEC mode of the PHY.
        BASE-R-enabled, RS-FEC-enabled, and
        RS-FEC-Int-enabled are only used by PHYs which
        support more than one type of FEC operation.";
    reference
        "IEEE Std 802.3, 30.5.1.1.16";
}
leaf rs-fec-bypass-enabled-status {
    type enabled-state;
    description
        "Is RS-FEC bypass indication enabled?";
    reference
        "IEEE Std 802.3, 30.5.1.1.32";
}
}
/* MultiGBASE-T SNR */
container baset-snr {
    if-feature "baset-snr-support";
    presence "The port supports BASE-T SNR measurements.";
    description
        "BASE-T SNR Status.";
    leaf snr-op-margin-chnl-a {
        type snr-value;
        description
            "The current SNR operating margin on channel A
            on this port for MultiGBASE-T PMAs.";
        reference
            "IEEE Std 802.3, 30.5.1.1.19";
    }
    leaf snr-op-margin-chnl-b {
        type snr-value;
        description
            "The current SNR operating margin on channel B
            on this port for MultiGBASE-T PMAs.";
        reference
            "IEEE Std 802.3, 30.5.1.1.20";
    }
    leaf snr-op-margin-chnl-c {
        type snr-value;
        description
            "The current SNR operating margin on channel C
            on this port for MultiGBASE-T PMAs.";
        reference
            "IEEE Std 802.3, 30.5.1.1.21";
    }
    leaf snr-op-margin-chnl-d {
        type snr-value;
        description
            "The current SNR operating margin on channel D
            on this port for MultiGBASE-T PMAs.";
        reference
            "IEEE Std 802.3, 30.5.1.1.22";
    }
}
```

```
}
/* TimeSync */
container time-sync {
  if-feature "time-sync-support";
  presence "The port supports Time Sync.";
  description
    "Time Sync Status.";
  leaf time-sync-ability {
    type bits {
      bit transmit {
        description
          "Transmit";
      }
      bit receive {
        description
          "Receive";
      }
    }
    description
      "The TimeSync capabilities of this port.";
    reference
      "IEEE Std 802.3, 30.13.1.1, 30.13.1.2";
  }
  leaf time-sync-delay-tx-max {
    type uint32;
    units "nano-seconds";
    description
      "The maximum transmit data delay between
        the xMII input and the MDI output.";
    reference
      "IEEE Std 802.3, 30.13.1.3";
  }
  leaf time-sync-delay-tx-min {
    type uint32;
    units "nano-seconds";
    description
      "The minimum transmit data delay between
        the xMII input and the MDI output.";
    reference
      "IEEE Std 802.3, 30.13.1.4";
  }
  leaf time-sync-delay-rx-max {
    type int32;
    units "nano-seconds";
    description
      "The maximum receive data delay between
        the MDI input and the xMII output.";
    reference
      "IEEE Std 802.3, 30.13.1.5";
  }
  leaf time-sync-delay-rx-min {
    type int32;
    units "nano-seconds";
    description
      "The minimum receive data delay between
        the MDI input and the xMII output.";
    reference
      "IEEE Std 802.3, 30.13.1.6";
  }
}
```

```
}
}
container statistics {
  config false;
  description
    "Counters for the port.";
  leaf phy-media-losts {
    type yang:counter64;
    description
      "The number of times phy-media-available changed
        from 'available'.";
    reference
      "IEEE Std 802.3, 30.5.1.1.5 aLoseMediaCounter.";
  }
  leaf jabbering-state-enters {
    type yang:counter64;
    description
      "The number of times this interface detected JABBER.";
    reference
      "IEEE Std 802.3, 30.5.1.1.6, aJabber.jabberCounter.";
  }
  leaf false-carriers {
    type yang:counter64;
    description
      "The number of false carrier
        events during IDLE in 100BASE-X and
        1000BASE-X links.

        For all other interface types, this counter
        will always indicate zero.

        It can increment at a maximum rate of once
        per 100 ms for 100BASE-X and once per 10 us
        for 1000BASE-X.";
    reference
      "IEEE Std 802.3, 30.5.1.1.10, aFalseCarriers.";
  }
  leaf pcs-coding-violations {
    type yang:counter64;
    description
      "The number of PCS coding violations for 100 Mb/s and
        1000 Mb/s ports.

        This counter increments at a maximum rate of 25 thousand
        per second for 100 Mb/s implementations and
        125 thousand per second for 1000 Mb/s
        implementations.

        The contents of this
        attribute is undefined when FEC is operating.";
    reference
      "IEEE Std 802.3, 30.5.1.1.14";
  }
  leaf fast-retrain-count-local {
    type counter-1000-second;
    description
      "The number of fast retrains initiated
        by the local device.";
    reference
```

```
        "IEEE Std 802.3, 30.5.1.1.24";
    }
    leaf fast-retrain-count-peer {
        type counter-1000-second;
        description
            "The number of fast retrains initiated
             by the link partner.";
        reference
            "IEEE Std 802.3, 30.5.1.1.25";
    }
}
container pcs-lane-statistics {
    if-feature "pcs-lane-statistics-support";
    config false;
    description
        "Per PCS lane counters for the port.
         These counters apply to the following:
         100BASE-PX,
         10/25/40/50/100/200/400GBASE-R
         100GBASE-P
         10GBASE-PR
         10/1GBASE-PRX";
    list pcs-lane {
        key "pcs-lane-index";
        description
            "Per PCS lane counters for the port.";
        leaf pcs-lane-index {
            type uint32 {
                range "0..255";
            }
            description
                "PCS lane index.";
        }
        leaf fec-corrected-blocks {
            type counter-pcs-lane;
            description
                "The number of blocks that FEC corrected.";
            reference
                "IEEE Std 802.3, 30.5.1.1.17";
        }
        leaf fec-uncorrected-blocks {
            type counter-pcs-lane;
            description
                "The number of blocks that FEC could not correct.";
            reference
                "IEEE Std 802.3, 30.5.1.1.18";
        }
    }
}
}
container auto-neg-table {
    if-feature "auto-negotiation-support";
    description
        "Configuration and status objects for Auto-Negotiation
         on Ethernet interfaces.";
    container control {
        description
            "Auto-Negotiation control.";
        leaf auto-neg-control {
```



```
type boolean;
default "true";
description
  "Enable/disable Auto-Negotiation for this interface.";
reference
  "IEEE Std 802.3, 30.6.1.2.2, acAutoNegAdminControl.";
}
leaf auto-neg-restart {
  type boolean;
  description
    "Setting this to true restarts Auto-Negotiation if enabled.";
  reference
    "IEEE Std 802.3,
      30.6.1.2.1, acAutoNegRestartAutoConfig.";
}
}
container status {
  config false;
  description
    "Auto-Negotiation status.";
  leaf auto-neg-enabled {
    type boolean;
    description
      "Is Auto-Negotiation enabled for this interface.";
    reference
      "IEEE Std 802.3,
        30.6.1.1.2 aAutoNegAdminState.";
  }
  leaf auto-neg-remote-signaling {
    type boolean;
    description
      "Is the link partner using Auto-Negotiation?";
    reference
      "IEEE Std 802.3, 30.6.1.1.3, aAutoNegRemoteSignaling.";
  }
  leaf auto-neg-status {
    type enumeration {
      enum other {
        description
          "Unknown state.";
      }
      enum configuring {
        description
          "Auto-Negotiation is active.";
      }
      enum complete {
        description
          "Auto-Negotiation is complete.";
      }
      enum disabled {
        description
          "Auto-Negotiation s disabled.";
      }
      enum parallelDetectFail {
        description
          "Auto-Negotiation failed because the
            link partner does not support, or has
            disabled, Auto-Negotiation.";
      }
    }
  }
}
```

```
    }  
    description  
        "The current status of Auto-Negotiation.";  
    reference  
        "IEEE Std 802.3, 30.6.1.1.4, aAutoNegAutoConfig.";  
    }  
    leaf auto-neg-capability-bits {  
        type ieee802-phy:auto-neg-ability-bits;  
        description  
            "The capabilities of the local Auto-Negotiation entity.";  
        reference  
            "IEEE Std 802.3, 30.6.1.1.5,  
            aAutoNegLocalTechnologyAbility.";  
    }  
    leaf auto-neg-cap-advertised-bits {  
        type ieee802-phy:auto-neg-ability-bits;  
        description  
            "The capabilities advertised by the local  
            Auto-Negotiation entity.";  
        reference  
            "IEEE Std 802.3, 30.6.1.1.6,  
            aAutoNegAdvertisedTechnologyAbility.";  
    }  
    leaf auto-neg-cap-received-bits {  
        type ieee802-phy:auto-neg-ability-bits;  
        description  
            "Auto-Negotiation capabilities received from  
            the link partner.";  
        reference  
            "IEEE Std 802.3, 30.6.1.1.7,  
            aAutoNegReceivedTechnologyAbility.";  
    }  
    }  
    }  
    }  
    }  
}
```

5.3.2.6 Ethernet PHY type module

```
module ieee802-ethernet-phy-type {  
    yang-version 1.1;  
    namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-phy-type";  
    prefix ieee802-eth-phy-type;  
  
    organization  
        "IEEE Std 802.3 Ethernet Working Group  
        Web URL: http://www.ieee802.org/3/";  
    contact  
        "Web URL: http://www.ieee802.org/3/";  
    description  
        "This module contains YANG definitions for Physical Layer device  
        (PHY) and Physical Medium Dependent (PMD) identities and related  
        types.";  
  
    revision 2025-09-10 {  
        description  
            "Initial revision derived from IANA-MAU-MIB REVISION
```

```
    202309070000Z using smidump.";
  reference
    "IEEE Std 802.3-2022, unless dated explicitly";
}

/* PHY type identities */

identity phy-type {
  description
    "Base type for PHY types.";
  reference
    "IEEE Std 802.3, 30.3.2.1.2 aPhyType";
}

identity phy-type-other {
  base phy-type;
  description
    "Undefined";
}

identity phy-type-unknown {
  base phy-type;
  description
    "Initializing, true state or type not yet known";
}

identity phy-type-none {
  base phy-type;
  description
    "MII present and nothing connected";
}

identity phy-type-10BASE-T {
  base phy-type;
  description
    "Clause 14 10 Mb/s PAM3";
}

identity phy-type-10BASE-T1L {
  base phy-type;
  description
    "Clause 146 10 Mb/s PAM3";
}

identity phy-type-10BASE-T1S {
  base phy-type;
  description
    "Clause 147 10 Mb/s DME";
}

identity phy-type-100BASE-T1 {
  base phy-type;
  description
    "Clause 96 100 Mb/s PAM3";
}

identity phy-type-100BASE-X {
  base phy-type;
  description
```

```
    "Clause 24 or subclause 66.1 100 Mb/s 4B/5B";
}

identity phy-type-1000BASE-H {
    base phy-type;
    description
        "Clause 115 1000 Mb/s PAM16-THP";
}

identity phy-type-1000BASE-T {
    base phy-type;
    description
        "Clause 40 1000 Mb/s 4D-PAM5";
}

identity phy-type-1000BASE-T1 {
    base phy-type;
    description
        "Clause 97 1000 Mb/s PAM3";
}

identity phy-type-1000BASE-X {
    base phy-type;
    description
        "Clause 36 or subclause 66.2 1000 Mb/s 8B/10B";
}

identity phy-type-2.5GBASE-T {
    base phy-type;
    description
        "Clause 126 2.5 Gb/s PAM16";
}

identity phy-type-2.5GBASE-T1 {
    base phy-type;
    description
        "Clause 149 2.5 Gb/s PAM4";
}

identity phy-type-2.5GBASE-X {
    base phy-type;
    description
        "Clause 127 2.5 Gb/s 8B/10B";
}

identity phy-type-5GBASE-R {
    base phy-type;
    description
        "Clause 129 5 Gb/s 64/66B";
}

identity phy-type-5GBASE-T {
    base phy-type;
    description
        "Clause 126 5 Gb/s PAM16";
}

identity phy-type-5GBASE-T1 {
    base phy-type;
```

```
    description
      "Clause 149 5 Gb/s PAM4";
  }

  identity phy-type-10G-1GBASE-PRX {
    base phy-type;
    description
      "Clause 76 10/1G-EPON 10 Gb/s 64B/66B downstream and
        1 Gb/s 8B/10B upstream";
  }

  identity phy-type-10GBASE-PR {
    base phy-type;
    description
      "Clause 76 10/10G-EPON 10 Gb/s 64B/66B";
  }

  identity phy-type-10GBASE-R {
    base phy-type;
    description
      "Clause 49 10 Gb/s 64B/66B";
  }

  identity phy-type-10GBASE-T {
    base phy-type;
    description
      "Clause 55 10 Gb/s DSQ128";
  }

  identity phy-type-10GBASE-T1 {
    base phy-type;
    description
      "Clause 149 10 Gb/s PAM4";
  }

  identity phy-type-10GBASE-X {
    base phy-type;
    description
      "Clause 48 10 Gb/s 4 lane 8B/10B";
  }

  identity phy-type-10GPASS-XR {
    base phy-type;
    description
      "Clause 101 PCS up to 10 Gb/s 64B/66B OFDM downstream
        and up to 1.6 Gb/s 64B/66B OFDMA upstream";
  }

  identity phy-type-10G-2p5GGBASE-SP {
    base phy-type;
    description
      "Subclause 164.3 10 Gb/s downstream and 2.5 Gb/s upstream
        256B/257B";
  }

  identity phy-type-10GBASE-SP {
    base phy-type;
    description
      "Subclause 164.3 10 Gb/s 256B/257B";
  }
```

```
}

identity phy-type-25G-10GBASE-PQ {
  base phy-type;
  description
    "Clause 142 25/10G-EPON 256B/257B";
}

identity phy-type-25GBASE-PQ {
  base phy-type;
  description
    "Clause 142 25/25G-EPON 256B/257B";
}

identity phy-type-25GBASE-R {
  base phy-type;
  description
    "Clause 107 25 Gb/s 64B/66B";
}

identity phy-type-25GBASE-T {
  base phy-type;
  description
    "Clause 113 25 Gb/s DSQ128";
}

identity phy-type-40GBASE-R {
  base phy-type;
  description
    "Clause 82 40 Gb/s multi-PCS lane 64B/66B";
}

identity phy-type-40GBASE-T {
  base phy-type;
  description
    "Clause 113 40 Gb/s DSQ128";
}

identity phy-type-50G-10GBASE-PQ {
  base phy-type;
  description
    "Clause 142 50/10G-EPON 256B/257B";
}

identity phy-type-50G-25GBASE-PQ {
  base phy-type;
  description
    "Clause 142 50/25G-EPON 256B/257B";
}

identity phy-type-50GBASE-PQ {
  base phy-type;
  description
    "Clause 142 50/50G-EPON 256B/257B";
}

identity phy-type-50GBASE-R {
  base phy-type;
  description
```

```
    "Clause 133 50 Gb/s multi-PCS lane 64B/66B";
}

identity phy-type-100GBASE-P {
    base phy-type;
    description
        "Clause 82 100 Gb/s multi-PCS lane using >2-level PAM";
}

identity phy-type-100GBASE-R {
    base phy-type;
    description
        "Clause 82 100 Gb/s multi-PCS lane using 2-level PAM";
}

identity phy-type-200GBASE-R {
    base phy-type;
    description
        "Clause 119 200 Gb/s multi-PCS lane 64B/66B";
}

identity phy-type-400GBASE-R {
    base phy-type;
    description
        "Clause 119 400 Gb/s multi-PCS lane 64B/66B";
}

identity phy-type-800GBASE-R {
    base phy-type;
    description
        "Clause 172 800 Gb/s multi-PCS lane 64B/66B";
}

/* PMD identities */

identity pmd-type {
    description
        "Base type for Physical Medium Dependent (PMD) sublayer types.";
    reference
        "IEEE Std 802.3, 30.5.1.1.2 aMAUType";
}

identity pmd-type-global {
    base pmd-type;
    description
        "Undefined";
}

identity pmd-type-other {
    base pmd-type;
    description
        "Nonconformant PMD.";
}

identity pmd-type-unknown {
    base pmd-type;
    description
        "Initializing, true state or type not yet known.";
}
```

```
identity pmd-type-10BASE-T {
  base pmd-type;
  description
    "Twisted-pair cabling MAU as specified in Clause 14";
}

identity pmd-type-10BASE-T1L {
  base pmd-type;
  description
    "Single balanced pair PHY as specified in Clause 146";
}

identity pmd-type-10BASE-T1S {
  base pmd-type;
  description
    "Single balanced pair PHY as specified in Clause 147, full
    duplex mode";
}

identity pmd-type-10BASE-T1SMD {
  base pmd-type;
  description
    "Single balanced pair PHY as specified in Clause 147,
    multidrop mode";
}

identity pmd-type-100BASE-BX10D {
  base pmd-type;
  description
    "One single-mode fiber OLT PHY as specified in Clause 58";
}

identity pmd-type-100BASE-BX10U {
  base pmd-type;
  description
    "One single-mode fiber ONU PHY as specified in Clause 58";
}

identity pmd-type-100BASE-FX {
  base pmd-type;
  description
    "100BASE-FX PMD as specified in Clause 26";
}

identity pmd-type-100BASE-LX10 {
  base pmd-type;
  description
    "Two fiber PHY as specified in Clause 58";
}

identity pmd-type-100BASE-T1 {
  base pmd-type;
  description
    "Single balanced twisted-pair copper cabling PHY as specified
    in Clause 96";
}

identity pmd-type-100BASE-TX {
```



```
    base pmd-type;
    description
      "Two-pair Category 5 twisted-pair cabling as specified in
       Clause 25";
  }

  identity pmd-type-1000BASE-BX10D {
    base pmd-type;
    description
      "One single-mode fiber OLT PHY as specified in Clause 59";
  }

  identity pmd-type-1000BASE-BX10U {
    base pmd-type;
    description
      "One single-mode fiber ONU PHY as specified in Clause 59";
  }

  identity pmd-type-1000BASE-KX {
    base pmd-type;
    description
      "1000BASE-KX PCS/PMA over an electrical backplane PMD as
       specified in Clause 70";
  }

  identity pmd-type-1000BASE-LX {
    base pmd-type;
    description
      "1000BASE-LX fiber over long-wavelength laser PMD as specified
       in Clause 38";
  }

  identity pmd-type-1000BASE-LX10 {
    base pmd-type;
    description
      "Two fiber 10 km PHY as specified in Clause 59";
  }

  identity pmd-type-1000BASE-PX10-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
       supporting a distance of at least 10 km, and a split of at
       least 1:16";
  }

  identity pmd-type-1000BASE-PX10-U {
    base pmd-type;
    description
      "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
       supporting a distance of at least 10 km, and a split of at
       least 1:16";
  }

  identity pmd-type-1000BASE-PX20-D {
    base pmd-type;
    description
      "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
       supporting a distance of at least 20 km, and a split of at
```

```
        least 1:16";
    }

    identity pmd-type-1000BASE-PX20-U {
        base pmd-type;
        description
            "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
            supporting a distance of at least 20 km, and a split of at
            least 1:16";
    }

    identity pmd-type-1000BASE-PX30-D {
        base pmd-type;
        description
            "One single-mode fiber OMP OLT PHY, as specified in Clause 60,
            supporting a distance of at least 20 km, and a split of at
            least 1:32";
    }

    identity pmd-type-1000BASE-PX30-U {
        base pmd-type;
        description
            "One single-mode fiber OMP ONU PHY, as specified in Clause 60,
            supporting a distance of at least 20 km, and a split of at
            least 1:32";
    }

    identity pmd-type-1000BASE-PX40-D {
        base pmd-type;
        description
            "One single-mode fiber OMP OLT PHY as specified in Clause 60,
            supporting a distance of at least 20 km, and a split of at
            least 1:64";
    }

    identity pmd-type-1000BASE-PX40-U {
        base pmd-type;
        description
            "One single-mode fiber OMP ONU PHY as specified in Clause 60,
            supporting a distance of at least 20 km, and a split of at
            least 1:64";
    }

    identity pmd-type-1000BASE-RHA {
        base pmd-type;
        description
            "Plastic optical fiber PHY as specified in Clause 115.";
    }

    identity pmd-type-1000BASE-RHB {
        base pmd-type;
        description
            "Plastic optical fiber PHY as specified in Clause 115.";
    }

    identity pmd-type-1000BASE-RHC {
        base pmd-type;
        description
            "Plastic optical fiber PHY as specified in Clause 115.";
```

```
}

identity pmd-type-1000BASE-SX {
  base pmd-type;
  description
    "1000BASE-SX fiber over short-wavelength laser PMD as
    specified in Clause 38";
}

identity pmd-type-1000BASE-T {
  base pmd-type;
  description
    "Four-pair Category 5 twisted-pair cabling PHY as specified in
    Clause 40";
}

identity pmd-type-1000BASE-T1 {
  base pmd-type;
  description
    "Single twisted-pair copper cable PHY as specified in Clause
    97";
}

identity pmd-type-1000BASE-X {
  base pmd-type;
  description
    "1000BASE-X PCS/PMA as specified in Clause 36 over undefined
    PMD";
}

identity pmd-type-2.5GBASE-KX {
  base pmd-type;
  description
    "2.5GBASE-X PMD as specified in Clause 128 over an electrical
    backplane as specified in Clause 128";
}

identity pmd-type-2.5GBASE-T {
  base pmd-type;
  description
    "Four-pair twisted-pair balanced copper cabling PHY as
    specified in Clause 126";
}

identity pmd-type-2.5GBASE-T1 {
  base pmd-type;
  description
    "Single balanced pair of conductors PHY as specified in Clause
    149";
}

identity pmd-type-2.5GBASE-X {
  base pmd-type;
  description
    "2.5GBASE-X PCS/PMA as specified in Clause 127 over undefined
    PMD";
}

identity pmd-type-5GBASE-KR {
```

```
base pmd-type;
description
  "5GBASE-KR PMD as specified in Clause 130 over an electrical
  backplane as specified in Clause 130";
}

identity pmd-type-5GBASE-R {
  base pmd-type;
  description
    "5GBASE-R PCS/PMA as specified in Clause 129 over undefined
    PMD";
}

identity pmd-type-5GBASE-T {
  base pmd-type;
  description
    "Four-pair twisted-pair balanced copper cabling PHY as
    specified in Clause 126";
}

identity pmd-type-5GBASE-T1 {
  base pmd-type;
  description
    "Single balanced pair of conductors PHY as specified in Clause
    149";
}

identity pmd-type-10G-1GBASE-PRX-D1 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
}

identity pmd-type-10G-1GBASE-PRX-D2 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
}

identity pmd-type-10G-1GBASE-PRX-D3 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
}

identity pmd-type-10G-1GBASE-PRX-D4 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
}

identity pmd-type-10G-1GBASE-PRX-U1 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
```

```
        1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
    }

    identity pmd-type-10G-1GBASE-PRX-U2 {
        base pmd-type;
        description
            "One single-mode fiber 10.3125 GBd continuous downstream /
            1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
    }

    identity pmd-type-10G-1GBASE-PRX-U3 {
        base pmd-type;
        description
            "One single-mode fiber 10.3125 GBd continuous downstream /
            1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
    }

    identity pmd-type-10G-1GBASE-PRX-U4 {
        base pmd-type;
        description
            "One single-mode fiber 10.3125 GBd continuous downstream /
            1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
    }

    identity pmd-type-10GBASE-BR10-D {
        base pmd-type;
        description
            "One single-mode fiber OLT PHY supporting a distance of at
            least 10 km as specified in Clause 158";
    }

    identity pmd-type-10GBASE-BR10-U {
        base pmd-type;
        description
            "One single-mode fiber ONU PHY supporting a distance of at
            least 10 km as specified in Clause 158";
    }

    identity pmd-type-10GBASE-BR20-D {
        base pmd-type;
        description
            "One single-mode fiber OLT PHY supporting a distance of at
            least 20 km as specified in Clause 158";
    }

    identity pmd-type-10GBASE-BR20-U {
        base pmd-type;
        description
            "One single-mode fiber ONU PHY supporting a distance of at
            least 20 km as specified in Clause 158";
    }

    identity pmd-type-10GBASE-BR40-D {
        base pmd-type;
        description
            "One single-mode fiber OLT PHY supporting a distance of at
            least 40 km as specified in Clause 158";
    }
```

```
identity pmd-type-10GBASE-BR40-U {
  base pmd-type;
  description
    "One single-mode fiber ONU PHY supporting a distance of at
    least 40 km as specified in Clause 158";
}

identity pmd-type-10GBASE-CX4 {
  base pmd-type;
  description
    "10GBASE-CX4 copper over 8 pair 100-Ohm balanced cable as
    specified in Clause 54";
}

identity pmd-type-10GBASE-ER {
  base pmd-type;
  description
    "10GBASE-ER fiber over 1550nm optics as specified in Clause 52";
}

identity pmd-type-10GBASE-EW {
  base pmd-type;
  description
    "10GBASE-EW fiber over 1550nm optics as specified in Clause 52";
}

identity pmd-type-10GBASE-KR {
  base pmd-type;
  description
    "10GBASE-KR PCS/PMA over an electrical backplane PMD as
    specified in Clause 72";
}

identity pmd-type-10GBASE-KX4 {
  base pmd-type;
  description
    "10GBASE-KX4 PCS/PMA over an electrical backplane PMD as
    specified in Clause 71";
}

identity pmd-type-10GBASE-LR {
  base pmd-type;
  description
    "10GBASE-LR fiber over 1310nm optics as specified in Clause 52";
}

identity pmd-type-10GBASE-LRM {
  base pmd-type;
  description
    "10GBASE-LRM fiber over 1310 nm optics as specified in
    Clause 68";
}

identity pmd-type-10GBASE-LW {
  base pmd-type;
  description
    "10GBASE-LW fiber over 1310nm optics as specified in Clause 52";
}
```

```
identity pmd-type-10GBASE-LX4 {
  base pmd-type;
  description
    "10GBASE-LX4 fiber over 4 lane 1310nm optics as specified in
    Clause 53";
}

identity pmd-type-10GBASE-PR-D1 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
}

identity pmd-type-10GBASE-PR-D2 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
}

identity pmd-type-10GBASE-PR-D3 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
}

identity pmd-type-10GBASE-PR-D4 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream OLT PHY as specified in Clause 75";
}

identity pmd-type-10GBASE-PR-U1 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
}

identity pmd-type-10GBASE-PR-U3 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
}

identity pmd-type-10GBASE-PR-U4 {
  base pmd-type;
  description
    "One single-mode fiber 10.3125 GBd continuous downstream /
    1.25 GBd burst mode upstream ONU PHY as specified in Clause 75";
}

identity pmd-type-10GBASE-R {
  base pmd-type;
  description
```

```
    "10GBASE-R PCS/PMA as specified in Clause 49 over undefined
    PMD";
}

identity pmd-type-10GBASE-SR {
    base pmd-type;
    description
        "10GBASE-SR fiber over 850nm optics as specified in Clause 52";
}

identity pmd-type-10GBASE-SW {
    base pmd-type;
    description
        "10GBASE-SW fiber over 850nm optics as specified in Clause 52";
}

identity pmd-type-10GBASE-T {
    base pmd-type;
    description
        "Four-pair twisted-pair balanced copper cabling PHY as
        specified in Clause 55";
}

identity pmd-type-10GBASE-T1 {
    base pmd-type;
    description
        "Single balanced pair of conductors PHY as specified in Clause
        149";
}

identity pmd-type-10GBASE-X {
    base pmd-type;
    description
        "10GBASE-X PCS/PMA as specified in Clause 48 over undefined
        PMD";
}

identity pmd-type-10GPASS-XR {
    base pmd-type;
    description
        "Coax cable distribution network PHY continuous
        downstream/burst mode upstream PHY as specified in Clause 100
        and Clause 101";
}

identity pmd-type-10G-2p5GGBASE-SP1-Dx {
    base pmd-type;
    description
        "One single-mode fiber 10.3125 GBd continuous downstream /
        2.578125 GBd burst mode upstream OLT PHY supporting
        channel x of FSR set 1, as specified in 164.2,";
}

identity pmd-type-10G-2p5GGBASE-SP1-Uxy {
    base pmd-type;
    description
        "One single-mode fiber 10.3125 GBd continuous downstream /
        2.578125 GBd burst mode upstream ONU PHY supporting
        channels x to y of FSR set 1, as specified in 164.2";
}
```



```
}

identity pmd-type-10GBASE-SP {
  base pmd-type;
  description
    "Subclause 164.3 10 Gb/s 256B/257B";
}

identity pmd-type-25G-10GBASE-PQG-D2 {
  base pmd-type;
  description
    "One single mode fiber, 1 x 25.78125 GBd continuous
    transmission / 1 x 10.3125 GBd burst mode reception, medium
    power class, as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQG-D3 {
  base pmd-type;
  description
    "One single mode fiber, 1 x 25.78125 GBd continuous
    transmission / 1 x 10.3125 GBd burst mode reception, high
    power class, as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQG-U2 {
  base pmd-type;
  description
    "One single mode fiber, 1 x 25.78125 GBd continuous reception
    / 1 x 10.3125 GBd burst mode transmission, medium power class,
    as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQG-U3 {
  base pmd-type;
  description
    "One single mode fiber, 1 x 25.78125 GBd continuous reception
    / 1 x 10.3125 GBd burst mode transmission, high power class,
    as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQX-D2 {
  base pmd-type;
  description
    "One single mode fiber, 1 x 25.78125 GBd continuous
    transmission / 1 x 10.3125 GBd burst mode reception, medium
    power class, as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQX-D3 {
  base pmd-type;
  description
    "One single mode fiber, 1 x 25.78125 GBd continuous
    transmission / 1 x 10.3125 GBd burst mode reception, high
    power class, as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQX-U2 {
  base pmd-type;
  description
```

```
"One single mode fiber, 1 x 25.78125 GBd continuous reception
/ 1 x 10.3125 GBd burst mode transmission, medium power class,
as specified in Clause 141";
}

identity pmd-type-25G-10GBASE-PQX-U3 {
  base pmd-type;
  description
    "One single mode fiber, 1 x 25.78125 GBd continuous reception
    / 1 x 10.3125 GBd burst mode transmission, high power class,
    as specified in Clause 141";
}

identity pmd-type-25GBASE-BR10-D {
  base pmd-type;
  description
    "One single-mode fiber OLT PHY supporting a distance of at
    least 10 km as specified in Clause 159";
}

identity pmd-type-25GBASE-BR10-U {
  base pmd-type;
  description
    "One single-mode fiber ONU PHY supporting a distance of at
    least 10 km as specified in Clause 159";
}

identity pmd-type-25GBASE-BR20-D {
  base pmd-type;
  description
    "One single-mode fiber OLT PHY supporting a distance of at
    least 20 km as specified in Clause 159";
}

identity pmd-type-25GBASE-BR20-U {
  base pmd-type;
  description
    "One single-mode fiber ONU PHY supporting a distance of at
    least 20 km as specified in Clause 159";
}

identity pmd-type-25GBASE-BR40-D {
  base pmd-type;
  description
    "One single-mode fiber OLT PHY supporting a distance of at
    least 40 km as specified in Clause 159";
}

identity pmd-type-25GBASE-BR40-U {
  base pmd-type;
  description
    "One single-mode fiber ONU PHY supporting a distance of at
    least 40 km as specified in Clause 159";
}

identity pmd-type-25GBASE-CR {
  base pmd-type;
  description
    "25GBASE-R PCS/PMA over shielded balanced copper cable PMD as
```

```
        specified in Clause 110";
    }

    identity pmd-type-25GBASE-CR-S {
        base pmd-type;
        description
            "25GBASE-R PCS/PMA over shielded balanced copper cable PMD as
            specified in Clause 110 without support for RS-FEC ";
    }

    identity pmd-type-25GBASE-ER {
        base pmd-type;
        description
            "25GBASE-R PCS/PMA over single-mode fiber PMD, with extended
            reach, as specified in Clause 114";
    }

    identity pmd-type-25GBASE-KR {
        base pmd-type;
        description
            "25GBASE-R PCS/PMA over an electrical backplane PMD as
            specified in Clause 111";
    }

    identity pmd-type-25GBASE-KR-S {
        base pmd-type;
        description
            "25GBASE-R PCS/PMA over an electrical backplane PMD as
            specified in Clause 111 without support for RS-FEC";
    }

    identity pmd-type-25GBASE-LR {
        base pmd-type;
        description
            "25GBASE-R PCS/PMA over single-mode fiber PMD, with long
            reach, as specified in Clause 114";
    }

    identity pmd-type-25GBASE-PQG-D2 {
        base pmd-type;
        description
            "One single mode fiber, 1 x 25.78125 GBd continuous
            transmission / 1 x 25.78125 GBd burst mode reception, medium
            power class, as specified in Clause 141";
    }

    identity pmd-type-25GBASE-PQG-D3 {
        base pmd-type;
        description
            "One single mode fiber, 1 x 25.78125 GBd continuous
            transmission / 1 x 25.78125 GBd burst mode reception, high
            power class, as specified in Clause 141";
    }

    identity pmd-type-25GBASE-PQG-U2 {
        base pmd-type;
        description
            "One single mode fiber, 1 x 25.78125 GBd continuous reception
            / 1 x 25.78125 GBd burst mode transmission, medium power
```

```
        class, as specified in Clause 141";
    }

    identity pmd-type-25GBASE-PQG-U3 {
        base pmd-type;
        description
            "One single mode fiber, 1 x 25.78125 GBd continuous reception
            / 1 x 25.78125 GBd burst mode transmission, high power class,
            as specified in Clause 141";
    }

    identity pmd-type-25GBASE-PQX-D2 {
        base pmd-type;
        description
            "One single mode fiber, 1 x 25.78125 GBd continuous
            transmission / 1 x 25.78125 GBd burst mode reception, medium
            power class, as specified in Clause 141";
    }

    identity pmd-type-25GBASE-PQX-D3 {
        base pmd-type;
        description
            "One single mode fiber, 1 x 25.78125 GBd continuous
            transmission / 1 x 25.78125 GBd burst mode reception, high
            power class, as specified in Clause 141";
    }

    identity pmd-type-25GBASE-PQX-U2 {
        base pmd-type;
        description
            "One single mode fiber, 1 x 25.78125 GBd continuous reception
            / 1 x 25.78125 GBd burst mode transmission, medium power
            class, as specified in Clause 141";
    }

    identity pmd-type-25GBASE-PQX-U3 {
        base pmd-type;
        description
            "One single mode fiber, 1 x 25.78125 GBd continuous reception
            / 1 x 25.78125 GBd burst mode transmission, high power class,
            as specified in Clause 141";
    }

    identity pmd-type-25GBASE-R {
        base pmd-type;
        description
            "PCS as specified in Clause 107 with PMA as specified in
            Clause 109 over undefined PMD";
    }

    identity pmd-type-25GBASE-SR {
        base pmd-type;
        description
            "25GBASE-R PCS/PMA over multimode fiber PMD as specified in
            Clause 112";
    }

    identity pmd-type-25GBASE-T {
        base pmd-type;
```

```
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 113";
  }

  identity pmd-type-40GBASE-CR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
       PMD as specified in Clause 85";
  }

  identity pmd-type-40GBASE-ER4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD, with
       extended reach, as specified in Clause 87";
  }

  identity pmd-type-40GBASE-FR {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over single mode fiber PMD as specified in
       Clause 89";
  }

  identity pmd-type-40GBASE-KR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over an electrical backplane PMD as
       specified in Clause 84";
  }

  identity pmd-type-40GBASE-LR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD, with
       long reach, as specified in Clause 87";
  }

  identity pmd-type-40GBASE-R {
    base pmd-type;
    description
      "Multi-lane PCS as specified in Clause 82 over undefined
       PMA/PMD";
  }

  identity pmd-type-40GBASE-SR4 {
    base pmd-type;
    description
      "40GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
       specified in Clause 86";
  }

  identity pmd-type-40GBASE-T {
    base pmd-type;
    description
      "Four-pair twisted-pair balanced copper cabling PHY as
       specified in Clause 113";
```

```
}

identity pmd-type-50G-10GBASE-PQG-D2 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous
     transmission / 1 x 10.3125 GBd burst mode reception, medium
     power class, as specified in Clause 141";
}

identity pmd-type-50G-10GBASE-PQG-D3 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous
     transmission / 1 x 10.3125 GBd burst mode reception, high
     power class, as specified in Clause 141";
}

identity pmd-type-50G-10GBASE-PQG-U2 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous reception
     / 1 x 10.3125 GBd burst mode transmission, medium power class,
     as specified in Clause 141";
}

identity pmd-type-50G-10GBASE-PQG-U3 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous reception
     / 1 x 10.3125 GBd burst mode transmission, high power class,
     as specified in Clause 141";
}

identity pmd-type-50G-10GBASE-PQX-D2 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous
     transmission / 1 x 10.3125 GBd burst mode reception, medium
     power class, as specified in Clause 141";
}

identity pmd-type-50G-10GBASE-PQX-D3 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous
     transmission / 1 x 10.3125 GBd burst mode reception, high
     power class, as specified in Clause 141";
}

identity pmd-type-50G-10GBASE-PQX-U2 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous reception
     / 1 x 10.3125 GBd burst mode transmission, medium power class,
     as specified in Clause 141";
}

identity pmd-type-50G-10GBASE-PQX-U3 {
```

```
base pmd-type;
description
  "One single mode fiber, 2 x 25.78125 GBd continuous reception
  / 1 x 10.3125 GBd burst mode transmission, high power class,
  as specified in Clause 141";
}

identity pmd-type-50G-25GBASE-PQG-D2 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous
    transmission / 1 x 25.78125 GBd burst mode reception, medium
    power class, as specified in Clause 141";
}

identity pmd-type-50G-25GBASE-PQG-D3 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous
    transmission / 1 x 25.78125 GBd burst mode reception, high
    power class, as specified in Clause 141";
}

identity pmd-type-50G-25GBASE-PQG-U2 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous reception
    / 1 x 25.78125 GBd burst mode transmission, medium power
    class, as specified in Clause 141";
}

identity pmd-type-50G-25GBASE-PQG-U3 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous reception
    / 1 x 25.78125 GBd burst mode transmission, high power class,
    as specified in Clause 141";
}

identity pmd-type-50G-25GBASE-PQX-D2 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous
    transmission / 1 x 25.78125 GBd burst mode reception, medium
    power class, as specified in Clause 141";
}

identity pmd-type-50G-25GBASE-PQX-D3 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous
    transmission / 1 x 25.78125 GBd burst mode reception, high
    power class, as specified in Clause 141";
}

identity pmd-type-50G-25GBASE-PQX-U2 {
  base pmd-type;
  description
    "One single mode fiber, 2 x 25.78125 GBd continuous reception
```

```
    / 1 x 25.78125 GBd burst mode transmission, medium power
    class, as specified in Clause 141";
}

identity pmd-type-50G-25GBASE-PQX-U3 {
    base pmd-type;
    description
        "One single mode fiber, 2 x 25.78125 GBd continuous reception
        / 1 x 25.78125 GBd burst mode transmission, high power class,
        as specified in Clause 141";
}

identity pmd-type-50GBASE-BR10-D {
    base pmd-type;
    description
        "One single-mode fiber OLT PHY supporting a distance of at
        least 10 km as specified in Clause 160";
}

identity pmd-type-50GBASE-BR10-U {
    base pmd-type;
    description
        "One single-mode fiber ONU PHY supporting a distance of at
        least 10 km as specified in Clause 160";
}

identity pmd-type-50GBASE-BR20-D {
    base pmd-type;
    description
        "One single-mode fiber OLT PHY supporting a distance of at
        least 20 km as specified in Clause 160";
}

identity pmd-type-50GBASE-BR20-U {
    base pmd-type;
    description
        "One single-mode fiber ONU PHY supporting a distance of at
        least 20 km as specified in Clause 160";
}

identity pmd-type-50GBASE-BR40-D {
    base pmd-type;
    description
        "One single-mode fiber OLT PHY supporting a distance of at
        least 40 km as specified in Clause 160";
}

identity pmd-type-50GBASE-BR40-U {
    base pmd-type;
    description
        "One single-mode fiber ONU PHY supporting a distance of at
        least 40 km as specified in Clause 160";
}

identity pmd-type-50GBASE-CR {
    base pmd-type;
    description
        "50GBASE-R PCS/PMA over shielded copper balanced cable PMD as
        specified in Clause 136 ";
}
```



```
}

identity pmd-type-50GBASE-ER {
    base pmd-type;
    description
        "50GBASE-R PCS/PMA over single-mode fiber PMD with reach up to
        at least 40 km as specified in Clause 139";
}

identity pmd-type-50GBASE-FR {
    base pmd-type;
    description
        "50GBASE-R PCS/PMA over single mode fiber PMD as specified in
        Clause 139";
}

identity pmd-type-50GBASE-KR {
    base pmd-type;
    description
        "50GBASE-R PCS/PMA over an electrical backplane PMD as
        specified in Clause 137";
}

identity pmd-type-50GBASE-LR {
    base pmd-type;
    description
        "50GBASE-R PCS/PMA over single mode fiber PMD as specified in
        Clause 139";
}

identity pmd-type-50GBASE-PQG-D2 {
    base pmd-type;
    description
        "One single mode fiber, 2 x 25.78125 GBd continuous
        transmission / 2 x 25.78125 GBd burst mode reception, medium
        power class, as specified in Clause 141";
}

identity pmd-type-50GBASE-PQG-D3 {
    base pmd-type;
    description
        "One single mode fiber, 2 x 25.78125 GBd continuous
        transmission / 2 x 25.78125 GBd burst mode reception, high
        power class, as specified in Clause 141";
}

identity pmd-type-50GBASE-PQG-U2 {
    base pmd-type;
    description
        "One single mode fiber, 2 x 25.78125 GBd continuous reception
        / 2 x 25.78125 GBd burst mode transmission, medium power
        class, as specified in Clause 141";
}

identity pmd-type-50GBASE-PQG-U3 {
    base pmd-type;
    description
        "One single mode fiber, 2 x 25.78125 GBd continuous reception
        / 2 x 25.78125 GBd burst mode transmission, high power class,
```

```
        as specified in Clause 141";
    }

    identity pmd-type-50GBASE-PQX-D2 {
        base pmd-type;
        description
            "One single mode fiber, 2 x 25.78125 GBd continuous
            transmission / 2 x 25.78125 GBd burst mode reception, medium
            power class, as specified in Clause 141";
    }

    identity pmd-type-50GBASE-PQX-D3 {
        base pmd-type;
        description
            "One single mode fiber, 2 x 25.78125 GBd continuous
            transmission / 2 x 25.78125 GBd burst mode reception, high
            power class, as specified in Clause 141";
    }

    identity pmd-type-50GBASE-PQX-U2 {
        base pmd-type;
        description
            "One single mode fiber, 2 x 25.78125 GBd continuous reception
            / 2 x 25.78125 GBd burst mode transmission, medium power
            class, as specified in Clause 141";
    }

    identity pmd-type-50GBASE-PQX-U3 {
        base pmd-type;
        description
            "One single mode fiber, 2 x 25.78125 GBd continuous reception
            / 2 x 25.78125 GBd burst mode transmission, high power class,
            as specified in Clause 141";
    }

    identity pmd-type-50GBASE-R {
        base pmd-type;
        description
            "Multi-lane PCS as specified in Clause 133 with PMA as
            specified in Clause 135 over undefined PMD";
    }

    identity pmd-type-50GBASE-SR {
        base pmd-type;
        description
            "50GBASE-R PCS/PMA over multimode fiber PMD as specified in
            Clause 138";
    }

    identity pmd-type-100GBASE-CR1 {
        base pmd-type;
        description
            "100GBASE-CR1 as specified in Clause 162 or 100GBASE-KR1 as
            specified in Clause 163";
    }

    identity pmd-type-100GBASE-CR2 {
        base pmd-type;
        description
```

```
    "100GBASE-R PCS/PMA over 2 lane shielded copper balanced cable  
    PMD as specified in Clause 136";  
}  
  
identity pmd-type-100GBASE-CR4 {  
    base pmd-type;  
    description  
        "100GBASE-R PCS/PMA over 4 lane shielded copper balanced cable  
        PMD as specified in Clause 92";  
}  
  
identity pmd-type-100GBASE-CR10 {  
    base pmd-type;  
    description  
        "100GBASE-R PCS/PMA over 10 lane shielded copper balanced  
        cable PMD as specified in Clause 85";  
}  
  
identity pmd-type-100GBASE-DR {  
    base pmd-type;  
    description  
        "100GBASE-R PCS/PMA over single mode fiber PMD as specified in  
        Clause 140";  
}  
  
identity pmd-type-100GBASE-ER4 {  
    base pmd-type;  
    description  
        "100GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD,  
        with extended reach, as specified in Clause 88";  
}  
  
identity pmd-type-100GBASE-FR1 {  
    base pmd-type;  
    description  
        "100GBASE-R PCS/PMA over single-mode fiber PMD with reach up  
        to at least 2 km as specified in Clause 140";  
}  
  
identity pmd-type-100GBASE-KP4 {  
    base pmd-type;  
    description  
        "100GBASE-P PCS/PMA over an electrical backplane PMD as  
        specified in Clause 94";  
}  
  
identity pmd-type-100GBASE-KR2 {  
    base pmd-type;  
    description  
        "100GBASE-R PCS/PMA over an electrical backplane PMD as  
        specified in Clause 137";  
}  
  
identity pmd-type-100GBASE-KR4 {  
    base pmd-type;  
    description  
        "100GBASE-R PCS/PMA over an electrical backplane PMD as  
        specified in Clause 93";  
}
```

```
identity pmd-type-100GBASE-LR1 {
  base pmd-type;
  description
    "100GBASE-R PCS/PMA over single-mode fiber PMD with reach up
     to at least 10 km as specified in Clause 140";
}

identity pmd-type-100GBASE-LR4 {
  base pmd-type;
  description
    "100GBASE-R PCS/PMA over 4 WDM lane single mode fiber PMD,
     with long reach, as specified in Clause 88";
}

identity pmd-type-100GBASE-R {
  base pmd-type;
  description
    "Multi-lane PCS as specified in Clause 82 over undefined
     100GBASE-R or 100GBASE-P PMA/PMD";
}

identity pmd-type-100GBASE-SR1 {
  base pmd-type;
  description
    "100GBASE-R PCS/PMA over multimode fiber PMD with
     reach up to at least 100 m as specified in Clause 167";
}

identity pmd-type-100GBASE-SR2 {
  base pmd-type;
  description
    "100GBASE-R PCS/PMA over 2 lane multimode fiber PMD as
     specified in Clause 138";
}

identity pmd-type-100GBASE-SR4 {
  base pmd-type;
  description
    "100GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
     specified in Clause 95";
}

identity pmd-type-100GBASE-SR10 {
  base pmd-type;
  description
    "100GBASE-R PCS/PMA over 10 lane multimode fiber PMD as
     specified in Clause 86";
}

identity pmd-type-100GBASE-VR1 {
  base pmd-type;
  description
    "100GBASE-R PCS/PMA over multimode fiber PMD with
     reach up to at least 50 m as specified in Clause 167";
}

identity pmd-type-100GBASE-ZR {
  base pmd-type;
```

```
    description
      "100GBASE-R PCS/100GBASE-ZR PMA over a PMD with reach up to at
      least 80 km as specified in Clause 154";
  }

  identity pmd-type-200GBASE-CR2 {
    base pmd-type;
    description
      "200GBASE-CR2 as specified in Clause 162 or 200GBASE-KR2 as
      specified in Clause 163";
  }

  identity pmd-type-200GBASE-CR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 lane shielded copper balanced cable
      PMD as specified in Clause 136 ";
  }

  identity pmd-type-200GBASE-DR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4-lane single-mode fiber PMD as
      specified in Clause 121";
  }

  identity pmd-type-200GBASE-ER4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
      reach up to at least 40 km as specified in Clause 122";
  }

  identity pmd-type-200GBASE-FR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
      reach up to at least 2 km as specified in Clause 122";
  }

  identity pmd-type-200GBASE-KR2 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over an electrical backplane PMD
      as specified in Clause 163";
  }

  identity pmd-type-200GBASE-KR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over an electrical backplane PMD as
      specified in Clause 137";
  }

  identity pmd-type-200GBASE-LR4 {
    base pmd-type;
    description
      "200GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
      reach up to at least 10 km as specified in Clause 122";
```

```
}

identity pmd-type-200GBASE-R {
  base pmd-type;
  description
    "Multi-lane PCS as specified in Clause 119 over undefined
    PMA/PMD";
}

identity pmd-type-200GBASE-SR2 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with
    reach up to at least 100 m as specified in Clause 167";
}

identity pmd-type-200GBASE-SR4 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over 4 lane multimode fiber PMD as
    specified in Clause 138";
}

identity pmd-type-200GBASE-VR2 {
  base pmd-type;
  description
    "200GBASE-R PCS/PMA over 2 lane multimode fiber PMD with
    reach up to at least 50 m as specified in Clause 167";
}

identity pmd-type-400GBASE-DR4 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 4-lane single-mode fiber PMD as
    specified in Clause 124";
}

identity pmd-type-400GBASE-DR4-2 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 4 single-mode fibers in each
    direction PMD with reach up to at least 2 km as specified
    in Clause 124";
}

identity pmd-type-400GBASE-ER8 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
    reach up to at least 40 km as specified in Clause 122";
}

identity pmd-type-400GBASE-FR4 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
    reach up to at least 2 km as specified in Clause 151";
}
```

```
identity pmd-type-400GBASE-FR8 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
     reach up to at least 2 km as specified in Clause 122";
}

identity pmd-type-400GBASE-LR4-6 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 4 WDM lane single-mode fiber PMD with
     reach up to at least 6 km as specified in Clause 151";
}

identity pmd-type-400GBASE-LR8 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 8 WDM lane single-mode fiber PMD with
     reach up to at least 10 km as specified in Clause 122";
}

identity pmd-type-400GBASE-R {
  base pmd-type;
  description
    "Multi-lane PCS as specified in Clause 119 over undefined
     PMA/PMD";
}

identity pmd-type-400GBASE-SR4 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with
     reach up to at least 100 m as specified in Clause 167";
}

identity pmd-type-400GBASE-SR4d2 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 8-lane multimode fiber PMD as
     specified in Clause 150";
}

identity pmd-type-400GBASE-SR8 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 8-lane multimode fiber PMD as
     specified in Clause 138";
}

identity pmd-type-400GBASE-SR16 {
  base pmd-type;
  description
    "400GBASE-R PCS/PMA over 16-lane multimode fiber PMD as
     specified in Clause 123";
}

identity pmd-type-400GBASE-VR4 {
  base pmd-type;
  description
```

```
    "400GBASE-R PCS/PMA over 4 lane multimode fiber PMD with
    reach up to at least 50 m as specified in Clause 167";
}

identity pmd-type-800GBASE-CR8 {
    base pmd-type;
    description
        "800GBASE-R PCS/PMA over 8-lane shielded balanced copper cable
        PMD as specified in Clause 162";
}

identity pmd-type-800GBASE-DR8 {
    base pmd-type;
    description
        "800GBASE-R PCS/PMA over 8 single-mode fibers in each
        direction PMD with reach up to at least 500 m as specified
        in Clause 124";
}

identity pmd-type-800GBASE-DR8-2 {
    base pmd-type;
    description
        "800GBASE-R PCS/PMA over 8 single-mode fibers in each
        direction PMD with reach up to at least 2 km as specified
        in Clause 124";
}

identity pmd-type-800GBASE-KR8 {
    base pmd-type;
    description
        "800GBASE-R PCS/PMA over an electrical backplane PMD as
        specified in Clause 163";
}

identity pmd-type-800GBASE-R {
    base pmd-type;
    description
        "Multi-lane PCS as specified in Clause 172 over
        undefined PMA/PMD";
}

identity pmd-type-800GBASE-SR8 {
    base pmd-type;
    description
        "800GBASE-R PCS/PMA over 8 multimode fibers in each
        direction PMD with reach up to at least 100 m as specified
        in Clause 167";
}

identity pmd-type-800GBASE-VR8 {
    base pmd-type;
    description
        "800GBASE-R PCS/PMA over 8 multimode fibers in each
        direction PMD with reach up to at least 50 m as specified
        in Clause 167";
}

typedef pmd-type-list {
    type bits {
```



```
/* 802.3-2022 */
bit bOther {
    description
        "Other";
}
bit b10baseT {
    description
        "10BASE-T";
}
bit b10baseFL {
    description
        "10BASE-FL";
}
bit b100baseTX {
    description
        "100BASE-TX";
}
bit b100baseFX {
    description
        "100BASE-FX";
}
bit b1000baseX {
    description
        "1000BASE-X";
}
bit b1000baseLX {
    description
        "1000BASE-LX";
}
bit b1000baseSX {
    description
        "1000BASE-SX";
}
bit b1000baseCX {
    description
        "1000BASE-CX";
}
bit b1000baseT {
    description
        "1000BASE-T";
}
bit b10GbaseX {
    description
        "10GBASE-X";
}
bit b10GbaseLX4 {
    description
        "10GBASE-LX4";
}
bit b10GbaseR {
    description
        "10GBASE-R";
}
bit b10GbaseER {
    description
        "10GBASE-ER";
}
bit b10GbaseLR {
    description
```

```
        "10GBASE-LR";
    }
    bit b10GbaseSR {
        description
            "10GBASE-SR";
    }
    bit b10GbaseSW {
        description
            "10GBASE-SW";
    }
    bit b10GbaseCX4 {
        description
            "10GBASE-CX4";
    }
    bit b100BaseBX10D {
        description
            "100BASE-BX10D";
    }
    bit b100BaseBX10U {
        description
            "100BASE-BX10U";
    }
    bit b100BaseLX10 {
        description
            "100BASE-LX10";
    }
    bit b1000BaseBX10D {
        description
            "1000BASE-BX10D";
    }
    bit b1000BaseBX10U {
        description
            "1000BASE-BX10U";
    }
    bit b1000BaseLX10 {
        description
            "1000BASE-LX10";
    }
    bit b1000BasePX10D {
        description
            "1000BASE-PX10D";
    }
    bit b1000BasePX10U {
        description
            "1000BASE-PX10U";
    }
    bit b1000BasePX20D {
        description
            "1000BASE-PX20D";
    }
    bit b1000BasePX20U {
        description
            "1000BASE-PX20U";
    }
    bit b10GbaseT {
        description
            "10GBASE-T";
    }
    bit b10GbaseLRM {
```

```
        description
            "10GBASE-LRM";
    }
    bit b1000baseKX {
        description
            "1000BASE-KX";
    }
    bit b10GbaseKX4 {
        description
            "10GBASE-KX4";
    }
    bit b10GbaseKR {
        description
            "10GBASE-KR";
    }
    bit b10G1GbasePRXD1 {
        description
            "10/1GBASE-PRXD1";
    }
    bit b10G1GbasePRXD2 {
        description
            "10/1GBASE-PRXD2";
    }
    bit b10G1GbasePRXD3 {
        description
            "10/1GBASE-PRXD3";
    }
    bit b10G1GbasePRXU1 {
        description
            "10/1GBASE-PRXU1";
    }
    bit b10G1GbasePRXU2 {
        description
            "10/1GBASE-PRXU2";
    }
    bit b10G1GbasePRXU3 {
        description
            "10/1GBASE-PRXU3";
    }
    bit b10GbasePRD1 {
        description
            "10GBASE-PRD1";
    }
    bit b10GbasePRD2 {
        description
            "10GBASE-PRD2";
    }
    bit b10GbasePRD3 {
        description
            "10GBASE-PRD3";
    }
    bit b10GbasePRU1 {
        description
            "10GBASE-PRU1";
    }
    bit b10GbasePRU3 {
        description
            "10GBASE-PRU3";
    }
}
```

```
bit b40GbaseKR4 {  
    description  
        "40GBASE-KR4";  
}  
bit b40GbaseCR4 {  
    description  
        "40GBASE-CR4";  
}  
bit b40GbaseSR4 {  
    description  
        "40GBASE-SR4";  
}  
bit b40GbaseFR {  
    description  
        "40GBASE-FR";  
}  
bit b40GbaseLR4 {  
    description  
        "40GBASE-LR4";  
}  
bit b100GbaseCR10 {  
    description  
        "100GBASE-CR10";  
}  
bit b100GbaseSR10 {  
    description  
        "100GBASE-SR10";  
}  
bit b100GbaseLR4 {  
    description  
        "100GBASE-LR4";  
}  
bit b100GbaseER4 {  
    description  
        "100GBASE-ER4";  
}  
bit b1000baseT1 {  
    description  
        "1000BASE-T1";  
}  
bit b1000basePX30D {  
    description  
        "1000BASE-PX30D";  
}  
bit b1000basePX30U {  
    description  
        "1000BASE-PX30U";  
}  
bit b1000basePX40D {  
    description  
        "1000BASE-PX40D";  
}  
bit b1000basePX40U {  
    description  
        "1000BASE-PX40U";  
}  
bit b10G1GbasePRXD4 {  
    description  
        "10/1GBASE-PRXD4";
```

```
}  
bit b10G1GbasePRXU4 {  
    description  
        "10/1GBASE-PRXU4";  
}  
bit b10GbasePRD4 {  
    description  
        "10GBASE-PRD4";  
}  
bit b10GbasePRU4 {  
    description  
        "10GBASE-PRU4";  
}  
bit b25GbaseCR {  
    description  
        "25GBASE-CR";  
}  
bit b25GbaseCRS {  
    description  
        "25GBASE-CRS";  
}  
bit b25GbaseKR {  
    description  
        "25GBASE-KR";  
}  
bit b25GbaseKRS {  
    description  
        "25GBASE-KRS";  
}  
bit b25GbaseR {  
    description  
        "25GBASE-R";  
}  
bit b25GbaseSR {  
    description  
        "25GBASE-SR";  
}  
bit b25GbaseT {  
    description  
        "25GBASE-T";  
}  
bit b40GbaseER4 {  
    description  
        "40GBASE-ER4";  
}  
bit b40GbaseR {  
    description  
        "40GBASE-R";  
}  
bit b40GbaseT {  
    description  
        "40GBASE-T";  
}  
bit b100GbaseCR4 {  
    description  
        "100GBASE-CR4";  
}  
bit b100GbaseKR4 {  
    description
```

```
        "100GBASE-KR4";
    }
    bit b100GbaseKP4 {
        description
            "100GBASE-KP4";
    }
    bit b100GbaseR {
        description
            "100GBASE-R";
    }
    bit b100GbaseSR4 {
        description
            "100GBASE-SR4";
    }
    bit b2p5GbaseT {
        description
            "2.5GBASE-T";
    }
    bit b5GbaseT {
        description
            "5GBASE-T";
    }
    bit b100baseT1 {
        description
            "100BASE-T1";
    }
    bit b1000baseRHA {
        description
            "1000BASE-RHA";
    }
    bit b1000baseRHB {
        description
            "1000BASE-RHB";
    }
    bit b1000baseRHC {
        description
            "1000BASE-RHC";
    }
    bit b2p5GbaseKX {
        description
            "2.5GBASE-KX";
    }
    bit b2p5GbaseX {
        description
            "2.5GBASE-X";
    }
    bit b5GbaseKR {
        description
            "5GBASE-KR";
    }
    bit b5GbaseR {
        description
            "5GBASE-R";
    }
    bit b10GpassXR {
        description
            "10GPASS-XR";
    }
    bit b25GbaseLR {
```

```
        description
            "25GBASE-LR";
    }
    bit b25GbaseER {
        description
            "25GBASE-ER";
    }
    bit b50GbaseR {
        description
            "50GBASE-R";
    }
    bit b50GbaseCR {
        description
            "50GBASE-CR";
    }
    bit b50GbaseKR {
        description
            "50GBASE-KR";
    }
    bit b50GbaseSR {
        description
            "50GBASE-SR";
    }
    bit b50GbaseFR {
        description
            "50GBASE-FR";
    }
    bit b50GbaseLR {
        description
            "50GBASE-LR";
    }
    bit b50GbaseER {
        description
            "50GBASE-ER";
    }
    bit b100GbaseCR2 {
        description
            "100GBASE-CR2";
    }
    bit b100GbaseKR2 {
        description
            "100GBASE-KR2";
    }
    bit b100GbaseSR2 {
        description
            "100GBASE-SR2";
    }
    bit b100GbaseDR {
        description
            "100GBASE-DR";
    }
    bit b200GbaseR {
        description
            "200GBASE-R";
    }
    bit b200GbaseDR4 {
        description
            "200GBASE-DR4";
    }
}
```

```
bit b200GbaseFR4 {  
    description  
        "200GBASE-FR4";  
}  
bit b200GbaseLR4 {  
    description  
        "200GBASE-LR4";  
}  
bit b200GbaseCR4 {  
    description  
        "200GBASE-CR4";  
}  
bit b200GbaseKR4 {  
    description  
        "200GBASE-KR4";  
}  
bit b200GbaseSR4 {  
    description  
        "200GBASE-SR4";  
}  
bit b200GbaseER4 {  
    description  
        "200GBASE-ER4";  
}  
bit b400GbaseR {  
    description  
        "400GBASE-R";  
}  
bit b400GbaseSR16 {  
    description  
        "400GBASE-SR16";  
}  
bit b400GbaseDR4 {  
    description  
        "400GBASE-DR4";  
}  
bit b400GbaseFR8 {  
    description  
        "400GBASE-FR8";  
}  
bit b400GbaseLR8 {  
    description  
        "400GBASE-LR8";  
}  
bit b400GbaseER8 {  
    description  
        "400GBASE-ER8";  
}  
bit b10baseT1L {  
    description  
        "10BASE-T1L";  
}  
bit b10baseT1S {  
    description  
        "10BASE-T1S";  
}  
bit b10baseT1SMD {  
    description  
        "10BASE-T1SMD";  
}
```



```
}  
bit b100GbaseFR1 {  
    description  
        "100GBASE-FR1";  
}  
bit b100GbaseLR1 {  
    description  
        "100GBASE-LR1";  
}  
bit b100GBASE-CR1 {  
    description  
        "100GBASE-LR1";  
}  
bit b400GbaseFR4 {  
    description  
        "400GBASE-FR4";  
}  
bit b400GbaseLR46 {  
    description  
        "400GBASE-LR46";  
}  
bit b400GbaseSR8 {  
    description  
        "400GBASE-SR8";  
}  
bit b400GbaseSR4p2 {  
    description  
        "400GBASE-SR4p2";  
}  
bit b2p5GbaseT1 {  
    description  
        "2.5GBASE-T1";  
}  
bit b5GbaseT1 {  
    description  
        "5GBASE-T1";  
}  
bit b10GbaseT1 {  
    description  
        "10GBASE-T1";  
}  
bit b25G10GbasePQGD2 {  
    description  
        "25/10GBASE-PQGD2";  
}  
bit b25G10GbasePQGD3 {  
    description  
        "25/10GBASE-PQGD3";  
}  
bit b25G10GbasePQGU2 {  
    description  
        "25/10GBASE-PQGU2";  
}  
bit b25G10GbasePQGU3 {  
    description  
        "25/10GBASE-PQGU3";  
}  
bit b25G10GbasePQXD2 {  
    description
```

```
    "25/10GBASE-PQXD2";
  }
  bit b25G10GbasePQXD3 {
    description
      "25/10GBASE-PQXD3";
  }
  bit b25G10GbasePQXU2 {
    description
      "25/10GBASE-PQXU2";
  }
  bit b25G10GbasePQXU3 {
    description
      "25/10GBASE-PQXU3";
  }
  bit b25GbasePQGD2 {
    description
      "25GBASE-PQGD2";
  }
  bit b25GbasePQGD3 {
    description
      "25GBASE-PQGD3";
  }
  bit b25GbasePQGU2 {
    description
      "25GBASE-PQGU2";
  }
  bit b25GbasePQGU3 {
    description
      "25GBASE-PQGU3";
  }
  bit b25GbasePQXD2 {
    description
      "25GBASE-PQXD2";
  }
  bit b25GbasePQXD3 {
    description
      "25GBASE-PQXD3";
  }
  bit b25GbasePQXU2 {
    description
      "25GBASE-PQXU2";
  }
  bit b25GbasePQXU3 {
    description
      "25GBASE-PQXU3";
  }
  bit b50G10GbasePQGD2 {
    description
      "50/10GBASE-PQGD2";
  }
  bit b50G10GbasePQGD3 {
    description
      "50/10GBASE-PQGD3";
  }
  bit b50G10GbasePQGU2 {
    description
      "50/10GBASE-PQGU2";
  }
  bit b50G10GbasePQGU3 {
```

```
        description
          "50/10GBASE-PQGU3";
      }
      bit b50G10GbasePQXD2 {
        description
          "50/10GBASE-PQXD2";
      }
      bit b50G10GbasePQXD3 {
        description
          "50/10GBASE-PQXD3";
      }
      bit b50G10GbasePQXU2 {
        description
          "50/10GBASE-PQXU2";
      }
      bit b50G10GbasePQXU3 {
        description
          "50/10GBASE-PQXU3";
      }
      bit b50G25GbasePQGD2 {
        description
          "50/25GBASE-PQGD2";
      }
      bit b50G25GbasePQGD3 {
        description
          "50/25GBASE-PQGD3";
      }
      bit b50G25GbasePQGU2 {
        description
          "50/25GBASE-PQGU2";
      }
      bit b50G25GbasePQGU3 {
        description
          "50/25GBASE-PQGU3";
      }
      bit b50G25GbasePQXD2 {
        description
          "50/25GBASE-PQXD2";
      }
      bit b50G25GbasePQXD3 {
        description
          "50/25GBASE-PQXD3";
      }
      bit b50G25GbasePQXU2 {
        description
          "50/25GBASE-PQXU2";
      }
      bit b50G25GbasePQXU3 {
        description
          "50/25GBASE-PQXU3";
      }
      bit b50GbasePQGD2 {
        description
          "50GBASE-PQGD2";
      }
      bit b50GbasePQGD3 {
        description
          "50GBASE-PQGD3";
      }
    }
```

```
bit b50GbasePQGU2 {  
    description  
        "50GBASE-PQGU2";  
}  
bit b50GbasePQGU3 {  
    description  
        "50GBASE-PQGU3";  
}  
bit b50GbasePQXD2 {  
    description  
        "50GBASE-PQXD2";  
}  
bit b50GbasePQXD3 {  
    description  
        "50GBASE-PQXD3";  
}  
bit b50GbasePQXU2 {  
    description  
        "50GBASE-PQXU2";  
}  
bit b50GbasePQXU3 {  
    description  
        "50GBASE-PQXU3";  
}  
bit b100GbaseZR {  
    description  
        "100GBASE-ZR";  
}  
bit b10GbaseBR10D {  
    description  
        "10GBASE-BR10D";  
}  
bit b10GbaseBR10U {  
    description  
        "10GBASE-BR10U";  
}  
bit b10GbaseBR20D {  
    description  
        "10GBASE-BR20D";  
}  
bit b10GbaseBR20U {  
    description  
        "10GBASE-BR20U";  
}  
bit b10GbaseBR40D {  
    description  
        "10GBASE-BR40D";  
}  
bit b10GbaseBR40U {  
    description  
        "10GBASE-BR40U";  
}  
bit b25GbaseBR10D {  
    description  
        "25GBASE-BR10D";  
}  
bit b25GbaseBR10U {  
    description  
        "25GBASE-BR10U";
```

```
}
bit b25GbaseBR20D {
    description
        "25GBASE-BR20D";
}
bit b25GbaseBR20U {
    description
        "25GBASE-BR20U";
}
bit b25GbaseBR40D {
    description
        "25GBASE-BR40D";
}
bit b25GbaseBR40U {
    description
        "25GBASE-BR40U";
}
bit b50GbaseBR10D {
    description
        "50GBASE-BR10D";
}
bit b50GbaseBR10U {
    description
        "50GBASE-BR10U";
}
bit b50GbaseBR20D {
    description
        "50GBASE-BR20D";
}
bit b50GbaseBR20U {
    description
        "50GBASE-BR20U";
}
bit b50GbaseBR40D {
    description
        "50GBASE-BR40D";
}
bit b50GbaseBR40U {
    description
        "50GBASE-BR40U";
}
/* 802.3ck-2022 */
bit b100GBaseCR1 {
    description
        "100GBASE-CR1";
}
bit b100GBaseKR1 {
    description
        "100GBASE-KR1";
}
bit b200GBaseCR2 {
    description
        "200GBASE-CR2";
}
bit b200GBaseKR2 {
    description
        "200GBASE-KR2";
}
bit b400GBaseCR4 {
```

```
        description
            "400GBASE-CR4";
    }
    bit b400GBaseKR4 {
        description
            "400GBASE-KR4";
    }
    /* 802.3cs-2022 */
    bit b10G2p5GBaseSP1Dx {
        description
            "10/2.5GBASE-SP1-Dx";
    }
    bit b10G2p5GBaseSP1Uxy {
        description
            "10/2.5GBASE-SP1-Uxy";
    }
    bit b10GBaseSP1Dx {
        description
            "10GBASE-SP1-Dx";
    }
    bit b10GBaseSP1Uxy {
        description
            "10GBASE-SP1-Uxy";
    }
    /* 802.3db-2022 */
    bit b100GBaseSR1 {
        description
            "100GBASE-SR1";
    }
    bit b100GBaseVR1 {
        description
            "100GBASE-VR1";
    }
    bit bit200GBaseSR2 {
        description
            "200GBASE-SR2";
    }
    bit bit200GBaseVR2 {
        description
            "200GBASE-VR2";
    }
    bit bit400GBaseSR4 {
        description
            "400GBASE-SR4";
    }
    bit bit400GBaseVR4 {
        description
            "400GBASE-VR4";
    }
    /* 802.3cz-2023 */
    bit b2p5GbaseAU {
        description
            "2.5GBASE-AU";
    }
    bit b5GbaseAU {
        description
            "5GBASE-AU";
    }
    bit b10GbaseAU {
```

```
        description
            "10GBASE-AU";
    }
    bit b25GbaseAU {
        description
            "25GBASE-AU";
    }
    bit b50GbaseAU {
        description
            "50GBASE-AU";
    }
    /* 802.3cy-2023 */
    bit b25GbaseT1 {
        description
            "25GBASE-T1";
    }
    /* 802.3df-2024 */
    bit bit400GBaseDR4d2 {
        description
            "400GBASE-DR4-2";
    }
    bit bit800GBaseCR8 {
        description
            "800GBASE-CR8";
    }
    bit bit800GBaseDR8 {
        description
            "800GBASE-DR8";
    }
    bit bit800GBaseDR8d2 {
        description
            "800GBASE-DR8-2";
    }
    bit bit800GBaseKR8 {
        description
            "800GBASE-KR8";
    }
    bit bit800GBaseR {
        description
            "800GBASE-R";
    }
    bit bit800GBaseSR8 {
        description
            "800GBASE-SR8";
    }
    bit bit800GBaseVR8 {
        description
            "800GBASE-VR8";
    }
    }
    description
        "This data type is used to report the set of pmd-type values
        an interface can implement.";
    reference
        "IEEE Std 802.3, 30.5.1.1.3 aMAUTypeList";
}

typedef media-available-type {
    type enumeration {
```

```
/* 802.3-2022 */
enum other {
  description
    "Undefined";
}
enum unknown {
  description
    "initializing, true state not yet known";
}
enum available {
  description
    "link or light normal, loopback normal";
}
enum notAvailable {
  description
    "link loss or low light, no loopback";
}
enum remoteFault {
  description
    "remote fault with no detail";
}
enum remoteJabber {
  description
    "remote fault, reason known to be jabber";
}
enum remoteLinkLoss {
  description
    "remote fault, reason known to be far-end link loss";
}
enum remoteTest {
  description
    "remote fault, reason known to be test";
}
enum offline {
  description
    "offline, applies only to Clause 37 Auto-Negotiation";
}
enum autoNegError {
  description
    "Auto-Negotiation Error, applies only to Clause 37
    auto-Negotiation";
}
enum pmdLinkFault {
  description
    "PMD/PMA receive link fault";
}
enum pcsLinkFault {
  description
    "PCS receive link fault";
}
enum excessiveBER {
  description
    "PCS Bit Error Ratio monitor reporting excessive error
    ratio";
}
enum dxsLinkFault {
  description
    "DTE XGXS receive link fault, applies only to XAUI ";
}
```



```
enum pxsLinkFault {
    description
        "PHY XGXS transmit link fault, applies only to XAUI";
}
}
description
    "This data type is used to report media availability.";
reference
    "IEEE Std 802.3, 30.5.1.1.4 aMediaAvailable";
}

typedef auto-neg-ability-bits {
    type bits {
        /* Not PHY specific */
        bit bOther {
            description
                "Other abilities beyond the scope of this module.";
        }
        bit bHalfDuplex {
            description
                "Half duplex";
        }
        bit bFullDuplex {
            description
                "Full duplex";
        }
        bit bPause {
            description
                "PAUSE operation as defined in Annex 31B";
        }
        bit bAPause {
            description
                "Asymmetric PAUSE operation as defined in
                Clause 37, Annex 28B, and Annex 31B";
        }
        bit bSPause {
            description
                "Symmetric PAUSE operation as defined in
                Clause 37, Annex 28B, and Annex 31B";
        }
        bit bBPause {
            description
                "Asymmetric and Symmetric PAUSE operation
                as defined in Clause 37, Annex 28B, and Annex 31B";
        }
        bit bRemFault {
            description
                "Remote fault bit (RF) as specified in Clause 73
                and Clause 98";
        }
        bit bRemFault1 {
            description
                "Remote fault bit 1 (RF1) as specified in Clause 37";
        }
        bit bRemFault2 {
            description
                "Remote fault bit 2 (RF2) as specified in Clause 37";
        }
        bit bFECCapable {
```

```
    description
      "FEC ability as specified in Clause 73 (see 73.6.5)
      and Clause 74";
  }
  bit bFECRequested {
    description
      "FEC requested as specified in Clause 73 (see 73.6.5)
      and Clause 74";
  }
  bit bRSFEC25Greq {
    description
      "25G RS-FEC requested as specified in Clause 73 (see 73.6.5)
      and Clause 108";
  }
  bit bBaseFEC25Greq {
    description
      "25Gb/s BASE-R FEC ";
  }
  bit bForceMS {
    description
      "Force MASTER-SLAVE as specified in Clause 98 (see
      98.2.1.2.5)";
  }
  /* PHY/PMD */
  bit b10baseT {
    description
      "10BASE-T as defined in Clause 14";
  }
  bit b10baseT1L {
    description
      "10BASE-T1L as specified in Clause 146";
  }
  bit b10baseT1S {
    description
      "10BASE-T1S as specified in Clause 147";
  }
  bit b100baseTX {
    description
      "100BASE-TX as defined in Clause 25";
  }
  bit b1000baseX {
    description
      "1000BASE-X half duplex as specified in Clause 36";
  }
  bit b1000baseT {
    description
      "1000BASE-T PHY as specified in Clause 40";
  }
  bit b1000baseT1 {
    description
      "1000BASE-T1 as specified in Clause 97";
  }
  bit b2p5GbaseT {
    description
      "2.5GBASE-T PHY as specified in Clause 126";
  }
  bit b2p5GbaseT1 {
    description
      "2.5GBASE-T1 as specified in Clause 149";
```

```
}
bit b2p5GbaseKX {
  description
    "2.5GBASE-KX as specified in Clause 128";
}
bit b5GBaseT {
  description
    "5GBASE-T PHY as specified in Clause 126";
}
bit b5GbaseT1 {
  description
    "5GBASE-T1 as specified in Clause 149";
}
bit b5GbaseKR {
  description
    "5GBASE-KR as specified in Clause 130";
}
bit b10GbaseT {
  description
    "10GBASE-T PHY as specified in Clause 55";
}
bit b1000baseKX {
  description
    "1000BASE-KX as specified in Clause 70";
}
bit b10GbaseKX4 {
  description
    "10GBASE-KX4 as specified in Clause 71";
}
bit b10GbaseKR {
  description
    "10GBASE-KR as specified in Clause 72";
}
bit b25GbaseT {
  description
    "25GBASE-T as specified in Clause 113";
}
bit b25GbaseRS {
  description
    "25GBASE-CR-S as specified in Clause 110 or
    25GBASE-KR-S as specified in Clause 111";
}
bit b25GbaseR {
  description
    "25GBASE-CR as specified in Clause 110 or
    25GBASE-KR as specified in Clause 111";
}
bit b25GbaseT1 {
  description
    "25GBASE-T1 as specified in Clause 165";
}
bit b40GbaseKR4 {
  description
    "40GBASE-KR4 as specified in Clause 84";
}
bit b40GbaseCR4 {
  description
    "40GBASE-CR4 as specified in Clause 85";
}
```

```
    bit b40GbaseT {
      description
        "40GBASE-T as specified in Clause 113";
    }
    bit b50GbaseR {
      description
        "50GBASE-CR as specified in Clause 136 or
        50GBASE-KR as specified in Clause 137";
    }
    bit b100GbaseCR10 {
      description
        "100GBASE-CR10 as specified in Clause 85";
    }
    bit b100GbaseCR4 {
      description
        "100GBASE-CR4 as specified in Clause 92";
    }
    bit b100GbaseKR4 {
      description
        "100GBASE-KR4 as specified in Clause 93";
    }
    bit b100GbaseKP4 {
      description
        "100GBASE-KP4 as specified in Clause 94";
    }
    bit b100GbaseR2 {
      description
        "100GBASE-CR2 as specified in Clause 136 or
        100GBASE-KR2 as specified in Clause 137";
    }
    bit b200GbaseR4 {
      description
        "200GBASE-CR4 as specified in Clause 136
        or 200GBASE-KR4 as specified in Clause 137";
    }
  }
  description
    "This data type is used as the syntax of the
    ieee802-eth-phy-auto-neg-capability-bits,
    ieee802-eth-phy-auto-neg-cap-advertised-bits,
    and ieee802-eth-phy-auto-neg-cap-received-bits
    objects in ieee802-eth-phy-auto-neg-table.";
  reference
    "IEEE Std 802.3, 30.6.1.1.5";
}
```

6. YANG module for Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) and Power over Data Lines (PoDL)

6.1 Introduction

This clause defines a YANG module to manage power via MDI Power Sourcing Equipment (PSE) and Power over Data Lines (PoDL) PSE.

IEEE Std 802.3 defines the hardware registers that allow management interfaces to be built for a DTE power via MDI and Power over Data Lines device. The YANG module defined in this clause extends the Ethernet-interface YANG data modules defined in Clause 5 with the management objects required for the management of PoE and PoDL devices and ports.

6.2 YANG module structure

The deprecated *ieee802-ethernet-pse* superseded by *ieee802-ethernet-pse-2* YANG module of this clause is focused on the configuration and monitoring of the Power over Ethernet (PoE) function defined in IEEE Std 802.3, including power via MDI, as well as Power over Data Lines, which can also be considered as the single pair PoE. The module augments the *ieee802-ethernet-interface* YANG module with attributes for the PoE function. The module is partitioned into two major containers.

The PoE PSE container describes a multi-pair PSE, while the PoDL PSE container describes a single-pair PSE.

6.3 Security considerations for the Ethernet data terminal equipment (DTE) power via medium dependent interface (MDI) and Power over Data Lines module

There are a number of data nodes defined in this YANG module that are configurable as read-write. Such data nodes can be considered sensitive or vulnerable in some network environments. The support for configuration operations in a non-secure environment without proper protection can have a negative effect on network operations.

Setting the following data nodes to incorrect values can result in improper operation of the PSE, including the possibility that the Powered Device (PD) does not receive power from the PSE port:

- `pse-enable`
- `powering-pairs`

Some of the readable operational states in this module can be considered sensitive or vulnerable in some network environments. These are as follows:

- `pairs-control-ability`
- `classifications`
- `pd-power-class`
- `pse-type`
- `detected-pd-type`

It is thus important to control GET access to these data nodes and to possibly encrypt their values when sending them over the network.

6.4 Mapping of IEEE Std 802.3, Clause 30 managed objects

This subclause contains the mapping between YANG data nodes included in the *ieee802-ethernet-pse* (see Table 6–1) YANG module, and the managed objects and attributes defined in IEEE Std 802.3, Clause 30.

Table 6–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-pse* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-pse</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
N/A	N/A		interfaces/interface/ethernet/pse	supported-pse-type	R
oPSE	aPSEAdminState	30.9.1	interfaces/interface/ethernet/pse/multi-pair	pse-enable	R
	aPSEPowerPairs			powering-pairs	R/W
	aPSEPowerPairsControlAbility			pairs-control-ability	R
	aPSEPowerDetectionStatus			detection-status	R
	aPSEPowerClassification			classifications	R
	aPSEActualPower			actual-power	R
	aPSEPowerAccuracy			power-accuracy	R
	aPSEInvalidSignatureCounter		interfaces/interface/ethernet/pse/multi-pair/statistics	invalid-signature	R
	aPSEPowerDeniedCounter			power-denied	R
	aPSEOverLoadCounter			overload	R
	aPSEShortCounter			short	R
	aPSEMPSAbsentCounter			mps-absent	R
	aPSECumulativeEnergy			cumulative-energy	R

Table 6–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-pse* YANG data nodes (continued)

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-pse</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oPoDLPSE	aPoDLPSEAdminState	30.15	interfaces/interface/ethernet/pse/single-pair	pse-enable	R
	aPoDLPSEPowerDetectionStatus			detection-status	R
	aPoDLPSEType			podl-type	R
	aPoDLPSEDetectedPDType			detected-pd-type	R
	aPoDLPSEDetectedPDPowerClass			pd-power-class	R
	aPoDLPSEActualPower			actual-power	R
	aPoDLPSEPowerAccuracy			power-accuracy	R
	aPoDLPSEInvalidSignatureCounter		interfaces/interface/ethernet/pse/single-pair/statistics	invalid-signature	R
	aPoDLPSEInvalidClassCounter			invalid-class	R
	aPoDLPSEPowerDeniedCounter			power-denied	R
	aPoDLPSEOverLoadCounter			overload	R
	aPoDLPSEMaintainFullVoltageSignatureAbsentCounter			fvs-absent	R
	aPoDLPSECumulativeEnergy			cumulative-energy	R

6.5 YANG module definition¹⁵

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

6.5.1 Tree hierarchy

6.5.1.1 ieee802-ethernet-pse

module: ieee802-ethernet-pse

```
augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
  +--rw pse
    +--ro supported-pse-type?    identityref
    x--rw multi-pair!
      | +--rw pse-enable?        boolean
      | +--rw powering-pairs?    identityref
      | +--ro pairs-control-ability? boolean
      | x--ro detection-status?   multi-pair-detection-state
      | x--ro classifications?    power-class
      | x--ro statistics
      | | +--ro power-denied?     yang:counter64
      | | +--ro invalid-signature? yang:counter64
      | | +--ro mps-absent?       yang:counter64
      | | x--ro overload?         yang:counter64
      | | x--ro short?            yang:counter64
      | | +--ro cumulative-energy? yang:counter64
      | +--ro actual-power?       decimal64
      | +--ro power-accuracy?     int64
    x--rw single-pair!
      x--rw pse-enable?          boolean
      x--ro detection-status?     single-pair-detection-state
      +--ro pod1-type?            enumeration
      +--ro detected-pd-type?     enumeration
      x--ro pd-power-class?       power-class
      x--ro statistics
      | +--ro power-denied?       yang:counter64
      | +--ro invalid-signature?  yang:counter64
      | +--ro invalid-class?      yang:counter64
      | +--ro overload?           yang:counter64
      | +--ro fvs-absence?        yang:counter64
      | +--ro cumulative-energy?   yang:counter64
      +--ro actual-power?         decimal64
      +--ro power-accuracy?       int64
```

6.5.1.2 ieee802-ethernet-pse-2

module: ieee802-ethernet-pse-2

```
augment /if:interfaces/if:interface/ieee802-eth-if:ethernet:
  +--rw pse-2
    +--ro supported-pse-type?    pse-support
    +--rw multi-pair! {multi-pair-pse}?
      | +--rw pse-enable?        boolean
```

¹⁵ Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.


```

| +--ro pse-state?                boolean
| +--rw multi-pair-powering-pairs? multi-pair-powering-pairs
| +--ro pairs-control-ability?    boolean
| +--ro detection-status?          multi-pair-detection-state
| +--ro classifications?          multi-pair-power-class
| +--ro statistics
| | +--ro power-denied?           yang:counter64
| | +--ro invalid-signature?      yang:counter64
| | +--ro mps-absent?             yang:counter64
| | x--ro overload?               yang:counter64
| | x--ro short?                  yang:counter64
| | +--ro cumulative-energy?       yang:counter64
| +--ro actual-power?             uint32
| +--ro power-accuracy?           int64
+--rw single-pair! {single-pair-pse}?
  +--rw pse-enable?               boolean
  +--ro detection-status?         single-pair-detection-state
  +--ro single-pair-type?         single-pair-pse-type
  +--ro detected-pd-type?         single-pair-pd-type
  +--ro pd-power-class?           single-pair-power-class
  +--ro statistics
  | +--ro power-denied?           yang:counter64
  | +--ro invalid-signature?      yang:counter64
  | +--ro invalid-class?          yang:counter64
  | +--ro overload?               yang:counter64
  | +--ro fvs-absence?            yang:counter64
  | +--ro cumulative-energy?       yang:counter64
  +--ro actual-power?             uint32
  +--ro power-accuracy?           int64

```

6.5.2 YANG module

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 6.2 through 6.5 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the YANG module can be found at the following URL:
<https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3>.¹⁶

6.5.2.1 Ethernet PSE module

```

module ieee802-ethernet-pse {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pse";
  prefix ieee802-pse;

  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ietf-yang-types {
    prefix yang;
  }

```

¹⁶ Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
reference
  "IETF RFC 6991";
}
import ieee802-ethernet-interface {
  prefix ieee802-eth-if;
}

organization
  "IEEE 802.3 Ethernet Working Group
  Web URL: http://www.ieee802.org/3/";
contact
  "Web URL: http://www.ieee802.org/3/";
description
  "This module is deprecated and superseded by
  ieee802-ethernet-pse-2.
  This module contains YANG definitions for configuring and
  managing ports with Power over Ethernet features defined by
  IEEE Std 802.3. It provides functionality roughly equivalent to
  that of the POWER-ETHERNET-MIB defined in IETF RFC 3621.";

revision 2025-09-10 {
  description
    "Deprecated in IEEE Std 802.3.2-2025, superseded by
    ieee802-ethernet-pse-2";
  reference
    "IEEE Std IEEE Std 802.3.2-2025, unless dated explicitly";
}

identity pse-type {
  description
    "Base type for PSE.";
}

identity all {
  base powering-pairs;
  description
    "All pairs are in use.";
}

identity four-pair {
  base pse-type;
  description
    "PSE supports IEEE Std 802.3, Clause 145.";
}

identity two-pair {
  base pse-type;
  description
    "PSE supports IEEE Std 802.3, Clause 33.";
}

identity single-pair {
  base pse-type;
  description
    "PSE supports IEEE Std 802.3, Clause 104.";
}

identity powering-pairs {
  description
```

```
    "Base type for powering pairs.";
}

identity signal {
    base powering-pairs;
    description
        "The signal pairs are in use.";
}

identity spare {
    base powering-pairs;
    description
        "The spare pairs are in use.";
}

typedef multi-pair-detection-state {
    type enumeration {
        enum disabled {
            value 1;
            description
                "PSE disabled.";
        }
        enum searching {
            value 2;
            description
                "PSE is searching.";
        }
        enum deliveringPower {
            value 3;
            description
                "PSE is delivering power.";
        }
        enum fault {
            value 4;
            description
                "PSE fault detected.";
        }
        enum test {
            value 5;
            description
                "PSE test mode.";
        }
        enum otherFault {
            value 6;
            description
                "PSE implementation specific fault detected.";
        }
    }
}

status deprecated;
description
    "Deprecated in IEEE Std 802.3.2-2025, superseded by
    ieee802-ethernet-pse-2
    Detection state of a multi-pair PSE.";
reference
    "IEEE Std 802.3, 30.9.1.1.5";
}

typedef single-pair-detection-state {
    type enumeration {
```

```
enum unknown {
    value 1;
    description
        "True detection state unknown.";
}
enum disabled {
    value 2;
    description
        "PoDL PSE is disabled.";
}
enum searching {
    value 3;
    description
        "PoDL PSE is searching.";
}
enum deliveringPower {
    value 4;
    description
        "PoDL PSE is delivering power.";
}
enum sleep {
    value 5;
    description
        "PoDL PSE is in sleep state.";
}
enum idle {
    value 6;
    description
        "PoDL PSE is idle.";
}
enum error {
    value 7;
    description
        "PoDL PSE error.";
}
}
status deprecated;
description
    "Deprecated in IEEE Std 802.3.2-2025, superseded by
    ieee802-ethernet-pse-2
    Detection state of a PoDL PSE.";
reference
    "IEEE Std 802.3, 30.15.1.1.3";
}

typedef power-class {
    type enumeration {
        enum class0 {
            value 1;
            description
                "Class 0";
        }
        enum class1 {
            value 2;
            description
                "Class 1";
        }
        enum class2 {
            value 3;
        }
    }
}
```

```
        description
            "Class 2";
    }
    enum class3 {
        value 4;
        description
            "Class 3";
    }
    enum class4 {
        value 5;
        description
            "Class 4";
    }
    enum class5 {
        value 6;
        description
            "Class 5 (for PoDL-only)";
    }
    enum class6 {
        value 7;
        description
            "Class 6 (for PoDL-only)";
    }
    enum class7 {
        value 8;
        description
            "Class 7 (for PoDL-only)";
    }
    enum class8 {
        value 9;
        description
            "Class 8 (for PoDL-only)";
    }
    enum class9 {
        value 10;
        description
            "Class 9 (for PoDL-only)";
    }
    enum unknown {
        value 11;
        description
            "Initializing, true Power Class not yet known
            (only for PoDL PSE).";
    }
}

status deprecated;
description
    "Deprecated in IEEE Std 802.3.2-2025, superseded by
    ieee802-ethernet-pse-2
    The power class.";
reference
    "IEEE Std 802.3, 30.9.1.1.8 aPSEPowerClassification and
    IEEE Std 802.3, 30.15.1.1.6 aPoDLPSEDetectedPDPowerClass.";
}

augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
    status deprecated;
    description
        "Deprecated in IEEE Std 802.3.2-2025, superseded by
```

```
ieee802-ethernet-pse-2
Augments ethernet interface configuration model with
nodes specific to DTE Power via MDI devices and ports";
container pse {
  description
    "DTE Power via MDI port configuration";
  reference
    "IEEE Std 802.3, 30.9.1 PoE PSE & IEEE Std 802.3, 30.15.1
    PoDL PSE";
  leaf supported-pse-type {
    type identityref {
      base ieee802-pse:pse-type;
    }
    config false;
    description
      "PSE supports one or more of IEEE Std 802.3 Clause 33,
      Clause 104, or Clause 145.";
  }
  container multi-pair {
    presence "PSE port supports IEEE Std 802.3, Clause 33.";
    status deprecated;
    description
      "Deprecated in IEEE Std 802.3.2-2025, superseded by
      ieee802-ethernet-pse-2
      PSE port configuration in IEEE Std 802.3, 30.9.1.";
    leaf pse-enable {
      type boolean;
      default "false";
      description
        "When true enables the PSE function on the interface,
        when false disables the PSE function on the
        interface.";
      reference
        "IEEE Std 802.3, 30.9.1.1.2 aPSEAdminState";
    }
    leaf powering-pairs {
      type identityref {
        base powering-pairs;
      }
      description
        "Describes or controls the PSE pairs in use. If the
        value of pairs-control-ability is true, this object
        is writeable.";
      reference
        "IEEE Std 802.3, 30.9.1.1.4 aPSEPowerPairs";
    }
  }
  leaf pairs-control-ability {
    type boolean;
    default "true";
    config false;
    description
      "Describes the ability to control switching the
      power sourcing pins of the PSE.";
    reference
      "IEEE Std 802.3, 30.9.1.1.3
      aPSEPowerPairsControlAbility";
  }
  leaf detection-status {
    type multi-pair-detection-state;
  }
}
```

```

config false;
status deprecated;
description
  "Deprecated in IEEE Std 802.3.2-2025, superseded by
  ieee802-ethernet-pse-2
  Describes the operational status of the port
  PD detection.";
reference
  "IEEE Std 802.3, 30.9.1.1.5 aPSEPowerDetectionStatus";
}
leaf classifications {
  when "../detection-status = 'deliveringPower'" {
    description
      "Deprecated in IEEE Std 802.3.2-2025, superseded by
      ieee802-ethernet-pse-2
      This node only applies when the detection status is
      delivering power.";
  }
  type power-class;
  config false;
  status deprecated;
  description
    "The power class of the PSE port.";
  reference
    "IEEE Std 802.3, 30.9.1.1.8 aPSEPowerClassification";
}
container statistics {
  config false;
  status deprecated;
  description
    "Deprecated in IEEE Std 802.3.2-2025, superseded by
    ieee802-ethernet-pse-2
    Statistics information of the multi-pair
    port.";
  leaf power-denied {
    type yang:counter64;
    description
      "This counter is incremented when the PSE state
      diagram enters the POWER_DENIED state, per
      IEEE Std 802.3, Figure 33-9.";
    reference
      "IEEE Std 802.3, 30.9.1.1.14";
  }
  leaf invalid-signature {
    type yang:counter64;
    description
      "This counter is incremented when the PoDL PSE state
      diagram enters the SIGNATURE_INVALID state pervariable mr\_in-
      valid\_signature transitions
IEEE Std 802.3, Figure 33-9from false to true.";
    reference
      "IEEE Std 802.3, 30.9.1.1.11";
  }
  leaf mps-absent {
    type yang:counter64;
    description
      "This counter is incremented when the PSE
      transitions directly from the POWER_ON state to the
      IDLE state due to tmpdo_timer_done being asserted,

```



```
        per IEEE Std 802.3, Figure 33-9.";
    reference
        "IEEE Std 802.3, 30.9.1.1.20";
}
leaf overload {
    type yang:counter64;
    status deprecated;
    description
        "This counter is incremented when the PSE state
        diagram enters the ERROR_DELAY state due to the
        ovld_detected variable being TRUE, per
        IEEE Std 802.3, Figure 33-9.";
    reference
        "IEEE Std 802.3, 30.9.1.1.17";
}
leaf short {
    type yang:counter64;
    status deprecated;
    description
        "This Yang object is deprecated as its not defined in
        base standard.
        This counter is incremented when the PSE state
        diagram enters the ERROR_DELAY state due to the
        short_detected variable being TRUE, per
        IEEE Std 802.3, Figure 33-9.";
    reference
        "IEEE Std 802.3-2018, 30.9.1.1.10 aPSEShortCounter";
}
leaf cumulative-energy {
    type yang:counter64;
    units "millijoules";
    description
        "The cumulative energy supplied by the PSE as
        measured at the MDI in millijoules.";
    reference
        "IEEE Std 802.3, 30.9.1.1.25";
}
}
leaf actual-power {
    type decimal64 {
        fraction-digits 4;
    }
    units "milliwatts";
    config false;
    description
        "The actual power drawn by a PD over the port.";
    reference
        "IEEE Std 802.3, 30.9.1.1.23";
}
leaf power-accuracy {
    type int64;
    units "milliwatts";
    config false;
    description
        "An integer value indicating the accuracy
        associated with power-accuracy in +/- milliwatts.";
    reference
        "IEEE Std 802.3, 30.9.1.1.24";
}
```



```
}
container single-pair {
  presence "PSE port working in PoDL.";
  status deprecated;
  description
    "Deprecated in IEEE Std 802.3.2-2025, superseded by
     ieee802-ethernet-pse-2
     PoDL PSE configuration as defined in
     IEEE Std 802.3, 30.15.1.";
  leaf pse-enable {
    type boolean;
    default "false";
    status deprecated;
    description
      "Deprecated in IEEE Std 802.3.2-2025, superseded by
       ieee802-ethernet-pse-2
       When true enables the PSE function on the interface,
       when false disables the PSE function on the
       interface.";
    reference
      "IEEE Std 802.3, 30.15.1.1.2 aPoDLPSEAdminState";
  }
  leaf detection-status {
    type single-pair-detection-state;
    config false;
    status deprecated;
    description
      "Deprecated in IEEE Std 802.3.2-2025, superseded by
       ieee802-ethernet-pse-2
       Indicates the current status of the PoDL PSE.";
    reference
      "IEEE Std 802.3, 30.15.1.1.3
       aPoDLPSEPowerDetectionStatus";
  }
  leaf podl-type {
    type enumeration {
      enum unknown {
        description
          "Unknown PSE type.";
      }
      enum typeA {
        description
          "TypeA PSE";
      }
      enum typeB {
        description
          "TypeB PSE";
      }
      enum typeC {
        description
          "Type PSEC";
      }
      enum typeD {
        description
          "TypeD PSE";
      }
      enum typeE {
        description
          "TypeE PSE";
      }
    }
  }
}
```

```
    }
    enum typeF {
      description
        "TypeF PSE";
    }
  }
  config false;
  description
    "PSE type specified in and
    IEEE Std 802.3, 30.15.1.1.4.";
}
leaf detected-pd-type {
  when "../detection-status = 'deliveringPower'" {
    description
      "This node only applies when the detection status is
      delivering power.";
  }
  type enumeration {
    enum unknown {
      description
        "Unknown PD type";
    }
    enum typeA {
      description
        "TypeA PD";
    }
    enum typeB {
      description
        "TypeB PD";
    }
    enum typeC {
      description
        "TypeC PD";
    }
    enum typeD {
      description
        "TypeD PD";
    }
    enum typeE {
      description
        "TypeE PD";
    }
    enum typeF {
      description
        "TypeF PD";
    }
  }
}
config false;
description
  "Indicates the Type of the detected PoDL PD as
  specified in IEEE Std 802.3, 104.5.1.";
reference
  "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
}
leaf pd-power-class {
  when "../detection-status = 'deliveringPower'" {
    description
      "This node only applies when the detection status is
      delivering power.";
```

```
}
type power-class;
config false;
status deprecated;
description
    "Deprecated in IEEE Std 802.3.2-2025, superseded by
    ieee802-ethernet-pse-2
    Power class of the PD detected on the PSE port.";
reference
    "IEEE Std 802.3, 30.15.1.1.6
    aPoDLPSEDetectedPDPowerClass";
}
container statistics {
    config false;
    status deprecated;
    description
        "Deprecated in IEEE Std 802.3.2-2025, superseded by
        ieee802-ethernet-pse-2
        Discontinuities in the values of counters in this
        container can occur at re-initialization of the
        management system, and at other times as indicated by
        the value of the 'discontinuity-time' leaf defined in
        the ietf-interfaces YANG module (IETF RFC 8343).";
    leaf power-denied {
        type yang:counter64;
        description
            "This counter is incremented when the PoDL PSE state
            diagram variable power_available transitions from
            true to false (see IEEE Std 802.3, 104.4.4.3).";
        reference
            "IEEE Std 802.3, 30.15.1.1.9
            aPoDLPSEPowerDeniedCounter";
    }
    leaf invalid-signature {
        type yang:counter64;
        description
            "This counter is incremented when the PSE state
            diagram enters the SIGNATURE_INVALID state per
            IEEE Std 802.3, Figure 33-9.";
        reference
            "IEEE Std 802.3, 30.15.1.1.7
            aPoDLPSEInvalidSignatureCounter";
    }
    leaf invalid-class {
        type yang:counter64;
        description
            "This counter is incremented when the PoDL PSE state
            diagram variable tclass_timer_done transitions from
            false to true or when the valid_class variable
            transitions from true to false
            (see IEEE Std 802.3, 104.4.4.3).";
        reference
            "IEEE Std 802.3, 30.15.1.1.8
            aPoDLPSEInvalidClassCounter";
    }
    leaf overload {
        type yang:counter64;
        description
            "This counter is incremented when the PSE state
```


6.5.2.2 Ethernet PSE-2 Module

```
module ieee802-ethernet-pse-2 {
  yang-version 1.1;
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-pse-2";
  prefix ieee802-pse-2;

  import ietf-interfaces {
    prefix if;
    reference
      "IETF RFC 8343";
  }
  import ietf-yang-types {
    prefix yang;
    reference
      "IETF RFC 6991";
  }
  import ieee802-ethernet-interface {
    prefix ieee802-eth-if;
  }

  organization
    "IEEE 802.3 Ethernet Working Group
     Web URL: http://www.ieee802.org/3/";
  contact
    "Web URL: http://www.ieee802.org/3/";
  description
    "This module contains YANG definitions for configuring and
     managing ports with Power over Ethernet features defined by
     IEEE Std 802.3-2022.
     This module is based on, and supersedes ieee802-ethernet-pse.yang
     from IEEE Std 802.3.2-2019.";

  revision 2025-09-10 {
    description
      "Updates under IEEE Std 802.3.2-2025";
    reference
      "IEEE Std IEEE Std 802.3.2-2025, unless dated explicitly";
  }

  feature multi-pair-pse {
    description
      "This object indicates the support for IEEE Std 802.3
       Clause 33 and/or Clause 145.";
  }

  feature single-pair-pse {
    description
      "This object indicates the support for IEEE Std 802.3
       Clause 104.";
  }

  typedef multi-pair-detection-state {
    type enumeration {
      enum disabled {
        description
          "PSE disabled.";
      }
    }
  }
```

```
enum searching {
    description
        "PSE is searching.";
}
enum deliveringPower {
    description
        "PSE is delivering power.";
}
enum fault {
    description
        "PSE fault detected.";
}
enum test {
    description
        "PSE test mode.";
}
enum otherFault {
    description
        "PSE implementation specific fault detected.";
}
}
description
    "Detection state of a multi-pair PSE.";
reference
    "IEEE Std 802.3, 30.9.1.1.5";
}

typedef single-pair-detection-state {
    type enumeration {
        enum unknown {
            description
                "True detection state unknown.";
        }
        enum disabled {
            description
                "PSE is disabled.";
        }
        enum searching {
            description
                "PSE is searching.";
        }
        enum deliveringPower {
            description
                "PSE is delivering power.";
        }
        enum sleep {
            description
                "PSE is in sleep state.";
        }
        enum idle {
            description
                "PSE is idle.";
        }
        enum error {
            description
                "PSE error.";
        }
    }
}
description
```

```
    "Detection state of a single-pair PSE.";
reference
    "IEEE Std 802.3, 30.15.1.1.3";
}

typedef multi-pair-power-class {
    type enumeration {
        enum class0 {
            description
                "Class 0";
        }
        enum class1 {
            description
                "Class 1";
        }
        enum class2 {
            description
                "Class 2";
        }
        enum class3 {
            description
                "Class 3";
        }
        enum class4 {
            description
                "Class 4";
        }
        enum class5 {
            description
                "Class 5";
        }
        enum class6 {
            description
                "Class 6";
        }
        enum class7 {
            description
                "Class 7";
        }
        enum class8 {
            description
                "Class 8";
        }
    }
    description
        "Multi-pair PoE power class.";
reference
    "IEEE Std 802.3, 30.9.1.1.8 aPSEPowerClassification ";
}

typedef pse-support {
    type enumeration {
        enum two-pair {
            description
                "PSE port supports IEEE Std 802.3, Clause 33.";
        }
        enum four-pair {
            description
                "PSE port supports IEEE Std 802.3, Clause 33 and
```

```
        Clause 145.";
    }
    enum single-pair {
        description
            "PSE port supports IEEE Std 802.3, Clause 104.";
    }
}
description
    "PSE support.";
}

typedef multi-pair-powering-pairs {
    type enumeration {
        enum signal {
            description
                "Using signal pairs.";
        }
        enum spare {
            description
                "Using spare pairs.";
        }
        enum both {
            description
                "Using signal and spare pairs.";
        }
    }
    description
        "Powering pairs.";
}

typedef single-pair-pd-type {
    type enumeration {
        enum unknown {
            description
                "Unknown PD type";
        }
        enum typeA {
            description
                "TypeA PD";
        }
        enum typeB {
            description
                "TypeB PD";
        }
        enum typeC {
            description
                "TypeC PD";
        }
        enum typeD {
            description
                "TypeD PD";
        }
        enum typeE {
            description
                "TypeE PD";
        }
        enum typeF {
            description
                "TypeF PD";
        }
    }
}
```



```
    }
  }
  description
    "Single-pair PoE PD type.";
  reference
    "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
}

typedef single-pair-pse-type {
  type enumeration {
    enum unknown {
      description
        "Unknown PSE type.";
    }
    enum typeA {
      description
        "TypeA PSE";
    }
    enum typeB {
      description
        "TypeB PSE";
    }
    enum typeC {
      description
        "TypeC PSE";
    }
    enum typeD {
      description
        "TypeD PSE";
    }
    enum typeE {
      description
        "TypeE PSE";
    }
    enum typeF {
      description
        "TypeF PSE";
    }
  }
  description
    "Single-pair PoE PSE type.";
  reference
    "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
}

typedef single-pair-power-class {
  type enumeration {
    enum class0 {
      description
        "Class 0";
    }
    enum class1 {
      description
        "Class 1";
    }
    enum class2 {
      description
        "Class 2";
    }
  }
}
```

```
enum class3 {  
    description  
        "Class 3";  
}  
enum class4 {  
    description  
        "Class 4";  
}  
enum class5 {  
    description  
        "Class 5";  
}  
enum class6 {  
    description  
        "Class 6";  
}  
enum class7 {  
    description  
        "Class 7";  
}  
enum class8 {  
    description  
        "Class 8";  
}  
enum class9 {  
    description  
        "Class 9";  
}  
enum class10 {  
    description  
        "Class 10";  
}  
enum class11 {  
    description  
        "Class 11";  
}  
enum class12 {  
    description  
        "Class 12";  
}  
enum class13 {  
    description  
        "Class 13";  
}  
enum class14 {  
    description  
        "Class 14";  
}  
enum class15 {  
    description  
        "Class 15";  
}  
enum unknown {  
    description  
        "Initializing, true Power Class not yet known.";  
}  
}  
description  
    "Single-pair PoE power class.";
```

```
reference
  "IEEE Std 802.3, 30.15.1.1.6 aPoDLPSEDetectedPDPowerClass";
}

augment "/if:interfaces/if:interface/ieee802-eth-if:ethernet" {
  description
    "Augments ethernet interface configuration model with
     nodes specific to Multi-pair and Single-pair PoE.";
  container pse-2 {
    description
      "Multi-pair and Single-pair PoE port configuration.

      Discontinuities in the values of counters in this container
      can occur at re-initialization of the management system,
      and at other times as indicated by the value of the
      'discontinuity-time' leaf defined in the ietf-interfaces
      YANG module (IETF RFC 8343).";
    reference
      "IEEE Std 802.3, 30.9.1 multi-pair PoE PSE and
       IEEE Std 802.3, 30.15.1 single-pair PoE PSE";
    leaf supported-pse-type {
      type pse-support;
      config false;
      description
        "The PSE types are supported.";
    }
    /* Multi-pair objects */
    container multi-pair {
      if-feature "multi-pair-pse";
      presence "PSE port supports IEEE Std 802.3 Clause 33
        and/or Clause 145.";
      description
        "PSE port configuration in IEEE Std 802.3, 30.9.1.";
      leaf pse-enable {
        type boolean;
        default "false";
        description
          "When true enables the PSE function on the port,
           when false disables the PSE function on the port.";
        reference
          "IEEE Std 802.3, 30.9.1.2.1 acPSEAdminControl";
      }
      leaf pse-state {
        type boolean;
        config false;
        description
          "The PSE enabled state on the port.";
        reference
          "IEEE Std 802.3, 30.9.1.1.2 aPSEAdminState";
      }
      leaf multi-pair-powering-pairs {
        type multi-pair-powering-pairs;
        description
          "The PSE pairs in use. If the value of
           pairs-control-ability is true, this object is
           writeable.";
        reference
          "IEEE Std 802.3, 30.9.1.1.4 aPSEPowerPairs";
      }
    }
  }
}
```

```
leaf pairs-control-ability {
  type boolean;
  default "true";
  config false;
  description
    "Can the PSE Port control switching the
    power sourcing pins.";
  reference
    "IEEE Std 802.3, 30.9.1.1.3 aPSEPowerPairsControlAbility";
}
leaf detection-status {
  type multi-pair-detection-state;
  config false;
  description
    "The operational status of the port PD detection.";
  reference
    "IEEE Std 802.3, 30.9.1.1.5 aPSEPowerDetectionStatus";
}
leaf classifications {
  when "../detection-status = 'deliveringPower'" {
    description
      "This node only applies when the detection status is
      deliveringPower.";
  }
  type multi-pair-power-class;
  config false;
  description
    "The power class of the detected PD.";
  reference
    "IEEE Std 802.3, 30.9.1.1.8 aPSEPowerClassification";
}
/* Multi-pair statistics */
container statistics {
  config false;
  description
    "Multi-pair PoE Statistics information for the port.";
  leaf power-denied {
    type yang:counter64;
    description
      "The number of times the PSE denied power.";
    reference
      "IEEE Std 802.3, 30.9.1.1.14";
  }
  leaf invalid-signature {
    type yang:counter64;
    description
      "The number of times the PSE detected an
      invalid signature.";
    reference
      "IEEE Std 802.3, 30.9.1.1.11";
  }
  leaf mps-absent {
    type yang:counter64;
    description
      "The number of times the PSE
      lost the 'Maintain Power Signature' (MPS).";
    reference
      "IEEE Std 802.3, 30.9.1.1.20";
  }
}
```

```
leaf overload {
  type yang:counter64;
  status deprecated;
  description
    "The number of times the PSE detected
    an output current overload.";
  reference
    "IEEE Std 802.3, 30.9.1.1.17";
}
leaf short {
  type yang:counter64;
  status deprecated;
  description
    "This object is deprecated as it's not defined in base
    standard.";
}
leaf cumulative-energy {
  type yang:counter64;
  units "millijoules";
  description
    "The cumulative energy supplied by the PSE.";
  reference
    "IEEE Std 802.3, 30.9.1.1.25";
}
}
/* Multi-pair measurement */
leaf actual-power {
  type uint32;
  units "milliwatts";
  config false;
  description
    "The actual power drawn by a PD over the port.";
  reference
    "IEEE Std 802.3, 30.9.1.1.23";
}
leaf power-accuracy {
  type int64;
  units "milliwatts";
  config false;
  description
    "The power accuracy of the port.";
  reference
    "IEEE Std 802.3, 30.9.1.1.24";
}
}
/* Single-pair objects */
container single-pair {
  if-feature "single-pair-pse";
  presence "PSE port supports IEEE Std 802.3 Clause 104.";
  description
    "Single-pair PSE configuration as defined in
    IEEE Std 802.3, 30.15.1.";
  leaf pse-enable {
    type boolean;
    default "false";
    description
      "When true enables the PSE function on the port,
      when false disables the PSE function on the port.";
    reference
```

```
    "IEEE Std 802.3, 30.15.1.1.2 aPoDLPSEAdminState";
}
leaf detection-status {
    type single-pair-detection-state;
    config false;
    description
        "Indicates the current status of the Single-pair PSE.";
    reference
        "IEEE Std 802.3,
        30.15.1.1.3 aPoDLPSEPowerDetectionStatus";
}
leaf single-pair-type {
    type single-pair-pse-type;
    config false;
    description
        "Single-pair PSE type.";
    reference
        "IEEE Std 802.3, 30.15.1.1.4 aPoDLPSEType";
}
leaf detected-pd-type {
    when "../detection-status = 'deliveringPower'" {
        description
            "This node only applies when the detection status is
            delivering power.";
    }
    type single-pair-pd-type;
    config false;
    description
        "Indicates the detected type of the single-pair PD.";
    reference
        "IEEE Std 802.3, 30.15.1.1.5 aPoDLPSEDetectedPDType";
}
leaf pd-power-class {
    when "../detection-status = 'deliveringPower'" {
        description
            "This node only applies when the detection status is
            delivering power.";
    }
    type single-pair-power-class;
    config false;
    description
        "The power class of the detected PD.";
    reference
        "IEEE Std 802.3,
        30.15.1.1.6 aPoDLPSEDetectedPDPowerClass";
}
/* Single-pair statistics */
container statistics {
    config false;
    description
        "Single-pair PoE Statistics information for the port.";
    leaf power-denied {
        type yang:counter64;
        description
            "The number of times the PSE denied power.";
        reference
            "IEEE Std 802.3,
            30.15.1.1.9 aPoDLPSEPowerDeniedCounter";
    }
}
```

```
leaf invalid-signature {
  type yang:counter64;
  description
    "The number of times the PSE detected an
    invalid signature.";
  reference
    "IEEE Std 802.3, 30.15.1.1.7
    aPoDLPSEInvalidSignatureCounter";
}
leaf invalid-class {
  type yang:counter64;
  description
    "The number of times PD classification timed out
    or did not get valid class information from SCCP.";
  reference
    "IEEE Std 802.3,
    30.15.1.1.8 aPoDLPSEInvalidClassCounter";
}
leaf overload {
  type yang:counter64;
  description
    "The number of times the PSE detected
    an output current overload.";
  reference
    "IEEE Std 802.3, 30.15.1.1.10 aPoDLPSEOverLoadCounter";
}
leaf fvs-absence {
  type yang:counter64;
  description
    "The number of times the PSE
    lost the 'Maintain Full Voltage Signature' (MFVS).";
  reference
    "IEEE Std 802.3, 30.15.1.1.11
    aPoDLPSEMaintainFullVoltageSignatureAbsentCounter";
}
leaf cumulative-energy {
  type yang:counter64;
  units "millijoules";
  description
    "The cumulative energy supplied by the PSE as
    measured at the MDI.";
  reference
    "IEEE Std 802.3, 30.15.1.1.14 aPoDLPSECumulativeEnergy";
}
}
/* Single-pair measurement */
leaf actual-power {
  type uint32;
  units "milliwatts";
  config false;
  description
    "The actual power drawn by a PD over the port.";
  reference
    "IEEE Std 802.3, 30.15.1.1.12 aPoDLPSEActualPower";
}
leaf power-accuracy {
  type int64;
  units "milliwatts";
  config false;
```

```
description
  "The power accuracy of the port.";
reference
  "IEEE Std 802.3, 30.15.1.1.13 aPoDLPSEPowerAccuracy";
}
}
}
}
```


8. YANG module for Ethernet link OAM

8.1 Introduction

IEEE Std 802.3, Clause 57 includes management capabilities for Ethernet-like interfaces to provide some basic operations, administration, and maintenance (OAM) functions. The defined functionality includes discovery, error signaling, loopback, and link monitoring. This clause defines a portion of the YANG module for use with NETCONF or RESTCONF to manage these Ethernet-like interface capabilities.

8.2 Overview

Ethernet OAM is composed of a core set of functions and a set of optional functional groups as described in Clause 57 of IEEE Std 802.3. The core functions include discovery operations (determining if the other end of the link is OAM capable and what OAM functions it supports), state machine implementation, and some critical event flows.

Ethernet OAM provides single-hop functionality in that it works only between two directly connected Ethernet stations. Ethernet OAM has three functional objectives, which are detailed in 8.2.1 through 8.2.3. The definition of a basic Ethernet OAM protocol data unit is given in 8.2.4.

8.2.1 Remote fault indication

Remote fault indication provides a mechanism for one end of an Ethernet link to signal to the other end that the receive path is non-operational. Some Ethernet Physical Layers offer mechanisms to signal this condition at the Physical Layer. Ethernet OAM added a mechanism so that some Ethernet Physical Layers can operate in unidirectional mode, allowing frames to be transmitted in one direction even when the other direction is non-operational. Traditionally, Ethernet PHYs do not allow frame transmission in one direction if the other direction is not operational. Using this mode, Ethernet OAM allows frame-based signaling of remote fault conditions while still not allowing higher layer applications to be aware of the unidirectional capability. This clause includes mechanisms for capturing that fault information and reflecting such information in data nodes and notifications within the NETCONF management framework.

8.2.2 Link monitoring

Ethernet OAM includes event signaling capability so that one end of an Ethernet link can indicate the occurrence of certain important events to the other end of the link. This happens via layer 2 protocols. This clause defines methods for incorporating the occurrence of these events, at both the local end and the far end of the link, into the YANG-based management framework.

Ethernet OAM also includes mechanisms for one Ethernet station to query another directly connected Ethernet station about the status of its Ethernet interface variables and status. This clause does not include mechanisms for controlling how one Ethernet endpoint may use this functionality to query the status or statistics of a peer Ethernet entity.

8.2.3 Remote loopback

Remote loopback is a link state where the peer Ethernet entity echoes every received packet (without modifications) back onto the link. Remote loopback is intrusive in that the other end of the link is not forwarding traffic from higher layers out over the link. This clause defines data nodes controlling loopback operation and reading the status of the loopback state.

8.2.4 Ethernet OAM protocol data units

An Ethernet OAM protocol data unit (OAMPDU) is a valid Ethernet frame with a destination Media Access Control (MAC) address equal to the MAC address assigned for Slow Protocols (see IEEE Std 802.3, Annex 57A), a lengthOrType field equal to the Type assigned for Slow Protocols, and a Slow Protocols subtype equal to that of the subtype assigned for Ethernet OAM.

OAMPDU is used throughout this clause as an abbreviation for Ethernet OAM protocol data unit. OAMPDUs are the mechanism by which two directly connected Ethernet interfaces exchange OAM information.

8.3 Security considerations for the Ethernet operations, administration, and maintenance (OAM) module

The readable data nodes in this module can provide information about network traffic, and therefore, they may be considered sensitive. In particular, OAM provides mechanisms for reading the IEEE 802.3 Clause 30 management attributes from a link partner via a layer 3 protocol. IEEE 802.3 OAM does not include encryption or authentication mechanisms. It should not be used in environments where this interface information is considered sensitive, and where the facility terminations are unprotected. By default, OAM is disabled on Ethernet-like interfaces and is therefore not a risk.

IEEE 802.3 OAM is designed to support deployment in access and enterprise networks. In access networks, one end of a link is the central office (CO) side and the other is the CPE side, and the facilities are often protected in wiring cages or closets. In such deployments, it is often the case that the CO side is protected from access from the CPE side. Within IEEE 802.3 OAM, this protection from remote access is accomplished by configuring the CPE side in passive mode using the mode leaf. This prevents the CPE from accessing functions and information at the CO side of the connection. In enterprise networks, read-only interface information is often considered non-sensitive.

The frequency of OAM PDUs on an Ethernet interface does not adversely affect data traffic, as OAM is a slow protocol with very limited bandwidth potential, and it is not required for normal link operation. Although there are a number of objects in this module with read-write or read-create MAX-ACCESS, they have limited effects on user data.

The loopback capability of OAM can have potentially disruptive effects; when remote loopback is enabled, the remote station automatically transmits all received traffic back to the local station except for OAM traffic. This completely disrupts all higher layer protocols such as bridging, IP, and NETCONF/RESTCONF.

The administrative state and mode are also configuration nodes. Disabling OAM can interrupt management activities between peer devices, potentially causing serious problems. Setting the mode node to an undesired value can allow access to Ethernet monitoring, events, and functions that may not be acceptable in a particular deployment scenario. In addition to loopback functionality, Ethernet interface statistics and events can be accessed via the OAM protocol, which may not be desired in some circumstances.

OAM event configuration also contains configuration nodes. These nodes control whether events are sent, and at what thresholds. Note that the frequency of event communication is limited by the frequency limits of Slow Protocols on Ethernet interfaces. Also, the information available via OAM events is also available via OAM Variable Requests. Access to this information via either OAM events or Variable Requests is controlled by the admin and mode nodes. As mentioned previously, inadequate protection of these variables can result in access to link information and functions.

8.4 Mapping of IEEE 802.3 managed objects

This subclause contains the mapping between the YANG data nodes defined in this clause and the attributes defined in IEEE Std 802.3, Clause 30. Table 8–1 provides the mapping between the *ieee802-ethernet-link-oam* module data nodes and the OAM attributes of IEEE Std 802.3, Clause 30.

Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects and *ieee802-ethernet-link-oam* YANG data nodes

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-link-oam</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
oOAM	aOAMAdminState	30.3.6.1.2	interfaces/interface/ethernet/link-oam	admin	R/W
	dot3OamOperStatus aOAMDiscoveryState aOAMLocalFlagsField aOAMRemoteFlagsField	IEEE Std 802.3.1 30.3.6.1.4 30.3.6.1.10 30.3.6.1.11	interfaces/interface/ethernet/link-oam/discovery-info/local	operational-status	R
	aOAMLocalState	30.3.6.1.14		loopback-mode	R
	aOAMMode	30.3.6.1.3		mode	RW
	aOAMLocalRevision	30.3.6.1.12		revision	R
	aOAMLocalPDUConfiguration	30.3.6.1.8		oammtu	R
	aOAMRemoteConfiguration	30.3.6.1.7	interfaces/interface/ethernet/link-oam/discovery-info/remote/functions-supported	uni-directional-link-fault	R/W
				loopback	R/W
				mib-retrieval	R/W
	aOAMLocalConfiguration	30.3.6.1.6	interfaces/interface/ethernet/link-oam/discovery-info/local/functions-supported/link-monitor	link-monitoring	R/W
	aOAMLocalErrSymPeriodConfig aOAMLocalErrFrameConfig aOAMLocalErrFramePeriodConfig aOAMLocalErrFrameSecsSummaryConfig aOAMLocalErrSymPeriodConfig	30.3.6.1.34 30.3.6.1.36 30.3.6.1.38 30.3.6.1.40 30.3.6.1.42	interfaces/interface/ethernet/link-oam/link-monitor/event-type	threshold-type window threshold	R/W R/W R/W



**Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects
and *ieee802-ethernet-link-oam* YANG data nodes (continued)**

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-link-oam</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aOAMRemoteMACAddress	30.3.6.1.5	interfaces/interface/ethernet/link-oam/discovery-info/remote	mac-address	R
	aOAMRemoteVendorOUI	30.3.6.1.16		vendor-oui	R
	aOAMRemoteVendorSpecificInfo	30.3.6.1.17		vendor-info	R
	aOAMRemoteState	30.3.6.1.15		loopback-mode	R
	aOAMMode	30.3.6.1.3		mode	R
	aOAMRemoteRevision	30.3.6.1.13		revision	R
	aOAMRemotePDUConfiguration	30.3.6.1.9		oammtu	R
	aOAMLocalConfiguration	30.3.6.1.6	interfaces/interface/ethernet/link-oam/discovery-info/local/functions-supported	uni-directional-link-fault	R
				loopback	R
				link-monitoring	R
				mib-retrieval	R
	dot3OamEventLogEntry	IEEE Std 802.3.1	interfaces-state/ interface/ethernet/link-oam/event-log/event-log-entry	index	R
				oui	R
				timestamp	R
				location	R
				event-type	R
				running-total	R
				event-total	R

**Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects
and *ieee802-ethernet-link-oam* YANG data nodes (continued)**

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-link-oam</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aOAMLocalErrSymPeriodEvent aOAMLocalErrFrameEvent aOAMLocalErrFramePeriodConfig aOAMLocalErrFrameSecsSummaryEvent aOAMRemoteErrSymPeriodEvent aOAMRemoteErrFrameEvent aOAMRemoteErrFramePeriodEvent aOAMRemoteErrFrameSecsSummaryEvent	30.3.6.1.35 30.3.6.1.37 30.3.6.1.38 30.3.6.1.41 30.3.6.1.42 30.3.6.1.43 30.3.6.1.44 30.3.6.1.45	interfaces/interface/ethernet/link-oam/event-log/event-log-entry/threshold	threshold-event-type window threshold value	R R R R
	Dot3OamStatsEntry	RFC 4878	interfaces/interface/ethernet/link-oam/statistics	out-information	R
	aOAMInformationTx	30.3.6.1.20			
	aOAMInformationRx	30.3.6.1.21		in-information	R
	aOAMUniqueEventNotificationTx	30.3.6.1.22		out-unique-event-notification	R
	aOAMUniqueEventNotificationRx	30.3.6.1.24		in-unique-event-notification	R
	aOAMDuplicateEventNotificationTx	30.3.6.1.23		out-duplicate-event-notification	R
	aOAMDuplicateEventNotificationRx	30.3.6.1.25		in-duplicate-event-notification	R
	aOAMLoopbackControlTx	30.3.6.1.26		out-loopback-control	R
	aOAMLoopbackControlRx	30.3.6.1.27		in-loopback-control	R
	aOAMVariableRequestTx	30.3.6.1.28		out-variable-request	R
	aOAMVariableRequestRx	30.3.6.1.29		in-variable-request	R
	aOAMVariableResponseTx	30.3.6.1.30		out-variable-response	R
	aOAMVariableResponseRx	30.3.6.1.31		in-variable-response	R
	aOAMOrganizationSpecificTx	30.3.6.1.32		out-org-specific	R

**Table 8–1—Mapping between IEEE Std 802.3, Clause 30 managed objects
and *ieee802-ethernet-link-oam* YANG data nodes (continued)**

IEEE Std 802.3, Clause 30		Reference	Corresponding <i>ieee802-ethernet-link-oam</i> YANG data nodes		
Managed object(s)	Attribute(s)		Container(s)	Data node(s)	R/W
	aOAMOrganizationSpecificRx	30.3.6.1.33		in-org-specific	R
	aOAMUnsupportedCodesTx	30.3.6.1.18		out-unsupported-codes	R
	aOAMUnsupportedCodesRx	30.3.6.1.19		in-unsupported-codes	R
	aFramesLostDueToOAMError	30.3.6.1.46		frames-lost-due-to-oam	R
	aOAMLocalErrSymPeriodEvent, Errored Symbols	30.3.6.1.35		local-error-symbol-period-log-entries	R
	aOAMLocalErrFrameEvent, Errored Frames	30.3.6.1.37		local-error-frame-log-entries	R
	aOAMLocalErrFramePeriodEvent, Errored Frames	30.3.6.1.39		local-error-frame-period-log-entries	R
	aOAMLocalErrFrameSecsSummaryEvent, Errored Frame Seconds Summary	30.3.6.1.41		local-error-frame-second-log-entries	R
	aOAMRemoteErrSymPeriodEvent, Errored Symbols	30.3.6.1.42		remote-error-symbol-period-log-entries	R
	aOAMRemoteErrFrameEvent, Errored Frames	30.3.6.1.43		remote-error-frame-log-entries	R
	aOAMRemoteErrFramePeriodEvent, Errored Frames	30.3.6.1.44		remote-error-frame-period-log-entries	R
	aOAMRemoteErrFrameSecsSummaryEvent, Errored Frame Seconds Summary	30.3.6.1.45		remote-error-frame-second-log-entries	R

8.5 YANG module definition¹⁹

The YANG module tree hierarchy uses terms defined in IETF RFC 8407.

8.5.1 Tree hierarchy

8.5.1.1 ieee802-ethernet-link-oam

module: ieee802-ethernet-link-oam

```
augment /if:interfaces/if:interface:
  +--rw link-oam!
    +---x remote-loopback {remote-loopback-initiate}?
      | +---w input
      | | +---w enable      boolean
      | +---ro output
      |   +--ro success      boolean
      |   +--ro error-message? string
    +---x reset-stats
      | +--ro output
      |   +--ro success      boolean
      |   +--ro error-message? string
    +---n non-threshold-event
      | +-- oui              vendor-oui
      | +-- timestamp        uint64
      | +-- location         event-location
      | +-- event-type       identityref
      | +-- running-total    yang:counter64
      | +-- event-total      yang:counter64
    +---n threshold-event {link-monitoring-local or link-monitoring-remote}?
      | +-- oui              vendor-oui
      | +-- timestamp        uint64
      | +-- location         event-location
      | +-- event-type       identityref
      | +-- running-total    yang:counter64
      | +-- event-total      yang:counter64
```

¹⁹ Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.


```

|   +-- threshold {link-monitoring-local or link-monitoring-remote}?
|       +-- threshold-event-type      threshold-event-enum
|       +-- window                    uint64
|       +-- threshold                 uint64
|       +-- value                     uint64
+--rw admin?                          admin-state
+--rw discovery-info
|   +--rw local
|       | +--ro operational-status      operational-state
|       | +--ro loopback-mode          loopback-status {remote-loopback-initiate or remote-loopback-respond}?
|       | +--rw mode?                  mode
|       | +--rw functions-supported
|       | | +--rw uni-directional-link-fault?  boolean {uni-directional-link-fault}?
|       | | +--rw loopback?                  boolean {remote-loopback-initiate}?
|       | | +--rw link-monitor {link-monitoring-remote or link-monitoring-local}?
|       | | | +--rw link-monitoring?  boolean
|       | | | +--rw event-type* [threshold-type] {link-monitoring-local}?
|       | | |   +--rw threshold-type  threshold-event-enum
|       | | |   +--rw window?         uint64
|       | | |   +--rw threshold?      uint64
|       | | x--rw mib-retrieval?      boolean
|       | | | {remote-mib-retrieval-initiate or remote-mib-retrieval-respond}?
remote-mib-retrieval-respond?
|       | | +--rw data-retrieval?      boolean
|       | | | {remote-data-retrieval-initiate or remote-mib-retrieval-respond}?
remote-mib-retrieval-respond?
|       | +--ro revision?              uint64
|       | +--ro oammtu?                uint16
|       +--ro remote
|           +--ro mac-address?         ieee:mac-address
|           +--ro vendor-oui?          vendor-oui
|           +--ro vendor-info?         uint64
|           +--ro loopback-mode        loopback-status
|           +--ro mode?                mode
|           +--ro functions-supported
|           | +--ro uni-directional-link-fault?  boolean
|           | +--ro loopback?                  boolean
|           | +--ro link-monitoring?            boolean
|           | +--ro mib-retrieval?              boolean
|           +--ro revision?                    uint64

```

```

|      +--ro oammtu?                uint16
+--ro event-log
|  +--ro event-log-entry* [index]
|  |  +--ro index                uint64
|  |  +--ro oui                  vendor-oui
|  |  +--ro timestamp            uint64
|  |  +--ro location             event-location
|  |  +--ro event-type           identityref
|  |  +--ro running-total        yang:counter64
|  |  +--ro event-total          yang:counter64
|  |  +--ro threshold {link-monitoring-local or link-monitoring-remote}?
|  |  |  +--ro threshold-event-type  threshold-event-enum
|  |  |  +--ro window                uint64
|  |  |  +--ro threshold             uint64
|  |  |  +--ro value                 uint64
+--ro statistics
|  +--ro out-information            yang:counter64
|  +--ro in-information            yang:counter64
|  +--ro out-unique-event-notification  yang:counter64 {link-monitoring-local}?
|  +--ro in-unique-event-notification  yang:counter64 {link-monitoring-remote}?
|  +--ro out-duplicate-event-notification  yang:counter64 {link-monitoring-local}?
|  +--ro in-duplicate-event-notification  yang:counter64 {link-monitoring-remote}?
|  +--ro out-loopback-control        yang:counter64 {remote-loopback-initiate}?
|  +--ro in-loopback-control         yang:counter64 {remote-loopback-respond}?
|  +--ro out-variable-request        yang:counter64 {remote-data-retrieval-initiate}?
|  +--ro in-variable-request         yang:counter64 {remote-mib-retrieval-respond}?
|  +--ro out-variable-response       yang:counter64 {remote-mib-retrieval-respond}?
|  +--ro in-variable-response        yang:counter64 {remote-data-retrieval-initiate}?
|  +--ro out-org-specific            yang:counter64
|  +--ro in-org-specific            yang:counter64
|  +--ro out-unsupported-codes       yang:counter64
|  +--ro in-unsupported-codes        yang:counter64
|  +--ro frames-lost-due-to-oam     yang:counter64
|  +--ro local-error-symbol-period-log-entries  yang:counter64
|  +--ro local-error-frame-log-entries  yang:counter64
|  +--ro local-error-frame-period-log-entries  yang:counter64
|  +--ro local-error-frame-second-log-entries  yang:counter64
|  +--ro remote-error-symbol-period-log-entries  yang:counter64 {link-monitoring-remote}?
|  +--ro remote-error-frame-log-entries  yang:counter64 {link-monitoring-remote}?
|  +--ro remote-error-frame-period-log-entries  yang:counter64 {link-monitoring-remote}?

```

```
+--ro remote-error-frame-second-log-entries      yang:counter64 {link-  
monitoring-remote}?
```

8.5.2 YANG module

In the YANG module definition, should any discrepancy between the text of the description for individual YANG nodes in the YANG modules of this clause and the corresponding definition in 8.2 through 8.5 of this clause occur, the definitions in the YANG modules attached to this standard shall take precedence.

An ASCII text version of the YANG module can be found at the following URL:
<https://github.com/YangModels/yang/tree/main/standard/ieee/published/802.3>.²⁰

```
module ieee802-ethernet-link-oam {  
  yang-version 1.1;  
  namespace "urn:ieee:std:802.3:yang:ieee802-ethernet-link-oam";  
  prefix ieee802-link-oam;  
  
  import ieee802-types {  
    prefix ieee;  
    reference  
      "IEEE 802 types";  
  }  
  import ietf-yang-types {  
    prefix yang;  
    reference  
      "IETF RFC 6991";  
  }  
  import iana-if-type {  
    prefix ianaift;  
    reference  
      "http://www.iana.org/assignments/yang-parameters/  
      iana-if-type@2023-01-26.yang";  
  }  
  import ietf-interfaces {  
    prefix if;  
    reference  
      "IETF RFC 8343";  
  }  
  
  organization  
    "IEEE 802.3 Ethernet Working Group  
    Web URL: http://www.ieee802.org/3/";  
  contact  
    "Web URL: http://www.ieee802.org/3/";  
  description  
    "This module contains a collection of YANG definitions  
    for managing the Ethernet Link OAM feature defined by IEEE  
    802.3. It provides functionality roughly equivalent to that of  
    the DOT3-OAM-MIB defined in IETF RFC 4878."  
  
  revision 2025-09-10 {  
    description  
      "Updates under IEEE Std 802.3.2-2025";  
    reference
```

²⁰ Copyright release for YANG modules: Users of this standard may freely reproduce the YANG module contained in this subclause so that it can be used for its intended purpose.

```
    "IEEE Std 802.3-2022, unless dated explicitly";
}

feature uni-directional-link-fault {
  description
    "This feature means the device supports Uni Directional Link
    Fault detection.";
  reference
    "IEEE Std 802.3, 57.1.2 a), 30.3.6.1.6 aOAMLocalConfiguration
    and 30.3.6.1.7 aOAMRemoteConfiguration";
}

feature remote-loopback-initiate {
  description
    "This feature means the device supports being the initiator
    of remote loopback.";
  reference
    "IEEE Std 802.3, 57.1.2 b), 30.3.6.1.6
    aOAMLocalConfiguration";
}

feature remote-loopback-respond {
  description
    "This feature means the device supports responding to remote
    loopback control OAMPDUs received from the peer";
  reference
    "IEEE Std 802.3, 57.1.2 b), 30.3.6.1.7
    aOAMRemoteConfiguration";
}

feature link-monitoring-local {
  description
    "This feature means the device monitors the link at the local
    side and can generate Link Event OAMPDUs to the peer
    device.";
  reference
    "IEEE Std 802.3, 57.1.2 c) 1), 30.3.6.1.6
    aOAMLocalConfiguration, and 30.3.6.1.7
    aOAMRemoteConfiguration";
}

feature link-monitoring-remote {
  description
    "This feature means the device can process Link Event OAMPDUs
    received from the peer device and report itself about this
    event on its own management interface.";
  reference
    "IEEE Std 802.3, 57.1.2 c) 1),
    30.3.6.1.6 aOAMLocalConfiguration,
    and 30.3.6.1.7 aOAMRemoteConfiguration";
}

feature remote-mib-retrieval-initiate {
  status deprecated;
  description
    "remote-mib-retrieval-initiate is deprecated and changed name
    to feature remote-data-retrieval-initiate. This feature
    means the device supports data retrieval from the peer
    device. That is, the device can send Variable Request OAMPDUs
```

```
        to the peer side and process the received Variable Response
        OAMPDUs.";
    reference
        "IEEE Std 802.3, 57.1.2 c) 2),
        30.3.6.1.6 aOAMLocalConfiguration,
        and 30.3.6.1.7 aOAMRemoteConfiguration";
}

feature remote-data-retrieval-initiate {
    description
        "This feature means the device supports data retrieval from
        the peer device. That is, the device can send Variable
        Request OAMPDUs to the peer side and process the received
        Variable Response OAMPDUs.";
    reference
        "IEEE Std 802.3, 57.1.2 c) 2),
        30.3.6.1.6 aOAMLocalConfiguration,
        and 30.3.6.1.7 aOAMRemoteConfiguration";
}

feature remote-mib-retrieval-respond {
    description
        "This feature means the device allows the peer device to
        retrieve data from the managed device. That is, the device
        can process received Variable Requests OAMPDUs and respond
        with Variable Response OAMPDUs.";
    reference
        "IEEE Std 802.3, 57.1.2 c) 2),
        30.3.6.1.6 aOAMLocalConfiguration,
        and 30.3.6.1.7 aOAMRemoteConfiguration";
}

identity event-type {
    description
        "Base identity for all Link OAM event types.";
}

identity threshold-event-type {
    base event-type;
    description
        "Event type for a Link Monitoring threshold event.";
}

identity link-fault-event {
    if-feature "uni-directional-link-fault";
    base event-type;
    description
        "Event type for a uni-directional link fault event.";
    reference
        "IEEE Std 802.3, 57.2.10.1";
}

identity dying-gasp-event {
    base event-type;
    description
        "Event type for a dying gasp event.";
    reference
        "IEEE Std 802.3, 57.2.10.1";
}
```

```
identity critical-event {
  base event-type;
  description
    "Event type for a critical event.";
  reference
    "IEEE Std 802.3, 57.2.10.1";
}

typedef threshold-event-enum {
  type enumeration {
    enum symbol-period-event {
      value 1;
      description
        "Errored symbol period event.";
    }
    enum frame-period-event {
      value 2;
      description
        "Errored frame period event.";
    }
    enum frame-event {
      value 3;
      description
        "Errored frame event";
    }
    enum frame-seconds-event {
      value 4;
      description
        "Errored frame seconds event.";
    }
  }
  description
    "Enumeration of the valid threshold event types.";
  reference
    "IEEE Std 802.3, 57.5.3";
}

typedef mode {
  type enumeration {
    enum passive {
      value 0;
      description
        "Ethernet Link OAM Passive mode.";
    }
    enum active {
      value 1;
      description
        "Ethernet Link OAM Active mode.";
    }
  }
  description
    "Enumeration of the valid modes in which Link OAM may run.";
  reference
    "IEEE Std 802.3, 57.2.9 and 30.3.6.1.3.";
}

typedef event-location {
  type enumeration {
```

```
enum event-location-local {
    value 1;
    description
        "A local event.";
}
enum event-location-remote {
    value 2;
    description
        "A remote event.";
}
}
description
    "The location of the event that caused a log entry.";
}

typedef loopback-status {
    type enumeration {
        enum none {
            value 1;
            description
                "Loopback is not being performed.";
        }
        enum initiating {
            value 2;
            description
                "Initiating master loopback.";
        }
        enum master-loopback {
            value 3;
            description
                "In master loopback mode.";
        }
        enum terminating {
            value 4;
            description
                "Terminating master loopback mode.";
        }
        enum local-loopback {
            value 5;
            description
                "In slave loopback mode.";
        }
        enum unknown {
            value 6;
            description
                "Parser and multiplexer combination unexpected.";
        }
    }
}
description
    "The loopback mode of an OAM interface.";
reference
    "IEEE Std 802.3, 57.2.11";
}

typedef operational-state {
    type enumeration {
        enum disabled {
            value 1;
            description
```

```
        "IEEE Std 802.3 OAM is disabled.";
    }
    enum link-fault {
        value 2;
        description
            "IEEE Std 802.3 OAM has encountered a link fault.";
    }
    enum passive-wait {
        value 3;
        description
            "Passive OAM entity waiting to see if peer is
            OAM capable.";
    }
    enum active-send-local {
        value 4;
        description
            "Active OAM entity trying to determine if peer
            is OAM capable.";
    }
    enum send-local-and-remote {
        value 5;
        description
            "OAM discovered peer but still to accept or
            reject peer configuration.";
    }
    enum send-local-and-remote-ok {
        value 6;
        description
            "OAM peering is allowed by local device.";
    }
    enum peering-locally-rejected {
        value 7;
        description
            "OAM peering rejected by local device.";
    }
    enum peering-remotely-rejected {
        value 8;
        description
            "OAM peering rejected by remote device.";
    }
    enum operational {
        value 9;
        description
            "IEEE Std 802.3 OAM is operational.";
    }
    enum operational-half-duplex {
        value 10;
        description
            "IEEE Std 802.3 OAM is operating in half-duplex mode.";
    }
}
description
    "Operational state of an interface.";
reference
    "IEEE Std 802.3, 30.3.6.1.4,
    30.3.6.1.10, and 30.3.6.1.11";
}

typedef vendor-oui {
```



```
type string {
    length "6";
}
description
    "Organizationally Unique Identifier (OUI) as assigned to
    vendor by the IEEE Registration Authority";
reference
    "IEEE Std 802, 8.2.2";
}

typedef admin-state {
    type enumeration {
        enum enabled {
            value 1;
            description
                "IEEE Std 802.3, Clause 57 OAM is in the
                enabled admin state.";
        }
        enum disabled {
            value 2;
            description
                "IEEE Std 802.3, Clause 57 OAM is in the
                disabled admin state.";
        }
    }
}
description
    "Admin state of the OAM function on an interface.";
reference
    "IEEE Std 802.3, 30.3.6.1.2 and 30.3.6.2";
}

grouping event-details {
    description
        "Nodes describing an event, used in the event log and in
        notifications.";
    reference
        "IETF RFC 4878, Dot3OamEventLogEntry";
    leaf oui {
        type vendor-oui;
        mandatory true;
        description
            "Organizationally Unique Identifier for the device that
            generated the event.";
    }
    leaf timestamp {
        type uint64;
        units "milliseconds";
        mandatory true;
        description
            "Timestamp in milliseconds since Unix epoch for when the
            event occurred.";
    }
    leaf location {
        type event-location;
        mandatory true;
        description
            "Where the event occurred (local or remote).";
    }
    leaf event-type {
```

```
type identityref {
  base event-type;
}
mandatory true;
description
  "Type of event that occurred.";
reference
  "IEEE Std 802.3, 30.3.6.1.10 and 30.3.6.11";
}
leaf running-total {
  type yang:counter64;
  mandatory true;
  description
    "The running total number of errors seen since OAM was
    enabled on the interface. For threshold events, this is
    the total number of times that particular type of error
    (e.g., symbol error) has occurred, which may be greater
    than the number of threshold-crossing event notifications
    of that type generated during that time (which is conveyed
    by the event-total leaf).";
}
leaf event-total {
  type yang:counter64;
  mandatory true;
  description
    "Total number of times this event has occurred since OAM
    was enabled on the interface. For threshold events this is
    the number of events generated of this type (as opposed to
    the total number of errors of that type, which may be
    greater, and is conveyed by the running-total leaf.";
}
}

grouping threshold-event-details {
  description
    "Nodes describing a threshold event, used in the event
    log and in notifications";
  reference
    "IETF RFC 4878, Dot3OamEventLogEntry";
  container threshold {
    when
      "../event-type = 'ieee802-link-oam:threshold-event-type'" {
        description
          "These nodes only apply to threshold event types";
      }
    if-feature
      "link-monitoring-local or link-monitoring-remote";
    description
      "Nodes specific to threshold (link monitoring) events";
    leaf threshold-event-type {
      type threshold-event-enum;
      mandatory true;
      description
        "The type of threshold event";
      reference
        "IEEE Std 802.3, 57.5.3";
    }
    leaf window {
      type uint64;
    }
  }
}
```

```
        mandatory true;
        description
            "Size of the window in which the event was generated.
            Units are dependent on the threshold event type.";
    }
    leaf threshold {
        type uint64;
        mandatory true;
        description
            "Size of the threshold that was breached during the
            window. Units are dependent on the threshold
            event type.";
    }
    leaf value {
        type uint64;
        mandatory true;
        description
            "Breaching value. Units are dependent on the threshold
            event type, and match that of the threshold.";
    }
}

grouping statistics-common {
    description
        "Collection of Link OAM event/packet counters.";
    reference
        "IETF RFC 4878, Dot3OamStatsEntry";
    leaf out-information {
        type yang:counter64;
        mandatory true;
        description
            "Number of information OAMPDUs transmitted.";
        reference
            "IEEE Std 802.3, 30.3.6.1.20";
    }
    leaf in-information {
        type yang:counter64;
        mandatory true;
        description
            "Number of information OAMPDUs received.";
        reference
            "IEEE Std 802.3, 30.3.6.1.21";
    }
    leaf out-unique-event-notification {
        if-feature "link-monitoring-local";
        type yang:counter64;
        mandatory true;
        description
            "Number of unique event notification OAMPDUs transmitted.";
        reference
            "IEEE Std 802.3, 30.3.6.1.22";
    }
    leaf in-unique-event-notification {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
            "Number of unique event notification OAMPDUs received.";
```

```
        reference
            "IEEE Std 802.3, 30.3.6.1.24";
    }
    leaf out-duplicate-event-notification {
        if-feature "link-monitoring-local";
        type yang:counter64;
        mandatory true;
        description
            "Number of duplicate event notification OAMPDUs
            transmitted.";
        reference
            "IEEE Std 802.3, 30.3.6.1.23";
    }
    leaf in-duplicate-event-notification {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
            "Number of duplicate event notification OAMPDUs
            received.";
        reference
            "IEEE Std 802.3, 30.3.6.1.25";
    }
    leaf out-loopback-control {
        if-feature "remote-loopback-initiate";
        type yang:counter64;
        mandatory true;
        description
            "Number of loopback control OAMPDUs transmitted.";
        reference
            "IEEE Std 802.3, 30.3.6.1.26";
    }
    leaf in-loopback-control {
        if-feature "remote-loopback-respond";
        type yang:counter64;
        mandatory true;
        description
            "Number of loopback control OAMPDUs received.";
        reference
            "IEEE Std 802.3, 30.3.6.1.27";
    }
    leaf out-variable-request {
        if-feature "remote-data-retrieval-initiate";
        type yang:counter64;
        mandatory true;
        description
            "Number of variable request OAMPDUs transmitted.";
        reference
            "IEEE Std 802.3, 30.3.6.1.28";
    }
    leaf in-variable-request {
        if-feature "remote-mib-retrieval-respond";
        type yang:counter64;
        mandatory true;
        description
            "Number of variable request OAMPDUs received.";
        reference
            "IEEE Std 802.3, 30.3.6.1.29";
    }
}
```

```
leaf out-variable-response {
  if-feature "remote-mib-retrieval-respond";
  type yang:counter64;
  mandatory true;
  description
    "Number of variable response OAMPDUs transmitted.";
  reference
    "IEEE Std 802.3, 30.3.6.1.30";
}
leaf in-variable-response {
  if-feature "remote-data-retrieval-initiate";
  type yang:counter64;
  mandatory true;
  description
    "Number of variable response OAMPDUs received.";
  reference
    "IEEE Std 802.3, 30.3.6.1.31";
}
leaf out-org-specific {
  type yang:counter64;
  mandatory true;
  description
    "Number of organization specific OAMPDUs transmitted.";
  reference
    "IEEE Std 802.3, 30.3.6.1.32";
}
leaf in-org-specific {
  type yang:counter64;
  mandatory true;
  description
    "Number of organization specific OAMPDUs received.";
  reference
    "IEEE Std 802.3, 30.3.6.1.33";
}
leaf out-unsupported-codes {
  type yang:counter64;
  mandatory true;
  description
    "Number of OAMPDUs with unsupported codes transmitted.";
  reference
    "IEEE Std 802.3, 30.3.6.1.18";
}
leaf in-unsupported-codes {
  type yang:counter64;
  mandatory true;
  description
    "Number of OAMPDUs with unsupported codes received.";
  reference
    "IEEE Std 802.3, 30.3.6.1.19";
}
leaf frames-lost-due-to-oam {
  type yang:counter64;
  mandatory true;
  description
    "A count of the number of frames that were dropped by the
    OAM multiplexer. Since the OAM multiplexer has multiple
    inputs and a single output, there may be cases where
    frames are dropped due to transmit resource contention.
    This counter is incremented whenever a frame is dropped by
```

```
        the OAM layer.";
    reference
        "IEEE Std 802.3, 30.3.6.1.46";
}
}

grouping discovery-remote {
    description
        "Nodes describing the discovery process remote end of a
        link.";
    leaf mode {
        type mode;
        description
            "Mode (passive/active).";
        reference
            "IEEE Std 802.3, 30.3.6.1.3";
    }
    container functions-supported {
        description
            "The Link OAM functions supported by this interface.";
        reference
            "IEEE Std 802.3, 30.3.6.1.7";
        leaf uni-directional-link-fault {
            type boolean;
            description
                "Unidirectional link fault support.";
        }
        leaf loopback {
            type boolean;
            description
                "Remote Loopback support.";
        }
        leaf link-monitoring {
            type boolean;
            description
                "Link monitoring support.";
        }
        leaf mib-retrieval {
            type boolean;
            description
                "MIB variable retrieval support.";
        }
    }
}
leaf revision {
    type uint64;
    config false;
    description
        "Configuration revision.";
    reference
        "IEEE Std 802.3, 30.3.6.1.12 and 30.3.6.1.13";
}
leaf oammtu {
    type uint16;
    units "octets";
    config false;
    description
        "The maximum OAMPDU size for the remote node.
        The peer OAM entities exchange the maximum size they can
        support and negotiate to use the smaller of the two maximum
```

```
        OAMPDU sizes.";
    reference
        "IEEE Std 802.3, 30.3.6.1.8 and 30.3.6.1.9";
}
}

grouping discovery-local {
    description
        "Nodes describing the local end discovery process of a
        link.";
    leaf mode {
        type mode;
        description
            "This object configures the mode of OAM operation as active
            or passive. Active mode provides capabilities to initiate
            monitoring activities with the remote OAM peer entity,
            while passive mode waits for the peer to initiate actions
            with it. Changing this value results in incrementing the
            revision field of locally generated OAMPDUs
            (see IEEE Std 802.3, 30.3.6.1.12) and triggers the
            OAM discovery process if the operational state was
            already 'operational'. The default value is
            implementation dependent.";
        reference
            "IEEE Std 802.3, 30.3.6.1.3";
    }
    container functions-supported {
        description
            "The Link OAM functions supported by this interface.";
        reference
            "IEEE Std 802.3, 30.3.6.1.7";
        leaf uni-directional-link-fault {
            if-feature "uni-directional-link-fault";
            type boolean;
            description
                "Unidirectional link fault support.
                This affects the setting of the 'Unidirectional Support'
                bit in the OAM configuration field put in the
                Information OAMPDU.
                This bit indicates to the peer device that it can send
                OAM PDUs on links that are operating in unidirectional
                mode (traffic flowing in one direction only).";
        }
        leaf loopback {
            if-feature "remote-loopback-initiate";
            type boolean;
            default "true";
            description
                "Remote Loopback support.";
        }
    }
    container link-monitor {
        if-feature
            "link-monitoring-remote or link-monitoring-local";
        description
            "Configure link monitor parameters.";
        reference
            "IEEE Std 802.3, 57.1.2 c)";
        leaf link-monitoring {
            type boolean;
        }
    }
}
```

```
default "true";
description
  "Enable or disable monitoring.
  This affects the setting of the 'Link Events' bit in
  the OAM configuration field put in the Information
  OAMPDU. This bit indicates to the peer device that the
  OAM entity can send and receive Event Notification
  OAMPDUs."
}
list event-type {
  if-feature "link-monitoring-local";
  key "threshold-type";
  description
    "A list containing at most one entry for each of the
    threshold event types. If there is no entry for a
    particular event type, the default values are used for
    both window size and threshold."
  leaf threshold-type {
    type threshold-event-enum;
    description
      "The type of threshold event for which this list
      entry is specifying the configuration."
    reference
      "IEEE Std 802.3, 57.5.3";
  }
  leaf window {
    type uint64;
    description
      "The size of the window to use when monitoring for
      this threshold event. The units, default, and upper
      and lower bounds depend on the threshold type as
      follows:

      Symbol Period:
        Units:    number of symbols
        Default:  number of symbols in one second for the
                  underlying physical layer
        Min:      number of symbols in one second for the
                  underlying physical layer
        Max:      number of symbols in one minute for the
                  underlying physical layer

      Frame:
        Units:    deciseconds
        Default:  1 second
        Min:      1 second
        Max:      1 minute

      Frame Period:
        Units:    number of frames
        Default:  number of minFrameSize frames in one
                  second for the underlying physical layer
        Min:      number of minFrameSize frames in one
                  second for the underlying physical layer
        Max:      number of minFrameSize frames in one
                  minute for the underlying physical layer

      Frame Seconds:
        Units:    deciseconds
```



```

        Default: 60 seconds
        Min:      10 seconds
        Max:      900 seconds";
    reference
        "IEEE Std 802.3, 30.3.6.1.34, 30.3.6.1.36, 30.3.6.1.38,
        and 30.3.6.1.40";
}
leaf threshold {
    type uint64 {
        range "1..max";
    }
    default "1";
    description
        "The threshold value to use when determining whether
        to generate an event given the number of errors that
        occurred in a given window. The units depend on the
        threshold type as follows:

        Symbol Period: number of errored symbols
        Frame:         number of errored frames
        Frame Period:  number of errored frames
        Frame Seconds: number of seconds containing at least
        1 frame error";
    reference
        "IEEE Std 802.3, 30.3.6.1.34, 30.3.6.1.36, 30.3.6.1.38,
        and 30.3.6.1.40";
}
}
}
leaf mib-retrieval {
    if-feature
        "remote-mib-retrieval-initiate or
        remote-mib-retrieval-respond";
        "remote-mib-retrieval-initiate or remote-mib-retrieval-respond";
    type boolean;
    status deprecated;
    description
        "leaf mib-retrieval is deprecated and changed name to
        data-retrieval.
        MIB variable retrieval support.
        This affects the setting of the 'Variable Retrieval' bit
        in the OAM configuration field put in the Information
        OAMPDU. This bit indicates to the peer device that the
        OAM entity can send and receive Variable Request and
        Response OAMPDUs.";
}
leaf data-retrieval {
    if-feature
        "remote-data-retrieval-initiate or
        remote-mib-retrieval-respond";
        "remote-data-retrieval-initiate or remote-mib-retrieval-respond";
    type boolean;
    description
        "Variable retrieval support.
        This affects the setting of the 'Variable Retrieval' bit
        in the OAM configuration field put in the Information
        OAMPDU. This bit indicates to the peer device that the
        OAM entity can send and receive Variable Request and
        Response OAMPDUs.";
}

```

```
    }  
  }  
  leaf revision {  
    type uint64;  
    config false;  
    description  
      "Configuration revision.";  
    reference  
      "IEEE Std 802.3, 30.3.6.1.12 and 30.3.6.1.13";  
  }  
  leaf oammtu {  
    type uint16;  
    units "octets";  
    config false;  
    description  
      "The maximum OAMPDU size for the local node. The peer OAM  
      entities exchange the maximum size they can support and  
      negotiate to use the smaller of the two maximum OAMPDU  
      sizes.";  
    reference  
      "IEEE Std 802.3, 30.3.6.1.8 and 30.3.6.1.9";  
  }  
}  
  
grouping discovery-info {  
  description  
    "Information relating to the discovery process.";  
  container local {  
    description  
      "Properties of the local device.";  
    leaf operational-status {  
      type operational-state;  
      config false;  
      mandatory true;  
      description  
        "Operational status.";  
      reference  
        "IETF RFC 4878, dot3OamOperStatus; IEEE Std 802.3,  
        30.3.6.1.4, 30.3.6.1.10, and 30.3.6.1.11";  
    }  
    leaf loopback-mode {  
      if-feature  
        "remote-loopback-initiate or remote-loopback-respond";  
      type loopback-status;  
      config false;  
      mandatory true;  
      description  
        "The loopback mode the interface is in.";  
      reference  
        "IEEE Std 802.3, 30.3.6.1.14";  
    }  
    uses discovery-local;  
  }  
  container remote {  
    config false;  
    description  
      "Properties of the remote (peer) device.";  
    leaf mac-address {  
      type ieee:mac-address;
```

```
        description
            "Remote MAC address.";
        reference
            "IEEE Std 802.3, 30.3.6.1.5";
    }
    leaf vendor-oui {
        type vendor-oui;
        description
            "Remote vendor OUI.";
        reference
            "IEEE Std 802.3, 30.3.6.1.16";
    }
    leaf vendor-info {
        type uint64;
        description
            "Remote vendor info. The semantics of this value are
            proprietary and specific to the vendor.";
        reference
            "IEEE Std 802.3, 30.3.6.1.17";
    }
    leaf loopback-mode {
        type loopback-status;
        mandatory true;
        description
            "The loopback mode the interface is in.";
        reference
            "IEEE Std 802.3, 30.3.6.1.15";
    }
    uses discovery-remote;
}

augment "/if:interfaces/if:interface" {
    when "derived-from-or-self(if:type, 'ianaift:ethernetCsmacd') or
        derived-from-or-self(if:type, 'ianaift:ptm') " {
        description
            "Augments the interface model with nodes
            specific to Ethernet Link OAM.";
    }
    description
        "Augments Ethernet interface model with nodes
        specific to Ethernet Link OAM.";
    container link-oam {
        presence "Implies Link OAM is configured on the interface.";
        description
            "Interface operational state for Ethernet Link OAM.";
        action remote-loopback {
            if-feature "remote-loopback-initiate";
            description
                "Start/stop remote loopback on the specified interface.";
            reference
                "IEEE Std 802.3, 57.1.2 b)";
            input {
                leaf enable {
                    type boolean;
                    mandatory true;
                    description
                        "Whether to enable or disable remote loopback.";
                }
            }
        }
    }
}
```

```
}
output {
  leaf success {
    type boolean;
    mandatory true;
    description
      "True if the operation was successful,
       false otherwise.";
  }
  leaf error-message {
    type string;
    description
      "If the operation failed, optionally used to
       provide extra details.";
  }
}
}
action reset-stats {
  description
    "Reset Ethernet Link OAM statistics on this interface.";
  output {
    leaf success {
      type boolean;
      mandatory true;
      description
        "True if the operation was successful,
         false otherwise.";
    }
    leaf error-message {
      type string;
      description
        "If the operation failed, optionally used to provide
         extra details.";
    }
  }
}
}
notification non-threshold-event {
  description
    "This notification is sent when a local or remote
     non-threshold crossing event is detected.";
  uses event-details {
    refine "event-type" {
      must ". != 'ieee802-link-oam:threshold-event-type'" {
        description
          "This leaf is not set to
           'threshold-event-type'.";
      }
    }
  }
}
}
notification threshold-event {
  if-feature
    "link-monitoring-local or link-monitoring-remote";
  description
    "This notification is sent when a local or remote
     threshold crossing event is detected.";
  uses event-details {
    refine "event-type" {
      must ". = 'ieee802-link-oam:threshold-event-type'" {
```

```
        description
            "This leaf is set to 'threshold-event-type'.";
    }
}
}
uses threshold-event-details;
}
leaf admin {
    type admin-state;
    default "disabled";
    description
        "This object is used to provision the default
        administrative OAM mode for this interface. This object
        represents the desired state of OAM for this interface.
        It starts in the disabled state until an explicit
        management action or configuration information retained
        by the system causes a transition to the enabled(1)
        state. When enabled(1), Ethernet OAM will attempt to
        operate over this interface. The default value is
        implementation dependent.";
}
container discovery-info {
    description
        "Information relating to the discovery process.";
    uses discovery-info;
}
container event-log {
    config false;
    description
        "List of Ethernet Link OAM event log entries on the
        interface.";
    list event-log-entry {
        key "index";
        description
            "Ethernet Link OAM event log entry.";
        leaf index {
            type uint64;
            description
                "Index of this event in the event log.";
        }
        uses event-details;
        uses threshold-event-details;
    }
}
container statistics {
    config false;
    description
        "Statistics for an 802.3 OAM interface.";
    uses statistics-common;
    leaf local-error-symbol-period-log-entries {
        type yang:counter64;
        mandatory true;
        description
            "Number of local error symbol period log entries.";
    }
    leaf local-error-frame-log-entries {
        type yang:counter64;
        mandatory true;
        description
```

```
        "Number of local error frame log entries.";
    }
    leaf local-error-frame-period-log-entries {
        type yang:counter64;
        mandatory true;
        description
            "Number of local error frame period log entries.";
    }
    leaf local-error-frame-second-log-entries {
        type yang:counter64;
        mandatory true;
        description
            "Number of local error frame second log entries.";
    }
    leaf remote-error-symbol-period-log-entries {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
            "Number of remote error symbol period log entries.";
    }
    leaf remote-error-frame-log-entries {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
            "Number of remote error frame log entries.";
    }
    leaf remote-error-frame-period-log-entries {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
            "Number of remote error frame period log entries.";
    }
    leaf remote-error-frame-second-log-entries {
        if-feature "link-monitoring-remote";
        type yang:counter64;
        mandatory true;
        description
            "Number of remote error frame second log entries.";
    }
}
}
```

NOTE—IEEE Std 802.3, Annex K defines optional alternative terminology for “master” and “slave”.